M.Sc. COMPUTER SCIENCE

CHOICE BASED CREDIT SYSTEM(CBCS)

Curriculum and Syllabus

Regular (2026 – 2027)

Enable | Enlighten | Enrich
KARPAGAM
ACADEMY OF HIGHER EDUCATION
(Deemed to be University)
(Established Under Section 3 of UGC Act, 1956)

DEPARTMENT OF COMPUTER SCIENCE

FACULTY OF ARTS, SCIENCE, COMMERCE AND MANAGEMENT

KARPAGAM ACADEMY OF HIGHER EDUCATION

(Deemed to be University)

(Established Under Section 3 of UGC Act,1956)

(Accredited with A+ Grade by NAAC in the Second Cycle)
Eachanari (Post), Coimbatore – 641 021.
Tamil Nadu, India
PhoneNo.0422-2980011-14FaxNo:0422-2980022-23

Email ID: info@kahedu.edu.in
Web: www.kahedu.edu.in

KARPAGAM ACADEMY OF HIGHER EDUCATION

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FACULTY OF ARTS, SCIENCE, COMMERCE AND MANAGEMENT UNDER– GRADUATE PROGRAMMES

(REGULAR PROGRAMME)

REGULATIONS
(2026)

CHOICE BASED CREDIT SYSTEM (CBCS)

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FACULTY OF ARTS, SCIENCE, COMMERCE AND MANAGEMENT
POST GRADUATE PROGRAMMES
(M.Sc., M. Com, MBA and MCA)

REGULAR MODE
CHOICE BASED CREDIT SYSTEM (CBCS)

REGULATION- 2026 - 2027

The following regulations shall apply to candidates admitted to Postgraduate (PG) Programmes in the Faculty of Arts, Science, Commerce and Management, Karpagam Academy of Higher Education (KAHE) from the Academic Year 2026 -2027 onwards.

1. PROGRAMMES OFFERED, MODE OF STUDY AND ADMISSION REQUIREMENTS

1.1. PG PROGRAMMES OFFERED

The various PG Programmes offered by KAHE are listed in the table below.

S. No.	Programmes Offered
1	M.Sc. Microbiology
2	M.Sc. Biotechnology
3	M.Sc. Physics
4	M.Sc. Chemistry
5	M.Sc. Computer Science
6	M.Com.
7	MBA – Master of Business Administration
8	MCA – Master of Computer Applications

1.2. MODE OF STUDY

All Programmes are offered under Full-Time Regular mode.

1.3. ADMISSION REQUIREMENTS (ELIGIBILITY)

A candidate for admission to the First Semester of the Master's Degree Programme shall be required to have passed an appropriate Degree Examination of Karpagam Academy

of Higher Education or any other University recognized by KAHE as equivalent thereto. Admission shall be granted only to those candidates who fulfil the qualifications prescribed for each programme, as specified in the table below.

QUALIFICATION FOR ADMISSION

S.No.	Programme	Eligibility
1	M.Sc. Microbiology	B.Sc. Microbiology / Applied Microbiology / Industrial Microbiology / Medical Microbiology / Botany / Zoology / Biology / Biotechnology /Industrial Biotechnology / Molecular Biology / Genetic Engineering / Biochemistry / Agriculture / Forestry / Medical Lab Technology / Life Sciences
2	M.Sc. Biotechnology	B.Sc. Degree with Biotechnology/ Industrial Biotechnology/ Biology / Biochemistry / B.Sc Biology with Chemistry Ancillary / B.F.Sc. / Microbiology / Zoology / Botany / Plant Science /Plant Biotechnology / Animal Science /Animal Biotechnology / B.Pharm / Applied Microbiology / Medical Microbiology / Human Genetics / Medical Genetics / Molecular Biology / Genetics / Environmental Science / Environment Biotechnology / Genetics Engineering / Bioinformatics / Plant Biology & Biotechnology / Animal Cell & Biotechnology / Agriculture / B.Tech (Biotech)/ Medical Biochemistry
3	M.Sc. Physics	B.Sc. Physics, B.Sc. Physics (CA) / B.Sc. Applied Sciences
4	M.Sc. Chemistry	B.Sc. Chemistry, B.Sc. Industrial Chemistry, B.Sc. Polymer Chemistry, B.Sc. Applied Sciences
5	M.Sc. Computer Science	B.Sc. Computer Science / B.Sc Computer Science related programmes / BCA / B.Sc. Mathematics/ B.E/B.Tech Computer Science related programmes.
6	M.Com.	B.Com. / B.Com.(CA) /B.Com (PA) / B.Com (BPS) / B.Com (Finance & Insurance) / B.Com. (e-Commerce) / B.Com.(IT) / B.B.M. /B.B.M (CA) / B.B.A/ B.B.A (CA) / B.Com (CS), B.A. Co-operation / Bachelor's Degree in Bank Management/ B.A. Economics / B. Com Financial Analytics/ B. Com International Accounting and Finance
7	MBA - Master of Business Administration	Any 3 year or 4 year UG Degree
8	MCA - Master of Computer Applications	Any 3 year or 4 year UG Degree

2. DURATION OF THE PROGRAMMES

2.1 The minimum and maximum duration for the completion of the PG Programmes shall be as follows:

Programme	Min. No. of Semesters	Max. No. of Semesters
M.Sc., M.Com., MBA., MCA	4	8

2.2 Each semester normally consists of 90 working days or 450 instructional hours of study. Examination shall be conducted at the end of every semester for the respective courses.

3. CHOICE BASED CREDIT SYSTEM

A credit refers to the weightage assigned to each course, as determined by the respective Board of Studies. All PG programmes are offered under the Choice Based Credit System (CBCS) with a minimum of 90 credits and up to a maximum of 94 credits.

4. STRUCTURE OF THE PROGRAMME

Every programme shall have a curriculum and syllabus comprising Major Courses, Major Elective Courses, Self Study, Open Labs, Project Work, Internship, and Open Elective.

4.1 Major Courses

Major Courses shall consist of both theory and practical components pertaining to the respective discipline.

4.2 Major Elective Courses

Major Elective Courses shall be chosen by the students, with the approval of the Head of the Department concerned, from the list of electives prescribed in the curriculum.

4.3 Open Laboratory Courses

Each Open Laboratory Course shall be a **one-credit course** with **two/three contact periods per week**. Such courses shall be offered to equip students with practical knowledge and skills in emerging technologies, industry-demanded competencies, and contemporary developments relevant to their discipline.

4.4 Professional Certification Courses

Each Professional Certification Course shall be a **two-credit course** with **two/three contact periods per week**. Such courses shall be offered by the industry to equip students with practical knowledge and skills in emerging technologies, industry-demanded competencies and contemporary developments relevant to their discipline.

4.5 Self Study Courses

Each Self Study Course shall be a **one-credit course**. The Self-Study Course shall be completed through independent learning by the student under the guidance of a faculty mentor, without formal classroom instruction.

4.6 Project Work

The Project Work shall commence at the beginning of the Fourth Semester and may be carried out in the Department, Industry, or a Research Institute (National/International). The Project Report shall be submitted at the end of the Fourth Semester.

The Head of the Department concerned shall assign a Project Supervisor, who shall monitor the progress of the student throughout the project period.

If a student undertakes the Project Work outside the Department, a faculty member from the Department shall act as the Internal Supervisor, and a teacher/scientist from the host institution shall act as the Co-supervisor. The student shall submit an attendance certificate issued by the organization where the project work was carried out.

A Project Assessment Committee (PAC) shall be constituted, comprising the Head of the Department and two senior faculty members of the Department. The PAC shall schedule the dates for reviews and demonstrations. A total of three reviews shall be conducted as part of the Continuous Internal Assessment. The student shall present the progress and demonstrate the project before the PAC, in the presence of the Supervisor, on the scheduled dates.

4.7 Internship

The student shall undergo an internship of 15 days' duration at the end of the Second Semester. The Internship Report shall be evaluated, and marks shall be awarded in the Third Semester. The student shall earn 2 credits for the internship. The internship shall be assessed internally, and marks shall be awarded out of 100.

4.8 Open Elective

The student may select one Open Elective course offered by other departments in the Third Semester. The student shall earn **2 credits** from this course. The student shall not select a course offered by the parent department.

5. ADVANCED LEARNERS AND ON-DEMAND EXAMINATION

Students who fulfill the following criteria are classified as Advanced Learners:

- Secure a minimum CGPA of 7.5 and maintain at least 80% attendance in every semester;
- Pass all courses in their first attempt.

If a student fails to meet any of the above conditions at any point in time, they will no longer be considered an Advanced Learner.

Advanced Learners are permitted to opt for on-demand examinations for courses in their forthcoming semester(s). Upon prior registration, they may appear for these examinations in

advance and complete their courses ahead of the prescribed duration of study. This enables them to undertake a full-time Research Project, Internship, or Minor Project during the remaining period of their programme. Both Internal and External examinations for such courses will be conducted in the same manner as regular courses.

6. TRANSFER OF CREDITS EARNED THROUGH ONLINE PLATFORM / INTERNATIONAL STUDIES

Students are encouraged to enrol in courses offered through NPTEL, SWAYAM, SWAYAM Plus, and international institutions of higher learning, either in virtual or in-person mode. The equivalence of credits for such courses shall be determined by a *Course Equivalence Committee and Grade Equivalence Committee*.

The committee shall comprise the Dean of the Faculty as Chairperson, the Dean International Relations, the Head of the Department (HoD), and one faculty member nominated by the Vice-Chancellor as members. The recommendations of the committee shall be submitted for ratification and approval by the Academic Council.

7. MEDIUM OF INSTRUCTION

The medium of instruction and examination shall be **English** for all courses, including seminar presentations and project/thesis/dissertation reports.

8 SCHEME OF EXAMINATION

8.1 Evaluation: Evaluation of the course comprises two parts such as the Continuous Internal Assessment (CIA) and the End Semester Examination (ESE).

The theory and practical courses shall carry a maximum of 100 marks, out of which 40 percent of marks are awarded for Continuous Internal Assessment (CIA) and 60 percent of marks for End Semester Examinations (ESE). For Laboratory Integrated Theory (LIT) courses, the evaluation shall also carry a maximum of 100 marks., with 50 marks allotted for Continuous Internal Assessment and 50 marks for the End Semester Examination (ESE), in accordance with the assessment pattern prescribed for the respective LIT course.

8.2 Marks for Project work

S. No.	Programme	Maximum Marks	CIA	ESE
1	M.Sc., M.Com., MBA, MCA	200	80	120

9. FACULTY MENTOR

To assist students in planning their course of study and to provide general academic guidance, the Head of the Department (HoD) shall assign approximately twenty students to a faculty member, who will serve as their Faculty Mentor throughout the period of study.

The Faculty Mentor shall advise the students, monitor their academic performance and

behaviour, and provide periodic counselling as required. The mentor shall also keep the parents informed about the progress of their mentees.

Further, the Faculty Mentor shall periodically display (once every 15 working days) the cumulative attendance details of their mentees on the Notice Board to enable students to track their attendance status and to ensure compliance with Clause 13 of these regulations.

10. ONLINE COURSE COORDINATOR

To assist students in planning their online courses and to provide general orientation, the Head of the Department (HoD) shall nominate an Online Course Coordinator.
The coordinator shall identify suitable courses relevant to the programme from various online platforms and periodically inform students about the available options.

The coordinator shall also orient students on the selection and enrolment process and monitor their participation and progress in the online courses.

11. CLASS COMMITTEE

Every class shall have a class committee consisting of teachers of the class concerned, student representatives (usually 2 boys and 2 girls) and a chairperson who is not teaching the class. It is like the 'Quality Circle' (more commonly used in industries) with the overall goal of improving the teaching- learning process. The functions of the class committee include:

- Solving problems experienced by students in the classroom and in the laboratories.
- Informing the student representatives, the academic schedule including the dates of assessments and the syllabus coverage for each assessment.
- Informing the student representatives the details of Regulations regarding weightage used for each assessment. In the case of practical courses (laboratory/ project work/seminar etc.) the breakup of marks for each experiment/exercise/module of work, should be clearly discussed in the class committee meeting and informed to the students.
- Analyzing the performance of the students of the class after each test and finding the ways and means of solving problems, if any.
- Identifying the slow learners, if any, and requesting the teachers concerned to provide some additional academic support.
- The class committee shall be constituted within the first week of each semester.
- The Chairperson of the class committee may convene the meeting of the class committee.
- The chairperson is required to prepare the minutes of every meeting, submit the same to the Dean through HoD within two days of the meeting and arrange to circulate it among the students and teachers concerned. If there are some points in the minutes requiring action by the Executive Council, the same shall be brought to the notice of the Registrar/VC by the HoD through Dean.
- The first meeting of the class committee shall be held within one week from the date of commencement of the semester, in order to inform the students about the nature and

weightage of assessments within the framework of the Regulations. Two or three subsequent meetings may be held in a semester at suitable intervals.

- During these meetings the student members representing the entire class, shall meaningfully interact and express the opinions and suggestions of the other students of the class in order to improve the effectiveness of the teaching-learning process.

12. COURSE COMMITTEE FOR COMMON COURSES

Each common course offered to more than one programme or department shall have a "Course Committee" comprising all the teachers handling the common course with one of them nominated as Course Coordinator.

The nomination of the course coordinator shall be made by the Head of the Department / Dean depending upon whether all the teachers handling the common course belong to a single department or to various other departments.

The 'Course Committee' shall meet at least three times during the course in order to arrive at a common scheme of evaluation for the tests to ensure a uniform evaluation of the tests. If feasible, the course committee shall prepare a common question paper for the Internal Assessment Test(s).
Course codes for common courses shall be suffixed with the letter "**G**".

13. ATTENDANCE REQUIREMENTS FOR APPEARING FOR THE END SEMESTER EXAMINATION

a. Every student is expected to attend all classes and earn 100% attendance. However, a student shall secure not less than **75% attendance in each course**, taking into account the number of instructional hours prescribed for that course in the curriculum, to become eligible to appear for the End Semester Examination in that course.

b. A student who has secured attendance between **65.00% and 74.4%** (both included) in **a course or in all the courses of a semester**, due to medical reasons such as hospitalization, accident, or specific illness, may be granted exemption from the prescribed minimum attendance requirement and permitted to appear for the End Semester Examination, subject to the recommendation of the Head of the Department concerned and the approval of the Dean. The Head of the Department shall verify and certify the genuineness of the case before forwarding the recommendation. The student shall submit an undertaking jointly signed by the parent/guardian assuring that such a situation will not recur. This concession shall be granted only once during the entire period of study.

c. A student who has secured **less than 75% attendance in a course without a valid reason** shall not be permitted to appear for the End Semester Examination in that course. However, the student shall be promoted to the next semester and shall be required to **earn a minimum of 75% attendance in the concerned course** by attending special classes conducted during the summer term or through any other mode prescribed by the Institution. Upon satisfying the attendance requirement, the student shall become eligible to appear for the End Semester Examination of that course during a subsequent examination session.

d. A student who has secured **less than 55% attendance in more than 50% of the courses offered in a semester**, irrespective of the reason, shall not be permitted to appear for the End Semester Examinations of that semester. Such a student shall be required to redo the semester along with the subsequent batch of students, subject to the approval of the Dean concerned. However, the student shall be permitted to appear for arrear examinations, if any.

14. PROCEDURE FOR AWARDING MARKS FOR INTERNAL ASSESSMENT

14.1 Attendance and Assessment: Every Faculty is required to maintain an **Attendance and Assessment Record (Log book)** which consists of attendance marked in each theory or practical or project work class, the test marks and the record of class work (topic covered), separately for each course

14.2 Continuous Internal Assessment (CIA): The performance of students in each course will be continuously assessed by the respective teachers as per the guidelines given below:

(a) Theory Courses

S. No.	Category	Maximum Marks
1.	Internal Component	10
2.	Test – I (2 ½ Units)	15
3.	Test – II (2 ½ Units)	15
Total		**40**

**The internal assessment components for each course may be selected from the pool given below:*

Internal Components Pool: Assignments, solution to application-oriented problems using software, solving of GATE questions, projects, prototype development, flipped classroom activities, seminars, group activities, report writing, role play, quizzes, presentations, practical works, case studies, literature review, article review, etc.

(b) Practical Courses

S.No.	Category	Maximum Marks
1.	Observation	10
2.	Continuous Assessment of Experiments *	10
3.	Model Examination	15
4.	Viva – Voce	5
	Total	**40**

** Each practical exercise/experiment shall be evaluated on the day of practice, and 10 marks shall be awarded upon successful completion. The total marks will be scaled down to 10 at the end of the semester.*

Every practical Exercise / Experiment shall be evaluated based on the conduct of Exercise/ Experiment and records need to be maintained.

(c) Laboratory Integrated Theory Courses (LIT)

Weightage of Continuous Internal Assessment and End Semester Examination marks will be 50% each. The distribution of marks for the theory and laboratory components in the Continuous Internal Assessments.

Type I - Lab Integrated Theory (LIT) Courses (L-T-P-C: 3-0-2-4)

Theory Component

S. No.	Category	Maximum Marks
1.	Test – I (2 ½ Units)	15
3.	Test – II (2 ½ Units)	15
Total		**30**

Practical Component

S.No	Category	Maximum Marks
1	Observation	5
2	Continuous Assessment of Experiments *	10
3	Model Exam	5
Total		**20**

* *Each practical exercise/experiment shall be evaluated on the day of practice, and 10 marks shall be awarded upon successful completion. The total marks will be scaled down to 10 at the end of the semester.*

The End-Semester Evaluation for **Type I – Lab Integrated Theory (LIT) Courses** shall consist of both **Theory** and **Practical** components. The Theory component shall be evaluated through an End-Semester Examination conducted for 100 marks, and the marks obtained shall be scaled down to 30 marks. The Practical component shall be evaluated through an End-Semester Practical Examination conducted for 50 marks, and the marks obtained shall be scaled down to 20 marks.

Thus, the External Evaluation for Type I – Lab Integrated Theory (LIT) Courses shall be for a total of 50 marks, comprising 30 marks from the Theory component and 20 marks from the Practical component.

Type II - Lab Integrated Theory (LIT) Courses (L-T-P-C: 1-0-4-3)

Continuous Internal Assessment

Theory & Laboratory Component – 50 Marks		
S. No.	**Category**	**Marks**
1	Theory - Test	20
2	Observation	10
3	Continuous Assessment of Experiments *	15
4	Mini Project / Viva Voce	5
	Total	**50**

* *Each practical exercise/experiment shall be evaluated on the day of practice, and 10 marks shall be awarded upon successful completion. The total marks will be scaled down to 15 at the end of the semester.*

14.3 Pattern of Test Question Paper Theory Courses:

Maximum Marks : 60* **Duration:** 2 ½ Hours

Section	Description
Part – A	Short Answers - Answer ALL the Questions **(6 x 2 = 12 Marks)**
Part - B	Long Answers - 2 five-marks questions 'either – or' type Answer ALL the Questions **(3 x 6 = 18 Marks)**
Part - C	Essay type Answers– 3 ten-mark questions 'either – or 'type Answer ALL the Questions **(3 x 10 = 30 Marks)**

**The 60 Marks is scaled down to 15 Marks.*

15. ESE EXAMINATIONS

15.1 End Semester Examination (ESE): ESE will be conducted at the end of each semester for each course. The question paper is for a maximum of 100 marks.

15.2 Pattern of ESE Question Paper

Theory Courses:

Maximum Marks: 100* **Duration: 3 Hours**

Section	Marks
Part – A	Answer ALL the Questions (10 x 2 = 20 Marks)
Part - B	Answer ALL the Questions (5 x 6 = 30 Marks) ('either – or' type)
Part - C	Answer ALL the Questions (5 x 10 = 50 Marks) ('either – or' type)

*** The 100 Marks is scaled down to 60 Marks.*

Practical Courses: There shall be combined valuation by the Internal and External examiners. The pattern of distribution of marks shall be as given below.

S. No.	Category	Maximum Marks
1.	Experiment - 1	20
2.	Experiment – 2	20
3.	Viva-Voce	10
4	Record	10
Total		**60**

Type I - Lab Integrated Theory (LIT) Courses (L-T-P-C: 3-0-2-4)

S. No.	Category	Maximum Marks
1.	Experiment - 1	20
2.	Experiment – 2	20
3.	Viva-Voce	05
4.	Record	05
Total		**50**

**For Laboratory Integrated Theory Courses 50 Marks will be scaled down to 20 marks.*

Type II - Lab Integrated Theory (LIT) Courses (L-T-P-C: 1-0-4-3)

S.No.	Category	Maximum Marks
1.	Record	10
2.	Experiment - I	40
3.	Experiment – II	40
4.	Viva – Voce	10
	Total	**100**

**For Laboratory Integrated Theory Courses 100 Marks will be scaled down to 50 marks.*

Self Study Course: There valuation by the Internal and External examiners. The pattern of distribution of marks shall be as given below.

S. No.	Category	Maximum Marks
1.	Experiment - 1	15
2.	Experiment – 2	15
3.	Viva-Voce	10
4	Record	10
Total		**50**

Record Notebooks for Practical Examination

Candidates taking the practical examination should submit a prescribed Bonafide Record Notebook for the practical examination; failing which the candidate will not be permitted to take the practical examination.

In case of failures in Practical Examination, the marks awarded for the Record at the time of first appearance of the Practical Examination shall remain the same at the subsequent appearance also by the candidate.

Professional Certification Course

A student who successfully completes a Professional Certification Course approved by the Institution shall be awarded two credits upon submission of the course completion certificate

and successful verification by the department.

15.3. Evaluation of Project Work

15.3.1 The Project shall carry maximum marks as specified in Clause 8.2. The End Semester Examination (ESE) shall be evaluated jointly by the Internal and External Examiners.

The distribution of marks for the Continuous Internal Assessment (CIA) is as given below:

Maximum Marks: 80

S. No.	Category	Maximum Marks
1.	Problem Selection	20
2.	Progress of the work (3 reviews X 10 marks)	30
3.	Presentation of the work (3 reviews X 10 marks)	30
Total		**80**

The distribution of marks for the End Semester Examination is as given below:
Maximum Marks: 120*

S. No.	Category	Maximum Marks
1.	Project Report	60
2.	Project Presentation	40
3.	Viva Voce	20
Total		**120***

***Combined valuation of Internal and External Examiners.*

15.3.2 The Project Report, prepared in accordance with the approved guidelines and duly signed by the Supervisor(s), shall be submitted to the Head of the Department.

Guidelines for the Preparation of the Project Report:
The Project Report shall include the following components:
a. Cover Page
b. Bonafide Certificate
c. Declaration
d. Acknowledgement
e. Table of Contents
f. Chapters, comprising:
- Introduction
- Aim and Objectives
- Materials and Methods (Methodology)
- Results (Analysis of Data) and Discussion (Interpretation)
- Summary
- References

15.3.3 The evaluation of the project shall be based on the submitted project report and a viva voce examination conducted by a panel consisting of the supervisor, who shall serve as the Internal Examiner, and an External Examiner appointed by the Controller of Examinations. In the absence of the supervisor, the HoD shall act as the Internal Examiner.

15.3.4 If a candidate fails to submit the project report on or before the date specified by the Examination Section, the candidate shall be deemed to have failed in the Project Work and shall re-enrol for the same in a subsequent semester. If a candidate fails in the viva voce examination, he/she shall resubmit the project report within 30 days from the date of declaration of results. The resubmitted report shall be evaluated in the subsequent semester.

15.3.5 A copy of the approved Project Report, after the successful completion of the Viva-Voce Examination, shall be submitted to and retained in the KAHE Library.

16. PASSING REQUIREMENTS

16.1 Passing Minimum

A candidate needs to secure a minimum of **50% of the maximum marks** in the **Continuous Internal Assessment (CIA)** and **50% of the maximum marks** in the **End Semester Examination (ESE)** separately to be declared to have passed the course.

16.2 If a candidate fails to secure a pass in a particular course (either CIA or ESE or Both) as per clause 16.1, it is mandatory that the candidate has to register and reappear for the examination in that course during the subsequent semester when examination is conducted for the same till, he / she receives pass both in CIA and ESE (vide Clause 2.1).

16.3 The CIA marks secured by the candidate in the first passed attempt shall be retained by the Office of the Controller of Examinations and shall remain valid for all subsequent attempts till the candidate secures a pass in the ESE.

16.4 Candidates who fail the Internal Assessment shall be permitted to reappear for the Internal Assessment in the subsequent semesters by taking the prescribed tests and resubmitting the required internal assessment components.

The distribution of marks for the re-assessment shall be as follows:
Theory Courses :

S. No.	Category	Maximum Marks
1.	Internal Component	10
2.	Test*	30
	Total	**40**

** Test shall be conducted in the CIA pattern for 100 marks and scaled down to 30 marks.*

Type I - Lab Integrated Theory (LIT) Courses (L-T-P-C: 3-0-2-4)

S. No.	Category	Maximum Marks
1.	Test*	30
2.	Observation	5
3.	Model Examination	15
	Total	**50**

** Test shall be conducted in the ESE pattern for 60 marks and scaled down to 30 marks.*

Type II - Lab Integrated Theory (LIT) Courses (L-T-P-C: 1-0-4-3)

Theory & Laboratory Component – 50 Marks		
S. No.	**Category**	**Marks**
1	Test *	20
2	Observation	10
3	Model Examination	20
	Total	**50**

** Test shall be conducted in the CIA pattern for 60 marks and scaled down to 20 marks.*

The re-assessment shall be completed within 6 weeks from the date of reopening of the subsequent semester.

16.5 A student who is absent for the End Semester Examination (ESE) in a course, practical, or project work after having registered for the same shall be marked as **Absent (AAA)** in that examination.

17. ONLINE EXAMINATIONS

The students who are going for Project / Internship / Coursework at National level are permitted to write their CIA test through Online Mode and ESE in Offline/Online mode. When they go for an International Project / Internship / Coursework, both the CIA and ESE shall be conducted through online mode.

18. IMPROVEMENT OF MARKS IN THE COURSES ALREADY PASSED

Students who wish to improve the marks secured in a course passed in their first attempt may reappear once, only for the End Semester Examination (ESE), in the subsequent semester. **The improved marks shall be considered for classification but not for ranking.** If no improvement is achieved, the marks awarded earlier shall
remain unchanged.

19.AWARD OF LETTER GRADES

All assessments of a course will be done on an absolute marks' basis. However, for the purpose of reporting the performance of a student, letter grades, each carrying certain number of points, will be awarded as per the range of total marks (out of 100) obtained by the student in each course as detailed below:

Letter grade	Marks Range	Grade Point	Description
O	91-100	10	OUTSTANDING
A+	81 - 90	9	EXCELLENT
A	71 - 80	8	VERY GOOD
B+	61 - 70	7	GOOD
B	56 - 60	6	AVERAGE
C	50 - 55	5	PASS
RA	Below 50	-	REAPPEARANCE
AAA	-	-	ABSENT

20. GRADE SHEET

After the declaration of the results, Grade Sheets will be issued to each student which will contain the following details:

i. The list of courses enrolled during the semester and the grade scored.

ii. The Grade Point Average (**GPA**) for the semester and

iii. The Cumulative Grade Point Average (**CGPA**) of all courses enrolled from first semester onwards.

GPA of a Semester and CGPA of a programme will be calculated as follows.

$$GPA = \frac{\Sigma (Ci \times GPi)}{\Sigma Ci} \quad \frac{\text{Sum of (Credits for each course * Grade Points)}}{\text{Total Credits Registered in Semester}}$$

$$CGPA = \frac{\Sigma (\text{GPA of each Semester} * \text{Total Credits in that Semester})}{\text{Total Credits across ALL Semesters}}$$

were,

Ci is the credit fixed for the course 'i' in any semester

GPi is the grade point obtained for the course 'i' in any semester

'n' refers to the Semester in which such courses are credited.

Note: RA grade will be excluded for calculating **GPA** and **CGPA**.

21. REVALUATION

A candidate may apply for revaluation or re-totalling of his/her semester examination answer scripts (**theory courses only**) within two weeks from the date of declaration of results, on payment of the prescribed fee. The application, in the prescribed format, shall be submitted to the Controller of Examinations through the Head of the Department (HoD).

A candidate may apply for revaluation for a maximum of 5 courses at a time. The Controller of Examinations shall arrange for the revaluation, and the results shall be communicated to the candidate through the concerned HoD. Revaluation shall not be permitted for Supplementary Examinations.

22.TRANSPARENCY AND GRIEVANCE COMMITTEE

As per Clause 21, revaluation and re-totalling are permitted upon representation. A candidate may obtain a photocopy of the evaluated answer script on payment of the prescribed fee.

- If the candidate has any grievance, it may be represented to the Grievance Committee, which shall consist of:
- The Dean of the Faculty (if the Dean is also the HoD, a Dean from another Faculty nominated by the Vice Chancellor
- The Head of the Department concerned
- The faculty member handling the course
- A Dean from another discipline nominated by the Vice Chancellor, and
- The Controller of Examinations.

If the Committee is satisfied that the grievance is genuine, the answer script may be sent for external evaluation. The marks awarded by the External Examiner shall be final. The candidate shall pay the prescribed fee for the same.

23. ELIGIBILITY FOR THE AWARD OF THE DEGREE

A student shall be declared eligible for the conferment of the Degree if he/she satisfies the following requirements:

- Successfully completes all components prescribed and earns the minimum required credits as specified in the curriculum of the respective programme within the stipulated period (vide Clause 4).
- Has no pending disciplinary enquiry or action against him/her.
- The award of the Degree is approved by the Executive Council.

24. CLASSIFICATION OF THE DEGREE AWARDED

24.1 Students who qualify for the award of the Degree (vide Clause 23), having passed all courses in their first appearance within the specified minimum number of semesters and securing a **CGPA of not less than 8.0**, shall be declared to have passed with **First Class with Distinction**.

24.2 Students who qualify for the award of the Degree (vide Clause 23), having passed all courses within the maximum number of semesters specified (vide Clause 2.1) and securing a **CGPA of not less than 6.5**, shall be declared to have passed with **First Class.**

24.3 Students who qualify for the award of the Degree (vide Clause 23) but do not fall under Clauses 24.1 and 24.2 shall be declared to have passed with **Second Class**.

25. RANKING

Students who qualify for the PG Degree Programme by passing all examinations in the first attempt and within the minimum prescribed duration of the programme (from Semester I through Semester IV) shall be eligible for ranking.

Ranking shall be limited to 10% of the total number of candidates who have qualified in the particular programme, subject to a maximum of 10 ranks.

26. SUPPLEMENTARY EXAMINATION

Supplementary Examination will be conducted only for the final semester students within ten days from the date of publication of results for students who have failed in one theory course only. This is applicable only for the ESE component of the course which the student failed. Such students shall apply with prescribed fee to the Controller of Examinations within the stipulated time.

27. DISCIPLINE

27.1 If a student indulges in malpractice in any of the Internal / External Examinations he / she shall be liable for punitive action as prescribed by the KAHE from time to time.

27.2 Every student is required to observe discipline and decorous behavior both inside and outside the campus and not to indulge in any activity which will tend to bring down the prestige of the KAHE. The erring students will be referred to the disciplinary committee constituted by the KAHE, to enquire into acts of indiscipline and recommend the disciplinary action to be taken.

27.3 A student who is placed under suspension for involvement in any disciplinary misconduct shall be required to complete an additional 30 hours of service under NSS / Yoga as a prerequisite for revocation of suspension.

28. REVISION OF REGULATION AND CURRICULUM

Karpagam Academy of Higher Education may from time-to-time revise, amend or change the Regulations, Scheme of Examinations and Syllabi, if found necessary.

Karpagam Innovation and Incubation Council (KIIC)

(A Section 8 Company)

Based on the 2019 National Innovation and Startup Policy and the 2019–2023 Tamil Nadu Startup Policy, KIIC has recommended to the KAHE students who are affiliated with the KIIC that it be incorporated in the university Program Regulations 2023-24 and implement from this academic year.

Norms to Student Start-Ups

a) Any (UG/PG /(Ph.D.) Research scholars, student, right from the first year of their program is allowed to set a startup (or) work part time/ full time in a startup or work as intern in a startup

b) Any (UG/PG / (Ph.D.) Research scholars) student right from the first year of their program is allowed to earn credit for working on Innovative prototypes/business Models/ Pre incubation (case to case basis).

c) Start Up activities will be evaluated based on the guidelines being given by the expert committee of the KIIC

d) Student Entrepreneurs may use the address of incubation center (KIIC) to register their venture while studying in KAHE.

e) Students engaged in startups affiliated with the KIIC or those who work for them may be exempted from KAHE's attendance requirements for academic courses under current regulations, up to a maximum of 30% attendance per semester, including claims for ODs and medical emergencies Potential Students who have been incubated at KIIC may be permitted to take their University semester exams even if their attendance is below the minimum acceptable percentage, with the proper authorization from the head of the institution.
(On case-to-case basis depends upon the applicability strength, societal benefits and quality of the Innovation and Subsequent engagement of the students with the/ her business)

f) Any Students Innovators/entrepreneurs are allowed to opt their startup in place mini project /major project, /seminar and summer training etc. (In plant training, Internship, value added Course.). The area in which the student wishes to launch a Startup may be interdisciplinary or multidisciplinary.

g) Student's startups are to be evaluated by Expert committee, formed by KIIC and KAHE

Guide lines to award Credits/ Marks to a Student startup

Student's startup stages are divided into five phases and these startup phases can be considered equally in place of the course title as mentioned below with the same credits allotted to the course title in a University curriculum.

Sl. No.	Description/Startup phases	In place of the Subject / Course title	Grades/Credits /Marks
1	Idea stage/Problem Identification	Seminar	Same Marks/Credits can be awarded that are listed in the course title's curriculum for the respective startup phases.
2	Proof of Concept (POC) /Solution development	In-plant training /Internship	
3	Product Development (Lab scale) /Prototype Model/ Company Registered	Mini Project/ Value added Course	
4	Validation/Testing	Main Project phase I	
5	Business Model/Ready for Commercialization/Implementation	Main Project phase II,	

PREAMBLE

The Master of Computer Science programme is designed to provide students with advanced and comprehensive knowledge in Computer Science, enabling them to understand, analyze and address complex computational and technological challenges. The curriculum integrates advanced theoretical knowledge with practical applications, research-oriented learning, project-based experiences and independent learning, thereby developing the ability to design, evaluate and implement innovative solutions to real-world problems. Aligned with the principles of the National Education Policy (NEP) 2020 and Outcome-Based Education (OBE), the programme fosters advanced critical thinking, problem-solving, analytical reasoning, scientific reasoning, effective communication, research aptitude, teamwork, reflective thinking, digital literacy, ethical awareness, leadership, multicultural competence, self-directed learning and lifelong learning. The programme equips graduates with the knowledge, skills and professional competencies required to conduct research, contribute to innovation, communicate effectively, collaborate in diverse environments, make ethical and informed decisions and adapt to emerging developments in Computer Science. It prepares graduates to contribute responsibly and effectively to industry, academia, research, innovation and society while pursuing continuous professional and intellectual growth.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO I: Graduates will acquire an in-depth understanding of advanced computer science concepts, enabling them to design, develop, and implement innovative solutions to complex computational problems.

PEO II: Graduates will engage in cutting-edge research, contribute to advancements in computer science, and apply innovative techniques to emerging challenges in the field.

PEO III: Graduates will exhibit leadership and ethical decision-making in professional environments, contributing responsibly to technology development and societal impact.

PEO IV: Graduates will demonstrate a commitment to continuous learning, staying current with evolving technologies, and adapting to new trends in computer science and related fields.

PROGRAM OUTCOMES (POs):

By the end of the programme, our graduates will

1. **Disciplinary knowledge:** Possess a profound understanding of the foundational concepts, theories, methodologies, and practices within the discipline of Computer Science.
2. **Communication Skills:** Emerge as confident communicators capable of articulating complex concepts, advocating for their viewpoints, and engaging in meaningful discourse to address contemporary issues and drive positive change.
3. **Critical thinking:** Master advanced critical thinking skills, analyzing complex issues, and solving problems through evidence-based decision-making.
4. **Problem solving:** Excel in problem-solving, applying analytical techniques and creative thinking to address complex challenges in the field of Computer Science.
5. **Analytical reasoning:** Emerge as adept analytical thinkers, equipped to tackle challenging problems, make informed decisions, and contribute to the advancement of knowledge in the field of Computer Science.
6. **Research-related skills:** Demonstrate proficiency in data analysis, critical appraisal, and ethical research practices, contributing original insights to the advancements in Computer Science.
7. **Cooperation/Team work:** Develop strong cooperation and teamwork skills, collaborating effectively with diverse peers to achieve common goals.
8. **Scientific reasoning:** Excel in scientific reasoning, applying logic and evidence to analyze phenomena, solve problems, and advance knowledge in the area of Computer Science.
9. **Reflective thinking:** Master reflective thinking, fostering self-awareness and insight to evaluate experiences, perspectives, and actions critically.
10. **Information/digital literacy:** Excel information and digital literacy, adeptly locating, evaluating, and ethically using diverse sources of information
11. **Self-directed learning:** Be empowered individuals to take ownership of their educational journey, fostering autonomy, critical thinking, and adaptability.
12. **Multicultural competence:** Be enabled to effectively navigate diverse contexts, fostering empathy, understanding, and collaboration across cultures.
13. **Moral and ethical awareness/reasoning:** Possess the capacity to critically analyze ethical issues from various perspectives and apply ethical principles to real-world situations.
14. **Leadership readiness/qualities:** Develop the skills and attributes necessary to effectively lead and inspire others.
15. **Lifelong learning:** Foster a commitment to lifelong learning by cultivating curiosity, critical thinking, and a growth mindset.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO 1: Graduates will be proficient in conducting advanced research in computer science, contributing to the development of new theories, algorithms, or technologies. They will demonstrate the ability to apply research findings to solve complex real-world problems in innovative ways.

PSO 2: Graduates will possess specialized expertise in a focused area of computer science, such as artificial intelligence, cybersecurity, data science, or software engineering. They will demonstrate leadership qualities, guiding teams and projects towards successful outcomes in their specialized field.

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MAPPING of PEOs and POs

PEOs / POs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	PSO1	PSO2
PEO I	X	-	X	X	X	-	-	X	-	X	-	-	-	-	X	X	X
PEO II	X	X	X	X	X	X	-	X	X	X	X	-	-	-	X	X	X
PEO III	-	X	-	-	-	X	X	-	X	-	-	X	X	X	-	-	X
PEO IV	X	-	X	-	X	-	X	-	X	X	X	X	X	X	X	X	X

DEPARTMENT OF COMPUTER SCIENCE
FACULTY OF ARTS, SCIENCE, COMMERCE, MANAGEMENT
PG PROGRAM (CBCS) – M. Sc. COMPUTER SCIENCE
(2026 – 2027 Batch and onwards)

Semester I											
Code No.	**Course**	**L**	**T**	**P**	**C**	**Hours/ Week**	**Maximum Marks**			**Category**	**Page No.**
							CIA	**ES**	**Total**		
Theory											
26CSP101	Cloud Computing	4	0	0	4	4	40	60	100	Major	1
26CSP102A/ 26CSP102B/ 26CSP102C	Elective - I	4	0	0	4	4	40	60	100	Major	4
Theory Cum Laboratory Courses											
26CSP141	Python Programming	1	0	4	3	5	50	50	100	Major	13
26CSP142	Data Mining Techniques and Tools	3	0	2	4	5	50	50	100	Major	17
26CSP143	Advanced Data Structures and Analysis of Algorithms	3	0	2	4	5	50	50	100	Major	21
26CSP144	Advanced Java Programming	3	0	2	4	5	50	50	100	Major	24
Mandatory											
	Journal Paper Analysis and Presentation	2	0	0	0	2	-	-	-	-	27
Total		20	0	10	23	30	280	320	600		

Semester II											
Code No.	**Course**	**L**	**T**	**P**	**C**	**Hours/ Week**	**Maximum Marks**			**Category**	**Page No.**
							CIA	**ES**	**Total**		
Theory											
26CSP201/ noc26_cs161*	Internet of Things / Introduction to Internet of Things*	4	0	0	4	4	40	60	100	Major	28
26CSP202A/ 26CSP202B/ 26CSP202C	Elective - II	4	0	0	4	4	40	60	100	Major	34
Theory Cum Laboratory Courses											
26CSP241	Web Development using Angular JS & MongoDB	1	0	4	3	5	50	50	100	Major	43
26CSP242	Machine Learning Operations	3	0	2	4	5	50	50	100	Major	47
26CSP243	Cyber Security	3	0	2	4	5	50	50	100	Major	50
Two Credit Course											
26CSP203	Community Engagement and Social Responsibility	2	0	0	2	2	40	60	100	Major	53
26CSP244	Professional Certification – UI/UX Design	1	0	2	2	3	-	-	-	Major	56
One Credit Course											
26CSP211	Data Analytics using Power BI	0	0	0	1	0	0	50	50	Major	59
Mandatory											
	Journal Paper Analysis and Presentation	2	0	0	0	2	-	-	-	-	61
*** Course Code and Course Name subject to change by NPTEL / SWAYAM**											
Total		20	0	10	24	30	270	380	650		

DEPARTMENT OF COMPUTER SCIENCE
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Semester III											
Code No.	**Course**	**L**	**T**	**P**	**C**	**Hours/ Week**	**Maximum Marks**			**Category**	**Page No.**
							CIA	**ES**	**Total**		
Theory											
26CSP301	Blockchain Technology	4	0	0	4	4	40	60	100	Major	62
26CSP302/ noc26_ee178*	Digital Image Processing and Applications/ Digital Image Processing*	4	0	0	4	4	40	60	100	Major	65
26CSP303A / 26CSP303B / 26CSP303C	Elective - III	4	0	0	4	4	40	60	100	Major	71
26XXXOE301	Open Elective	3	0	0	3	3	40	60	100	Open Elective	79
Theory Cum Laboratory Courses											
26CSP341	Deep Learning	1	0	4	3	5	50	50	100	Major	102
26CSP342	Software Testing and Quality Assurance	3	0	2	4	5	50	50	100	Major	105
26CSP343	Natural Language Processing	3	0	2	4	5	50	50	100	Major	108
Two Credit Courses											
26CSP391	Internship	0	0	0	2	0	100	-	100	Internship	111
*** Course Code and Course Name subject to change by NPTEL / SWAYAM**											
Total		22	0	8	28	30	410	390	800		

Semester IV											
Code No.	**Course**	**L**	**T**	**P**	**C**	**Hours/ Week**	**Maximum Marks**			**Category**	**Page No.**
							CIA	**ES**	**Total**		
Project											
26CSP491	Project and Viva Voce	0	0	0	15	0	80	120	200	Project	112
*** Course Code and Course Name subject to change by NPTEL / SWAYAM**											
Total		0	0	0	15	0	80	120	200		
Grand Total (Upto IV Semester)		62	0	28	90	90	1040	1210	2250		

DEPARTMENT OF COMPUTER SCIENCE
FACULTY OF ARTS, SCIENCE, COMMERCE, MANAGEMENT
PG PROGRAM (CBCS) – M. Sc. COMPUTER SCIENCE
(2026 – 2027 Batch and onwards)

ELECTIVE COURSES

	Course Code	Name of the Course
ELECTIVE – I	26CSP102A	Quantum Computing
	26CSP102B	Cryptography and Network Security
	26CSP102C	Wireless Sensor Networks
ELECTIVE – II	26CSP202A	Statistical Data Analysis
	26CSP202B	Edge and Fog Computing
	26CSP202C	Soft Computing
ELECTIVE – III	26CSP303A	Quantum Artificial Intelligence
	26CSP303B	Robotic Process Automation
	26CSP303C	Intelligent Agents

OPEN ELECTIVE COURSES

S.No.	Name of the offering Department	Course Code	Name of the Course
1	English	26EGPOE301	English for Competitive Examinations
2	Commerce	26CMPOE301	Personal Finance and Planning
3	Management	26MBAPOE301	Organizational Behaviour
4	Computer Applications	26CAPOE301	Data Intelligence Systems
5	Computer Science	26CSPOE301	Cyber Forensics
6	Mathematics	26MMPOE301	Applied Statistics and Data Analysis
7	Physics	26PHPOE301	Electrical Appliances and Servicing
8	Chemistry	26CHPOE301	Industrial Chemistry
9	Microbiology	26MBPOE301	Probiotics
10	Biotechnology	26BTPOE301	Nutrition and Dietetics

M. Sc. Computer Science		2026-2027
Category: Major		**SEMESTER I**
26CSP101	**Cloud Computing**	**4H - 4C**
Instruction Hours/week: L:4 T:0 P:0		**Marks: Internal:40 External:60 Total:100**
		End Semester Exam: 3 Hours

PREREQUISTICS:

- Understanding basic networking concepts such as IP addresses, DNS, routers, and firewalls can be helpful in grasping cloud networking concepts.
- Familiarity with cloud services.

COURSE OBJECTIVES (CO):

- To provide a good understanding of the concepts, standards in Cloud computing
- To make the student understand about the cloud service providers and their usage.
- To learn how to secure the data in cloud depending.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

Cos	Course Outcomes	Blooms Level
CO1	Utilize the recent trends in the field of cloud computing and Providing exposures to some open source and commercial clouds.	K3
CO2	Identify the architecture of the cloud and the usage of clouds.	K3
CO3	Analyze the secure data from the security issues.	K4
CO4	Develop the students to work based on the various service level agreements.	K3
CO5	Apply the traditional cloud and Microsoft azure, etc.	K3
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I INTRODUCTION To CLOUD COMPUTING **12 HOURS**

Characteristics of Cloud Computing -Paradigm shift - Benefits of cloud computing - Disadvantages of cloud computing- Role of Open Standards-Cloud Computing Architecture: Cloud computing stack-Public cloud –Private cloud -Hybrid cloud -Community cloud

UNIT II SERVICE MODELS **12 HOURS**

Infrastructure as a Service (IaaS) -Platform as a Service (PaaS) -Software as a Service (SaaS) - Identity as a Service (IdaaS) -Compliance as a Service (CaaS)- Cloud storage- Advantages of Cloud Storage – Cloud Storage Providers.

UNIT III VIRTUALIZATION TECHNOLOGIES **12 HOURS**

Load Balancing and Virtualization -Advanced load balancing -The Google cloud - Hypervisors - Virtual machine types -VMware vSphere - Machine Imaging -Porting Applications -The Simple Cloud API - AppZero Virtual Application Appliance. Virtualization of CPU –Memory – I/O Devices –Virtualization Support and Disaster Recovery.

UNIT IV CLOUD INFORMATION SECURITY OBJECTIVES **12 HOURS**

Confidentiality Integrity and Availability -Cloud Security Services - Relevant Cloud Security Design Principles -Cloud Computing Risk Issues -The CIA Triad Privacy and Compliance Risks - Threats to Infrastructure Data and Access Control -Cloud Access Control Issues -Database Integrity Issues -Cloud Service Provider Risks Architectural Considerations General Issues- Trusted Cloud Computing -Identity Management and Access Control

UNIT V DATACENTER **12 HOURS**

Datacenter: Data center overview -Components - Provisions - Need of Data Center - Data Center Architecture - Different Racks - Data center architecture for cloud computing - role of data center in cloud computing.

Case Studies

Case Study on Open Source and Commercial Clouds: Microsoft Azure- Amazon EC2-Google Web services – Open Nebula.

TOTAL: 60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Amazon Web Services (AWS)	Provide scalable cloud services like computing, storage, and databases
Microsoft Azure	Build, deploy, and manage applications through Microsoft data centers
Google Cloud Platform (GCP)	Offer cloud storage, AI tools, and data analytics services

TEXT BOOKS:

1. Dr Kumar Saurabh. (2018), "Cloud Computing" .2ndediton. New Delhi: Wiley India.
2. Nikos Antonopoulos, Lee Gillam. (2018). "Cloud Computing: Principles Systems and Applications". Springer.
3. Giovanni Toraldo. (2017). "Open Nebula 3 Cloud Computing". 2nd edition, Wiley, New Delhi, India

REFERENCE BOOKS:

1. Rajkumar Buyya, James Broberg, & Andrzej, M. Goscinski. (2017). *"Cloud Computing Principles and Paradigms"*, 3rdediton, New Delhi: Tata Mc-Graw Hill.
2. Ronald, L. Krutz, Russell Dean Vines. (2016*). "Cloud Security: A Comprehensive Guide to Secure Cloud Computing",* 3rdediton, Wiley New Delhi, India,
3. Anthony T.Velte Toby J.Velte Robert Elsenpeter. (2016)." *Cloud Computing Practical Approach", first* edition, Tata McGraw Hill. New Delhi

WEBSITES:

1. en.wikipedia.org/wiki/Cloud_computing
2. www.ibm.com/cloud-computing/in/en/
3. www.oracle.com/CloudComputing

JOURNALS:

1. K.Naresh, P.Leela, P.Suneetha, Purandhar.N ," Research on Cloud Computing By using SaaS Model ", International Journal of Recent Technology and Engineering (IJRTE), Volume-8, Issue-1S4, June 2019

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**2**	**-**	**3**	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**3**	**-**	**-**	**-**	**-**	**3**	**-**
CO2	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2**
CO3	**-**	**-**	**3**	**3**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**-**	**-**	**-**	**-**	**-**	**-**	**3**	**3**	**2**	**-**	**3**	**-**	**-**	**2**	**-**	**-**	**-**
CO5	**-**	**-**	**3**	**-**	**-**	**-**	**3**	**-**	**-**	**-**	**3**	**-**	**-**	**-**	**-**	**-**	**-**
Average	**2**	**-**	**3**	**3**	**1**	**-**	**3**	**3**	**2**	**-**	**3**	**-**	**-**	**2**	**-**	**3**	**2**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

M. Sc. Computer Science		2026-2027
Category: Major		**SEMESTER I**
26CSP102A	**Quantum Computing**	**4H - 4C**
Instruction Hours/week: L:4 T:0 P:0		**Marks: Internal:40 External:60 Total:100**
		End Semester Exam: 3 Hours

PREREQUISTICS:

- Basic knowledge of classical computing concepts and basic programming skills.

COURSE OBJECTIVES (CO):

- To introduce the basic principles of quantum computing and its industrial applications.
- To understand and apply quantum algorithms and analyze complexity classes.
- To develop skills in quantum programming, error correction, communication, and explore future trends.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply the principles of quantum computing to represent quantum information using qubits and quantum states and implement basic quantum gate operations.	K3
CO2	Analyze various quantum algorithms and their applications.	K4
CO3	Apply quantum optimization programming techniques in problem-solving.	K3
CO4	Apply quantum communication protocols such as quantum teleportation, quantum key distribution (BB84), and explore their applications in quantum cryptography.	K3
CO5	Demonstrate practical skills in quantum computing in industry.	K3
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I INTRODUCTION **12 HOURS**

Classical vs Quantum Computing- Basic Concepts: Qubits, Superposition, Entanglement- Quantum States, Dirac Notation- Quantum Gates and Quantum Circuits- Applications of Quantum Computing in Industry

UNIT II QUANTUM ALGORITHMS AND COMPLEXITY **12 HOURS**

Quantum Parallelism-Quantum Speedup and Complexity classes (P, NP, BQP)-Deutsch-Jozsa Algorithm-Grover's Search Algorithm- Shor's Factorization Algorithm

UNIT III QUANTUM PROGRAMMING **12 HOURS**

Introduction to Quantum Programming Languages (Qiskit, Cirq, Microsoft Q#)-Installation of Qiskit and Quantum Lab Environment-Writing basic Quantum programs- Simulating Quantum Circuits

UNIT IV QUANTUM ERROR CORRECTION & COMMUNICATION **12 HOURS**

Quantum decoherence and errors- Quantum Error Correction Codes (Shor code, Steane code -

overview)- Quantum Teleportation Protocol- Quantum Key Distribution (BB84 Protocol) Quantum Cryptography applications

UNIT V: FUTURE TRENDS AND INDUSTRY APPLICATIONS 12 HOURS

Quantum Machine Learning basics- Quantum Computing in Cybersecurity, Drug discovery, Finance- Careers in Quantum Computing- Overview of Quantum Hardware (IBM Q, Google Sycamore, Rigetti, IonQ)

TOTAL: 60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
IBM Quantum Experience	Access real quantum computers and run quantum circuits online
Qiskit	Design, simulate, and execute quantum algorithms
Google Cirq	Build and simulate quantum circuits, especially for NISQ devices

TEXT BOOKS:

1. Chuck Easttom, (2021) "Quantum Computing Fundamentals", 1st edition, Published by Addison-Wesley Professional
2. Ronald de Wolf, (2023) "Quantum Computing and Communication Complexity", first Edition, University of Amsterdam.
3. Elias F. Combarro, Samuel Gonzalez-Castillo, and Alberto Di Meglio, (2023) "A Practical Guide to Quantum Machine Learning and Quantum Optimization: Hands-on Approach to Modern Quantum Algorithms", Packt Publishing Ltd.

REFERENCE BOOKS:

1. Kasirajan, Venkateswaran (2021), "Fundamentals of quantum computing", Springer International Publishing
2. Chris Bernhardt (2020), "Quantum Computing for Everyone", The MIT Press, Cambridge.
3. Nielsen, Michael A., and Isaac L. Chuang (2013), "Quantum Computation and Quantum Information" Cambridge University Press.

WEBSITES:

1. www.cse.iitk.ac.in/users/rkg/Talks/mobile_main.pdf
2. Qiskit basic Quantum Computation course - https://qiskit.org/learn/intro-qc-qh/
3. https://docs.quantum.ibm.com/

JOURNALS:

1. O. Crampton, P. Promponas, R. Chen, P. Polakos, L. Tassiulas, and L. Samuel, "A Genetic Approach to Minimising Gate and Qubit Teleportations for Multi-Processor Quantum Circuit Distribution," J. Quantum Comput., vol. 7, no. 1, pp. 1–15, 2025.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**2**	**-**	**-**	**-**	**3**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**2**	**-**
CO2	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**3**
CO3	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**2**	**-**	**-**	**-**	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO5	**-**	**-**	**3**	**1**	**3**	**-**	**1**	**-**	**-**	**-**	**1**	**-**	**-**	**-**	**-**	**-**	**-**
Average	**2**	**-**	**2.5**	**1**	**3**	**-**	**1**	**1**	**2**	**-**	**1**	**-**	**-**	**-**	**-**	**2**	**3**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

M. Sc. Computer Science		2026-2027
Category: Major		**SEMESTER I**
26CSP102B	**Cryptography and Network Security**	**4H - 4C**
Instruction Hours/week: L:4 T:0 P:0	**Marks: Internal:40 External:60 Total:100**	
	End Semester Exam: 3 Hours	

PREREQUISTICS:

- A strong understanding of mathematical principles, such as linear algebra, number theory, and combinatorics

COURSE OBJECTIVES (CO):

- To understand basics of Cryptography and Network Security.
- To understand theory of fundamental cryptography, encryption and decryption algorithms.
- To understand various Block Ciphers, DES and AES algorithms

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Analyze and design classical encryption techniques and block ciphers	K4
CO2	Organize various public key cryptographic techniques	K3
CO3	Make use of symmetric and asymmetric key algorithms for cryptography	K3
CO4	To develop secure authentication systems and evaluate the authentication and hash algorithms.	K3
CO5	Compare any network from the threats in the world.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I INTRODUCTION **12 HOURS**

Security Trends - The OSI Security Architecture – Security Attacks – Security Services – Security Mechanisms– A Model for Network Security. Classical Encryption Techniques – Symmetric Cipher Model – Substitution Techniques - Transposition Techniques – Rotor Machines - Steganography.

UNIT II BLOCK CIPHERS AND DATA ENCRYPTION STANDARDS **12 HOURS**

Block Cipher Principles – The Data Encryption Standard - The Strength of DES –Advanced Encryption Standard (AES) – Evaluation Criteria for AES – The AES Cipher – Multiple Encryption and Triple DES – Block Cipher Modes of Operation – Stream Ciphers and RC4- modular Arithmetic and Euclidean Algorithm

UNIT III CONFIDENTIALITY USING SYMMETRIC ENCRYPTION **12 HOURS**

Principles of Public Key Cryptosystems, Placement of Encryption Function – Traffic Confidentiality – Public key Cryptography and RSA – The RSA Algorithm- Basic prime numbers and Discrete Logarithms– Diffie Hellman Key Exchange, - Key Management: Digital Signatures – Digital Signature Standard.

UNIT IV AUTHENTICATION AND HASH FUNCTIONS **12 HOURS**

Authentication Functions, Message Authentication Codes (MAC's), User Authentication: Password, Certificate based & Biometric Authentication, Authentication Applications: Kerberos and X.509. Hashes, MD5 & SHA – Security of Hash Functions and MAC's Digital Signatures and Authentication Protocols.

UNIT V NETWORK SECURITY **12 HOURS**

System Security: Intruders - Viruses and related threats, Need for Security, Network Security-Firewalls, IP Security, VPN, Intrusion Detection, E-mail security, Web Security: Secure socket layer and transport layer security, Blockchain Technology concepts.

TOTAL: 60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Wireshark	Capture and analyze network traffic, detect attacks and vulnerabilities
Kali Linux	Perform penetration testing, ethical hacking, and security analysis
OpenSSL	Implement encryption, decryption, SSL/TLS protocols, and key generation

TEXT BOOKS:

1. Atul Kahate. (2019). "Cryptography and Network Security", 4th Edition. Tata McGraw Hill. New Delhi.
2. Charlie Kaufman, Radia Perlman, &Mike Speciner. (2017), "Network Security Private Communication in a Public World". 2nd Edition. Prentice-Hall of India. New Delhi.
3. William Stallings. (2017), "Cryptography and Network Security Principles and Practices". 6th Edition. Pearson Education. New Delhi.

REFERENCE BOOKS:

1. William Stallings (2020), "Cryptography and Network Security Principles and Practice", Fourth Edition, Pearson Education.
2. Wenbo Mao (2019), "Modern Cryptography: Theory and Practice", Fourth Edition, Prentice Hall PTR.
3. William Stallings (2021), "Network Security Essentials: Applications and Standards", Fourth Edition Prentice Hall.

WEBSITES:

1. williamstallings.com/Crypto3e.html
2. cryptofundamentals.com/algorithms

JOURNALS:

1. Yahia Alemami, Mohamad Afendee Mohamed, Saleh Atiewi, Research on Various Cryptography Techniques, International Journal of Recent Technology and Engineering (IJRTE), Volume-8, Issue-2S3, July 2019.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**1**	**-**	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**-**	**-**	**2**	**-**
CO2	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**
CO3	**-**	**-**	**3**	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**-**	**-**	**-**	**-**	**3**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO5	**-**	**-**	**3**	**-**	**3**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**
Average	**1**	**-**	**3**	**3**	**3**	**-**	**2**	**-**	**-**	**-**	**1**	**-**	**2**	**-**	**-**	**2**	**1**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

M. Sc. Computer Science		2026-2027
Category: Major		**SEMESTER I**
26CSP102C	**Wireless Sensor Networks**	**4H - 4C**
Instruction Hours/week: L:4 T:0 P:0	**Marks: Internal:40 External:60 Total:100**	
	End Semester Exam: 3 Hours	

PREREQUISITE:

- Basic knowledge of Data Communication Networks**.**

COURSE OBJECTIVES (CO):

- To understand the basics of Adhoc & Sensor Networks
- To learn various fundamental and emerging protocols of all layers in adhoc network
- To study about the issues pertaining to major obstacles in establishment and efficient management of adhoc and sensor networks

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Utilize a Sensor network environment for different type of applications	K3
CO2	Model the sensor network architectures using QoS and Congestion control mechanisms	K3
CO3	Apply appropriate routing algorithms for different network environments	K3
CO4	Analyze the working of ad-hoc and sensor network for various applications	K4
CO5	Classify security mechanisms in the wireless ad-hoc and sensor networks	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I ADHOC NETWORKS FUNDAMENTALS & COMMUNICATION PROTOCOLS **12 HOURS**

Fundamentals and Elements of WLANs – IEEE 802.11 Architecture –Self Configuration and Auto configuration – issues in Ad-Hoc Wireless Networks – MAC Protocols for Ad-Hoc Wireless Networks – Contention Based Protocols – TCP Over AdHoc Networks – TCP Protocol Overview TCP and MANETs – Solutions for TCP Over AdHoc Networks.

UNIT II ADHOC NETWORK ROUTING AND MANAGEMENT **12 HOURS**

Routing in Ad-Hoc Networks – Introduction – Topology based versus Position based Approaches Proactive Routing – DSDV, WRP, TBRPF Reactive Routing – DSR, AODV, Hybrid Routing Approach ZRP, CBRP- Location services – DREAM – Quorums based Location Service – Forwarding Strategies– Greedy Packet Forwarding, LAR.

UNIT III SENSOR NETWORK COMMUNICATION PROTOCOLS 12 HOURS

Introduction – Architecture – Single Node Architecture – Sensor Network Design Considerations Energy Efficient Design Principles for WSN‟s – Protocols for WSN – Physical Layer – Transceiver Design Considerations – MAC Protocols for wireless sensor network –IEEE 802.15.4 Zigbee – Link Layer and Error Control Issues – Routing Protocols – Challenges for Wireless Sensor Networks Gossiping and agent based unicast forwarding, Energy efficient unicast – Transport Protocols & QoS – Congestion Control Issues – Application specific Support – Target detection and tracking.

UNIT IV SENSOR NETWORK MANAGEMENT AND PROGRAMMING 12 HOURS

Sensor Management – Topology Control Protocols and Sensing Mode Selection Protocols – time Synchronization – Localization and Positioning – Operating Systems and Sensor Network Programming – Sensor Network Simulators - Case study: Industrial automation and tsunami early warning system with wireless sensor networks.

UNIT V ADHOC AND SENSOR NETWORK SECURITY 12 HOURS

Security in Ad-Hoc and Sensor Networks – Key Distribution and Management –Software based Anti – tamper Techniques – Water Marking techniques – Defence against Routing Attacks – Secure Adhoc Routing Protocols – Broad cast Authentication WSN Protocols – TESLA – Biba – Sensor Network Security Protocols – SPINS.

TOTAL: 60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
OMNeT++	Model and simulate WSN architectures and network behavior
MATLAB	Simulate sensor networks, analyze data, and visualize network performance
NS-3	Simulate WSN protocols, routing, and communication performance

TEXT BOOKS:

1. Khan Shafiullah Et. Al, September (2021). “Wireless Sensor Networks Current status and Future Trends”, Second Edition,Taylor & Francis, ISBN: 9781138582712.
2. Rohtash Ghuriya, April (2015). “Wireless AdHoc and Sensor Networks”, Third Edition, Gazelle Book Services, ISBN: 9789380090733.
3. Carlos De Morais Cordeiro, Dharma Prakash Agrawal,(2011). “Ad Hoc and Sensor Networks: Theory and Applications”, Second Edition, World Scientific Publishing

REFERENCE BOOKS:

1. TeiebZnati Kazem Sohraby, Daniel Minoli, (2010). “Wireless Sensor Networks: Technology, Protocols and Applications, Third Edition,Wiley, ISBN: 13 978-8126527304
2. Waltenegus Dargie, Christian Poellabauer, (2010). “Fundamentals of Wireless Sensor Networks Theory and Practice”, Third Edition, John Wiley and Sons,
3. Amiya Nayak,Ivan Stojmenovic,(2010). “Wireless Sensor and Actuator Networks: Algorithm and Protocols for Scalable Coordination and Data communication”, Third Edition, John Wiley & Sons.

WEBSITES:

1. www.winlab.rutgers.edu ›~crose›dimacs03›kumar
2. www.ibm.com/cloud-computing/in/en/
3. www.oracle.com/CloudComputing

JOURNALS:

1. K.Naresh, P.Leela, P.Suneetha, Purandhar.N ," Research on Cloud Computing By using SaaS Model, "International Journal of Recent Technology and Engineering (IJRTE), Volume-8, Issue-1S4, June 2019.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**2**	**-**	**3**	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**3**	**-**	**-**	**-**	**-**	**3**	**-**
CO2	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2**
CO3	**-**	**-**	**3**	**3**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**-**	**-**	**-**	**-**	**-**	**-**	**3**	**3**	**2**	**-**	**3**	**-**	**-**	**2**	**-**	**-**	**-**
CO5	**-**	**-**	**3**	**-**	**-**	**-**	**3**	**-**	**-**	**-**	**3**	**-**	**-**	**-**	**-**	**-**	**-**
Average	**2**	**-**	**3**	**3**	**1**	**-**	**3**	**3**	**2**	**-**	**3**	**-**	**-**	**2**	**-**	**3**	**2**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER I**

26CSP141 **Python Programming** **5H - 3C**

Instruction Hours/week: L:1 T:0 P:4 **Marks: Internal:50 External:50 Total:100**

End Semester Exam: 3 Hours

PREREQUISITES:

- Familiarity with using a computer, navigating file systems, and using software applications.
- Basic arithmetic and logic are useful, especially for algorithmic thinking.

COURSE OBJECTIVES(CO):

- To create the basic python programming using expressions and operators.
- To understand the fundamental principles of Object-Oriented Programming, as well as data and information processing techniques
- To define the python functions to facilitate code reusability and to create GUI based python applications using Object oriented programming.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply modules for reusability and the object-oriented principles for modeling.	K3
CO2	Make use of various data structures for solving real world problems.	K3
CO3	Analyze the concept of string and text files using python programming.	K4
CO4	To learn how to build and package Python modules for reusability.	K3
CO5	Develop simple Python programs for solving problems.	K3
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I PYTHON DATA TYPES, OPERATORS AND TYPE CONVERSION **3T+12P = 15 HOURS**

Introduction to Variables, and Data Types- Operators, and Type Conversion

PRACTICAL:

- Student Performance Analysis System Using Variables and Data Types
- Create a program to illustrate assignment operators
- Create a program to perform implicit type conversion between integer and float data types.
- Real-Time Data Validation and Computational Analysis Using Operators and Expressions

UNIT II FUNCTIONS, MODULES AND CONTROL STATEMENTS **3T+12P = 15 HOURS**

Functions, Modules, Selection Statements, and Iterative Control Structures in Python

PRACTICAL:

- Student Grade Evaluation System Using Functions and Multi-Way Selection
- Employee Payroll Management System Using Functions and Conditional Statements
- Menu-Driven Banking Transaction System Using Conditional Iteration
- Data Processing and Report Generation Using Functions and Looping Constructs

UNIT III STRINGS AND TEXT FILES **3T+12P = 15 HOURS**

String Processing and Data Encryption - Text File Processing and File System Management.

PRACTICAL:

- Secure Text Encryption and Decryption System Using String Manipulation
- Program to count the number of vowels, consonants, digits, and special characters in a given string.
- Program to tokenize a sentence and display individual words.
- Program to implement text compression by removing duplicate words from a sentence.

UNIT IV LISTS AND DICTIONARIES **3T+12P = 15 HOURS**

List Processing, Tuples, and Functional Programming- Dictionary-Based Data Management and Traversal Techniques.

PRACTICAL:

- Develop a program to perform list slicing, concatenation, and repetition operations.
- Inventory and Product Management Application Using List Operations and Sorting Techniques
- Employee Salary Processing and Payroll Analysis Using Lists and User-Defined Functions
- Hospital Patient Record Management System Using Dictionaries and Functions
- Library Book Catalog and Member Tracking System Using Dictionaries
- Develop a program to compare two dictionaries and identify common and unique entries.

UNIT V DESIGN OOPs, AND GRAPHICAL USER INTERFACE **3T+12P = 15 HOURS**

Object Oriented Programming- Database Programming and GUI Application Development.

PRACTICAL:

- Student Information Management System Using Classes and Objects
- Create a program to demonstrate method overloading and method overriding concepts.
- Hospital Patient Record Management System Using Encapsulation and Data Hiding
- Bank Account Simulation System Using Constructors and Object-Oriented Principles
- Library Management System Using Python Classes and Database Connectivity
- Retail Inventory Control System Using Python DB-API and SQL Tables
- Develop a program to demonstrate getter and setter methods.
- University Course Registration System Using Database Programming Techniques
- Program to implement radio buttons, checkboxes, and combo boxes.

TOTAL: 75 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS

AI Tools / Software Tools	Purpose
Codeium	Free alternative for code suggestions and automation.
Tabnine	Provides fast, privacy-focused Python code completion.
GitHub Copilot	Auto-completes Python code, suggests functions, and speeds up development
Claude	Strong for debugging, code explanation, and algorithm design.

TEXT BOOKS:

1. Liang, Y. D. (2026). Introduction to Python programming and data structures (3rd ed.). Pearson Education.
2. Horstmann, C. S., & Necaise, R. D. (2026). Python for everyone (3rd ed.). John Wiley & Sons.
3. Das, U., Lawson, A., Mayfield, C., & Norouzi, N. (2024). Introduction to Python programming. OpenStax. ISBN: 9781961584457.

REFERENCE BOOKS:

1. Kenneth A. Lambert, Martin Osborne, (2018) "Fundamentals of Python: First Programs, Cengage Learning", Second edition, Cengage Learning, Canada, ISBN13:978-1337560092.
2. Michal Jaworski, TarekZiade, (2016), "Expert Python Programming", Second Revised edition, Packt Publishing, ISBN-13: 978-1785886850.

WEBSITES:

1. www.python.org/about/gettingstarted/
2. www.tutorialspoint.com/python/index.htm
3. www.realpython.com/python-beginner-tips/
4. https://nptel.ac.in/courses/106/106/106106182/
5. https://nptel.ac.in/courses/106106145/

JOURNALS:

1. Kalyani Adawadkar, "Python Programming-Applications and Future", International Journal of Advance Engineering and Research Development, Special Issue SIEICON- 2017, April 2017.
2. K. R. Srinath, **"**Python – The Fastest Growing Programming Language**",** International Research Journal of Engineering and Technology (IRJET), Volume: 04 Issue:12, Dec-2017.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**3**	**-**	**-**	**1**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**2**	**-**
CO2	**-**	**-**	**3**	**-**	**-**	**-**	**2**	**3**	**-**	**-**	**2**	**-**	**-**	**-**	**3**	**-**	**-**
CO3	**-**	**-**	**3**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**-**	**-**	**-**	**1**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO5	**-**	**-**	**-**	**1**	**2**	**-**	**2**		**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
Average	**3**	**-**	**3**	**1**	**2**	**-**	**2**	**1**	**2**	**-**	**2**	**-**	**-**	**-**	**3**	**2**	**-**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER I**

26CSP142 **Data Mining Techniques and Tools** **5H - 4C**

Instruction Hours/week: L:3 T:0 P:2 **Marks: Internal:50 External:50 Total:100**

End Semester Exam: 3T+2P = 5 Hours

PREREQUISITES:

- A solid understanding of mathematics is essential for data mining. This includes knowledge of algebra, calculus, probability, and statistics.
- Mathematical concepts such as linear algebra and optimization are also important for understanding algorithms used in data mining.
- Proficiency in programming is crucial for working with data and implementing data mining algorithms.
- Common programming languages used in data mining include Python, R, and SQL.

COURSE OBJECTIVES(CO):

- To understand the concepts of data mining, KDD process, issues and applications.
- To know the working of different data mining techniques and its uses.
- To learn the usage of data mining tools WEKA and R and Understand the difference between Web mining, Text mining and Sequence mining.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Analyze data mining problems and reason about the most appropriate methods to apply to a given dataset and knowledge extraction need.	K4
CO2	Analyze the working of different clustering algorithms.	K4
CO3	Apply the basic concepts of Association Rule Mining and evaluate the working of various Association Rule Mining algorithms	K3
CO4	Distinguish the difference between Web mining, Text mining and Sequence mining	K4
CO5	Analyze the working of WEKA and R Tools	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I DATA MINING **9T+6P = 15 HOURS**

Introduction – Definitions - KDD vs. Data Mining – Data Mining Techniques – Issues and Challenges in Data Mining – Data Mining Application Areas. Classification Technique: Introduction – Decision Trees: Tree Construction Principle - Decision Tree Construction Algorithm – CART – ID3 – Rainforest – CLOUDS.

PRACTICAL:

- Decision Tree Classification using CART Algorithm

- Implementation of Decision Tree using ID3

UNIT II CLUSTERING TECHNIQUES **9T+6P = 15 HOURS**

Clustering Paradigms – Partitioning Algorithm - K-Means – K-Medoid Algorithms – CLARA – Hierarchical Clustering - DBSCAN – BIRCH – Categorical Clustering Algorithms – STIRR - Other Techniques. Introduction to Neural Networks - Learning in Neural Networks – Unsupervised Learning - Genetic Algorithm.

PRACTICAL:

- K-Means Clustering for Customer Segmentation
- Disease Prediction using Neural Network

UNIT III ASSOCIATION RULES **9T+6P = 15 HOURS**

Concepts - Methods to Discover Association Rules - Apriori Algorithm – Partition Algorithm - Dynamic Item Set Counting Algorithm - FP-Tree Growth Algorithm - Incremental Algorithm - Generalized Association Rule.

PRACTICAL:

- Apriori Algorithm for Market Basket Analysis
- Implementation of Partition Algorithm for Load Balancing in Distributed Systems

UNIT IV WEB MINING **9T+6P = 15 HOURS**

Basic Concepts – Web Content Mining – Web Structure Mining – Web Usage Mining – Text Mining: Text Clustering - Sequence Mining: The GSP Algorithm – SPADE.

PRACTICAL:

- Implementation of Text Clustering
- Demonstrate Web Usage mining
- Demonstrate Sequence mining

UNIT V TOOLS **9T+6P = 15 HOURS**

Tools: Need for Data Mining Tools - Introduction to WEKA – The Explorer – The Experimenter – Classification – Regression – Clustering - Nearest Neighbor - Introduction to R - Data Types-Variables Operators - Decision Making - Loop Control – Function – Strings - Vectors - Lists-Matrices – Arrays – Factors - Data Frames – Packages - Charts and graphs – Statistics - Discussion on Case Study - Expert Lectures - Online Seminars – Webinars – Workshops.

PRACTICAL:

- Implementation of Classification
- Demonstrate Clustering
- Demonstrate data frames and statistics

TOTAL: 75 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
RapidMiner	Drag-and-drop tool for data mining, predictive analytics, and machine learning.
KNIME	Open-source platform for data analysis, data mining, and workflow automation.
Apache Hadoop	Distributed storage and processing for large datasets.
Tableau	Converts mined data into interactive dashboards.

TEXT BOOKS:

1. Arun K. Pujari, (2009), *"Data Mining Techniques",* Third Edition, Universities Press (India) Limited. Hyderabad.
2. Margaret H. Dunham, (2004) *"Data Mining Introductory and Advanced Topics",* Second Edition, Pearson Education, Hyderabad.
3. Ian H. Witten, Eibe Frank, Mark A. Hall, (2011), "*Data Mining: Practical Machine Learning Tools and Techniques".* Elsevier.
4. Norman Matloff, (2011), *"The Art of R Programming a Tour of Statistical Software Design",* Third Edition, William Pollock, San Fransisco.
5. Emmanuel Paradis, (2005), *"R for Beginners",* Second Edition, Institutes Sciences Evolution.

REFERENCE BOOKS:

1. Pieter Adriaans, Dolf Zantinge, (2008), *"Data Mining",* Second Edition, Addison Wesley.
2. Jaiwei Han and Micheline Kamber, (2011), *"Data Mining Concepts and Techniques"*, 3rd Edition, Morgan Kaufmann Publishers.
3. Dr. Matthew A. North, (2012) *"Data Mining for the Masses",* 3rd Editoin, A Global Text Project Book, Pearson Education.
4. Roger D. Peng, (2015), *"R Programming for Data Science"*, Third Edition, Lean Publishing.

WEBSITES:

1. https://swayam.gov.in/nd2_cec20_cs12/preview
2. https://onlinecourses.nptel.ac.in/noc19_ma33/preview
3. https://www.futurelearn.com/courses/data-mining-with-weka
4. https://onlinecourses.nptel.ac.in/noc21_cs06/preview
5. https://www.coursera.org/specializations/data-mining
6. https://www.mygreatlearning.com/academy/learn-for-free/courses/data-mining

JOURNALS:

1. Dr. Malla Reddy Jogannagari, Mrs. Maheshwari Manchala," Data Mining: Techniques, Tools and its Challenges ", International Journal of Creative Research Thoughts (IJCRT) www.ijcrt.org, 2020 IJCRT | Volume 8, Issue 7 July 2020 | ISSN: 2320-2882.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**2**	**-**	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**-**	**-**	**3**	**-**
CO2	**-**	**-**	**-**	**3**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2**
CO3	**-**	**-**	**-**	**3**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**2**	**-**	**3**	**-**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO5	**2**	**-**	**3**	**3**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
Average	**2**	**-**	**3**	**3**	**1**	**-**	**-**	**-**	**1**	**-**	**1**	**-**	**-**	**-**	**-**	**3**	**2**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER I**

26CSP143 **Advanced Data Structures and Analysis of Algorithms** **5H - 4C**

Instruction Hours/week: L:3 T:0 P:2 **Marks: Internal:50 External:50 Total:100**

End Semester Exam: 3T+2P=5 Hours

PREREQUISITE:

- Knowledge of programming fundamentals, basic data structures, algorithm design techniques, asymptotic analysis, discrete mathematics, and problem-solving skills.

COURSE OBJECTIVES(CO):

- To introduce the basic concepts of algorithm analysis and advanced data structures.
- To give an insight into applications based on advanced data structures.
- To provide a broad perspective of different design techniques and their analysis.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Analyze algorithms using asymptotic notations and evaluate time and space complexities.	K4
CO2	Construct and apply advanced data structures such as trees, AVL trees, B-trees, B+ trees, and graphs.	K3
CO3	Implement and analyze graph algorithms including shortest path and minimum spanning tree techniques.	K3
CO4	Apply searching, sorting, hashing, and external file organization methods for efficient data processing.	K4
CO5	Apply algorithm design techniques and classify problems based on computational complexity (P, NP, NP-Complete).	K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I INTRODUCTION **9T+6P = 15 HOURS**

Algorithm Analysis, Space and Time Complexity Analysis - Asymptotic Notations (Big O, Omega, Theta) - (Analysing Control Structures) - Amortized Analysis - String Matching

PRACTICAL:

- To demonstrate time complexity analysis
- Implementation of String-Matching Algorithm

UNIT II DATA STRUCTURES AND ADT's **9T+6P = 15 HOURS**

List – Stacks – Queues - Priority Queues - Trees – Traversals - Binary Trees - Search Trees - AVL Trees, B-Trees - B+ Trees – Graphs – (Applications using Stacks)

PRACTICAL:

- Program to demonstrate Stack operations
- Program to implement Queue operations

UNIT III GRAPH ALGORITHMS **9T+6P = 15 HOURS**

Definition and Representations - Shortest Path Algorithms – Dijkstra's, All Pairs - Network Flow Problems - Minimum Spanning Tree – Prim's, Kruskal's Algorithms – (Depth First and Breadth First Applications)

PRACTICAL:

- To demonstrate Dijkstra's Algorithm
- Program for implementation of Prim's Algorithm

UNIT IV SORTING AND FILE STRUCTURES **9T+6P = 15 HOURS**

Searching Algorithms: Linear Search and Binary Search, Sorting Algorithms: Insertion Sort, Shell Sort, Heap Sort, Merge Sort, Quick Sort – Analysis - Establishing Lower Bounds - External Sorting - Model for External Sorting - Multiway - Polyphase Sorting - Direct Files - Indexed Sequential Files - External Searching - Hashing – Virtual, Linear – (ISAM), VSAM

PRACTICAL:

- Implementation of searching algorithms
- Program for Sorting Algorithms
- Create hash table using dictionary

UNIT V DESIGN TECHNIQUES AND COMPUTATIONAL COMPLEXITY **9T+6P = 15 HOURS**

Greedy Algorithms: Minimum Cost Spanning Trees - Divide and Conquer: Max-Min Dynamic Programming: (Bellman-Ford Algorithm) - Backtracking: N-Queens Problem - Branch-and-Bound: Travelling Salesperson Problem. (Parallel Algorithms) - Introduction to Complexity Classes: P and NP Problems, NP-Complete Problems

PRACTICAL:

- Program to demonstrate Minimum Cost Spanning Tree
- Program for implementation of Bellman-Ford Algorithm
- Program for implementation of N-Queens Problem

TOTAL: 75 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
VisuAlgo	Visualizes trees, graphs, and sorting algorithms for better understanding.
Apache Spark	Supports large-scale graph and optimization algorithms.
Wolfram Mathematica	Helps in mathematical modeling and complexity analysis of algorithms.
JGraphT	Library for advanced graph algorithms like shortest path, matching, and flow problems

TEXT BOOKS:

1. Abirami A., Priya R.L (2023), Advanced Data Structures and Algorithms, BPB Publisher.
2. Mark A. Weiss (2021), Data Structures and Algorithm Analysis in Java, Pearson.

REFERENCE BOOKS:

1. Marcello La Rocca (2021), Advanced Algorithms and Data Structures, Manning Publications.
2. Anany Levitin (2015), Introduction to the Design and Analysis of Algorithms, Pearson Education.

WEBSITES:

1. https://www.tutorialspoint.com/data_structures_algorithms/index.html
2. https://www.cs.unm.edu/~saia/classes/561-f09

JOURNALS:

1. Lade. Gunakar Rao; Erupaka Nitya; Jyoshna Devi Kusumba; K. Rajchandar; S. Narasimha Murthy, "Optimizing Algorithmic Efficiency: Exploring Advanced Data Structures for High-Performance Computing", IEEE Xplore, 2024.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**3**	**2**	**3**	**2**	**2**	**1**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO2	**3**	**3**	**2**	**2**	**2**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO3	**3**	**3**	**2**	**2**	**2**	**1**	**1**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**3**	**3**	**2**	**1**	**2**	**2**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO5	**3**	**1**	**1**	**1**	**2**	**1**	**1**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
Average	**3**	**2**	**2**	**2**	**2**	**1**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER I**

26CSP144 **Advanced Java Programming** **5H - 4C**

Instruction Hours/week: L:3 T:0 P:2 **Marks: Internal:50 External:50 Total:100**

End Semester Exam: 3T+2P=5 Hours

PREREQUISITES:

- Basic knowledge of programming concepts and logic building.
- Fundamental understanding of object-oriented programming and databases.

COURSE OBJECTIVES(CO):

- To learn Java string handling and file input/output operations.
- To develop GUI applications using Java Swing components.
- To understand JavaBeans and Servlets for web application development.
- To create JSP applications and implement network programming concepts.
- To perform database connectivity and data manipulation using JDBC.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply Java utility classes, string handling methods, and file processing techniques using byte and character streams to develop basic Java applications.	K3
CO2	Design and develop GUI-based applications using Java Swing components, containers, and event handling mechanisms.	K3
CO3	Analyze JavaBeans components and Servlet life cycle processes to implement efficient web applications with session tracking and cookie management.	K4
CO4	Develop dynamic web applications using JSP elements and implement client-server communication using Java networking concepts.	K3
CO5	Evaluate the effectiveness of JDBC mechanisms for database creation, record manipulation, and result set management in enterprise applications.	K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I UTILITY CLASSES **9T+6P = 15 HOURS**

String Handling: The String Constructors - String Length - Special String Operations - Character Extraction - String Comparison - Searching Strings - Modifying a String - Input/Output: The I/O Classes and Interfaces – File - Byte Streams - Character Streams.

PRACTICAL:

- Implement a Java program to perform various string handling operations such as string comparison, character extraction, substring searching, and string modification
- Demonstrate file handling operations using byte streams and character streams in Java

UNIT II SWING **9T+6P = 15 HOURS**

Introducing GUI Programming with Swing– Introducing Swing - Swing Is Built on the AWT Two Key Swing Features – The MVC Connection - Components and Containers - The Swing Packages - A Simple Swing Application - Exploring Swing.

PRACTICAL:

- Design a Java Swing application for student information entry and display using event handling.
- Implement a simple calculator using Java Swing components.

UNIT III JAVA BEANS AND SERVLETS **9T+6P = 15 HOURS**

Introduction - Advantages of Beans – Introspection - The JavaBeans API. Servlets: Life Cycle Simple Servlet-Servlet API-Packages-Cookies session tracking.

PRACTICAL:

- Illustrate the use of JavaBeans for storing and accessing student details using getter and setter methods.
- Demonstrate user login validation and session tracking using Servlet technology.

UNIT IV JSP & NETWORK PROGRAMMING **9T+6P = 15 HOURS**

JSP Elements: The JSP Development Model – Components of a JSP Page. Network Programming: Working with URLs- Working with Sockets - Remote Method Invocation.

PRACTICAL:

- Demonstrate student mark processing using JSP elements.
- Implement client-server communication using socket programming in Java.

UNIT V CONNECTING DATABASE **9T+6P = 15 HOURS**

Introduction to Database Management Systems - Tables, Rows, and Columns - Introduction to the SQL SELECT Statement - Inserting Rows - Updating and Deleting Existing Rows - Creating and Deleting Tables - Creating a New Database with JDBC - Scrollable Result Sets.

PRACTICAL:

- Demonstrate database connectivity and record manipulation using JDBC.
- Implement scrollable ResultSet operations for navigating and updating database records using JDBC.

TOTAL: 75 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Google Gemini Code Assist	AI-powered coding support with syntax suggestions and code completion
Codeium	Free AI code completion and auto-generation tool

TEXT BOOKS:

1. Herbert Schildt, (2017) "Java the Complete Reference", 9th Edition, McGraw Hill Publishing Company Ltd, New Delhi. (Unit I, II, III, IV)
2. Phil Hanna (2001), "JSP: The Complete Reference", McGraw-Hill Companies. (Unit – IV)
3. Tony Gaddis, (2016) "Starting out with Java from Control Structures Through Objects", 6th Edition, Pearson Education Limited. (Unit – V)

REFERENCE BOOKS:

1. Keogh, James, "J2ME: The complete reference", Osborne, 2003
2. McFarland, David Sawyer, "Javascript & jQuery: the missing manual", O'Reilly Media, Inc.", 2011.
3. Deitel, Paul J, "Java how to program", Pearson Education India, 2002
4. Campione, Mary, Kathy Walrath, and Alison Huml, "The Java tutorial: a short course on the basics", Addison-Wesley Professional, 2001.

WEBSITES:

1. https://www.javatpoint.com/servlet-tutorial
2. https://www.tutorialspoint.com/java/index.htm
3. https://onlinecourses.nptel.ac.in/noc19_cs84/preview
4. https://www.geeksforgeeks.org/java/java/
5. https://docs.oracle.com/javase/tutorial/?utm_source=chatgpt.com

JOURNALS:

1. A. Sharma and R. Gupta, "Advanced Java-Based Web Application Development Using JSP, Servlets, and JDBC," *IEEE Access*, vol. 12, pp. 45231–45245, 2025.
2. S. Karthikeyan and P. Ramesh, "Efficient Client–Server Communication and Database Connectivity Using Java Socket and JDBC Technologies," *International Journal of Information Technology*, vol. 17, no. 2, pp. 1189–1198, 2024.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**3**	**2**	**2**	**3**	**2**	**-**	**-**	**2**	**-**	**2**	**2**	**-**	**-**	**-**	**2**	**3**	**2**
CO2	**3**	**2**	**3**	**3**	**2**	**-**	**2**	**2**	**-**	**2**	**2**	**-**	**-**	**1**	**2**	**3**	**3**
CO3	**3**	**2**	**3**	**3**	**2**	**1**	**2**	**2**	**-**	**2**	**2**	**-**	**1**	**1**	**2**	**3**	**3**
CO4	**3**	**2**	**3**	**3**	**3**	**1**	**2**	**3**	**-**	**2**	**2**	**-**	**1**	**1**	**2**	**3**	**3**
CO5	**3**	**2**	**3**	**3**	**3**	**2**	**2**	2	**-**	**3**	**2**	**-**	**1**	**1**	**2**	**3**	**3**
Average	**3**	**2**	**2.8**	**3**	**2.4**	**1.3**	**2**	**2.2**	**-**	**2.2**	**2**	**-**	**1**	**1**	**2**	**3**	**2.8**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

M.Sc Computer Science	**2026-2027**
Category: -	**SEMESTER I**
Journal Paper Analysis and Presentation	**2H - 0C**
Instruction Hours/week: L:2 T:0 P:0	**Marks: Internal: - External: - Total: -**
	End Semester Exam: -

26CSP201 **Internet of Things** **4H - 4C**

Instruction Hours / Week: L: 4 T: 0 P: 0 **Marks: Internal: 40 External: 60 Total: 100**

End Semester Exam: 3 Hours

PREREQUISITE:

- To understand the basic cybersecurity concepts and best practices for securing IoT devices and networks is crucial.

COURSE OBJECTIVES (CO):

- To introduce the solution of different real time problems in Healthcare.
- To explore smart cities using IoT.
- To explore the smart sensing healthcare and power challenge.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply the fundamentals of IoT architecture, networking, and communication protocols.	K3
CO2	Analyze Arduino, Python, sensors, actuators, and Raspberry Pi for IoT application development.	K4
CO3	Apply SDN, cloud computing, fog computing, and data analytics in IoT systems.	K3
CO4	Apply IoT frameworks and Industrial IoT solutions for real-world applications.	K3
CO5	Analyze IoT-based solutions for smart cities, healthcare, agriculture, and industrial automation.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I – IoT FUNDAMENTALS AND NETWORKING **12 HOURS**

Introduction to Internet of Things (IoT)- IoT Architecture and Components- Sensing and Actuation- Basics of Networking- Network Models- Internet Protocol Suite- Addressing and Routing- Wireless Communication Technologies- IoT Communication Requirements.

UNIT II - COMMUNICATION PROTOCOLS AND SENSOR NETWORKS **12 HOURS**

Communication Protocols in IoT- Sensor Networks- Wireless Sensor Networks (WSN), Sensor Node Architecture, Routing in Sensor Networks, Energy-Efficient Communication-Machine-to-Machine (M2M) Communications- Interoperability in IoT

UNIT III – IoT HARDWARE PLATFORMS AND PROGRAMMING **12 HOURS**

Introduction to Arduino Programming- Arduino Architecture and Development Environment- Integration of Sensors and Actuators with Arduino- Introduction to Python Programming- Introduction to Raspberry Pi- IoT Application Development using Raspberry Pi- Implementation of IoT Systems with Raspberry Pi.

UNIT IV – SDN, CLOUD AND DATA ANALYTICS FOR IoT **12 HOURS**

Introduction to Software Defined Networking (SDN)- SDN Architecture and Components- SDN for IoT- Data Handling and Analytics in IoT- Cloud Computing Fundamentals- IoT Cloud Platforms- Sensor-Cloud Integration- Fog Computing and Edge Computing.

UNIT V –SMART APPLICATIONS AND INDUSTRIAL IOT **12 HOURS**

Smart Cities- Smart Homes- Connected Vehicles- Smart Grid- Industrial Internet of Things (IIoT)- Industry 4.0 Concepts- Case Studies: Smart Agriculture- Healthcare Systems- Activity Monitoring Systems- Challenges and Future Trends in IoT.

TOTAL:60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Arduino IDE	Code writing and uploading to boards
ThingSpeak	Patient monitoring data visualization

TEXT BOOKS:

1. Misra, S., Mukherjee, A., & Roy, A. (2020). Introduction to Internet of Things. Cambridge University Press.
2. Misra, S., Roy, C., & Mukherjee, A. (2020). Introduction to Industrial Internet of Things and Industry 4.0. CRC Press.
3. Bahga, A., & Madisetti, V. K. (2015). Internet of Things: A hands-on approach. Universities Press.

REFERENCE BOOKS:

1. Minoli, D., Sohraby, K., & Occhiogrosso, B. (2017). IoT security (Internet of Things security): Advances in authentication. Wiley.
2. Misra, S., Roy, C., & Mukherjee, A. (2021). Introduction to industrial Internet of Things and Industry 4.0. CRC Press.
3. Veneri, G., & Capasso, A. (2018). Hands-on industrial Internet of Things. Packt Publishing.

WEBSITES:

1. https://onlinecourses.nptel.ac.in/noc19_cs65
2. https://onlinecourses.nptel.ac.in/noc21_cs20/preview
3. The Internet of Things in the Cloud, A Middleware Perspective, Honbo (taylorfrancis.com)

JOURNALS:

1. Yaqoob, I., Salah, K., Jayaraman, R., & Omar, M., "Blockchain for Healthcare Data Management and Sharing: A Survey of Recent Advances", Future Generation Computer Systems, 141: 1–16, 2023.
2. Nguyen, D. C., Pathirana, P. N., Ding, M., & Seneviratne, A., "IoT and Artificial Intelligence Convergence: Applications and Challenges", IEEE Access, 12: 45231–45248, 2024.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**3**	**-**	**-**	**-**	**3**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**3**	**-**
CO2	**-**	**-**	**2**	**1**	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2**
CO3	**3**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**
CO5	**-**	**-**	**2**	**-**	**3**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
Average	**3**	**-**	**2**	**1**	**3**	**2**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**3**	**2**

1 - Low, 2 - Medium, 3 - High, '-'; - No Correlation

noc26_cs161 **Introduction To Internet of Things** **4H - 4C**

Instruction Hours / Week: L: 4 T: 0 P: 0 **Marks: Internal: 40 External: 60 Total: 100**

End Semester Exam: 3 Hours

PREREQUISITE:

- Basic knowledge of computer networks, programming concepts, and embedded systems.

COURSE OBJECTIVES (CO):

- To understand the fundamental concepts, architectures, communication protocols, and networking technologies used in IoT systems.
- To develop IoT applications by integrating sensors, actuators, Arduino, Raspberry Pi, and cloud-based services.
- To analyze Industrial IoT solutions, data analytics techniques, and smart applications in real-world domains such as healthcare, agriculture, and smart cities.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply the architecture, components, communication protocols, and networking concepts of IoT systems.	K3
CO2	Apply Arduino, Python programming, and sensor-actuator interfacing techniques to develop IoT applications.	K3
CO3	Apply IoT-based monitoring and control systems using Raspberry Pi and related hardware platforms.	K3
CO4	Analyze IoT data management techniques, including SDN, cloud computing, sensor-cloud, and fog computing architectures.	K3
CO5	Evaluate Industrial IoT solutions and smart applications for healthcare, agriculture, smart cities, connected vehicles, and Industry 4.0 environments.	K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I –FUNDAMENTALS OF INTERNET OF THINGS AND NETWORKING 12 HOURS

Introduction to Internet of Things (IoT) - Evolution and Characteristics of IoT- IoT Architecture and Components, Basics of Networking- Communication Models in IoT- Communication Protocols- Sensor Networks- Machine-to-Machine (M2M) Communications- Interoperability in IoT.

UNIT II - IoT HARDWARE PLATFORMS AND PROGRAMMING 12 HOURS

Introduction to Embedded Systems- Introduction to Arduino Platform- Arduino Programming Fundamentals- Interfacing Sensors and Actuators with Arduino- Data Acquisition using Arduino- Introduction to Python Programming- Python for IoT Applications- IoT Device Communication and Control.

UNIT III –RASPBERRY PI AND IOT IMPLEMENTATION 12 HOURS

Introduction to Raspberry Pi- Raspberry Pi Architecture and Features- Operating System Installation and Configuration- GPIO Programming- Sensor and Actuator Interfacing with Raspberry Pi- IoT Application

Development using Raspberry Pi- Implementation of IoT Solutions with Raspberry Pi- Real-Time Data Collection and Monitoring.

UNIT IV – IOT DATA MANAGEMENT AND EMERGING TECHNOLOGIES 12 HOURS

Introduction to Software Defined Networking (SDN)- SDN Architecture and Components- SDN for IoT Networks- Data Handling and Analytics in IoT- Big Data Concepts for IoT- Cloud Computing Fundamentals- Sensor-Cloud Architecture- Fog Computing and Edge Computing- Comparison of Cloud, Fog, and Edge Paradigms.

UNIT V –INDUSTRIAL IOT AND SMART APPLICATIONS 12 HOURS

Introduction to Industrial IoT (IIoT)- Industry 4.0 Concepts and Technologies- Smart Cities and Smart Homes- Connected Vehicles- Smart Grid Systems- Industrial IoT Applications- Case Studies: Smart Agriculture, Healthcare Monitoring Systems, Activity Monitoring and Wearable Systems- Challenges and Future Trends in IoT and IIoT.

TOTAL: 60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Arduino IDE	Programming Arduino boards, interfacing sensors and actuators, and developing IoT prototypes.
Node-Red	Connecting IoT devices, APIs, cloud services, and visualizing data with minimal coding.

TEXT BOOKS:

1. Misra, S., Mukherjee, A., & Roy, A. (2020). Introduction to Internet of Things. Cambridge University Press.
2. Misra, S., Roy, C., & Mukherjee, A. (2020). Introduction to Industrial Internet of Things and Industry 4.0. CRC Press.
3. Bahga, A., & Madisetti, V. K. (2015). Internet of Things: A hands-on approach. Universities Press.

REFERENCE BOOKS:

1. Minoli, D., Sohraby, K., & Occhiogrosso, B. (2017). IoT security (Internet of Things security): Advances in authentication. Wiley.
2. Buyya, R., & Dastjerdi, A. V. (2016). Internet of Things: Principles and paradigms. Morgan Kaufmann.
3. Holler, J., Tsiatsis, V., Mulligan, C., Karnouskos, S., Avesand, S., & Boyle, D. (2014). From machine-to-machine to the Internet of Things: Introduction to a new age of intelligence. Academic Press.

WEBSITES:

1. https://onlinecourses.nptel.ac.in/noc19_cs65
2. https://onlinecourses.nptel.ac.in/noc21_cs20/preview
3. https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs161

JOURNALS:

1. Alaa, M., Zaidan, A. A., Zaidan, B. B., Talal, M., & Kiah, M. L. M., "A Review of Smart Home Applications Based on Internet of Things", Journal of Network and Computer Applications, 97: 48–65, 2022.

2 Rjoub, G., Al-Migbal, D., Alqahtani, M., & Alruwaili, M., “Industrial Internet of Things (IIoT): Applications, Challenges and Future Directions”, Sensors, 23(4): 1856, 2023.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	3	-	-	-	3	-	-	-	-	-	2	-	-	-	-	3	-
CO2	-	-	2	1	3	-	-	-	-	-	-	-	-	-	-	-	2
CO3	3	-	-	-	-	2	-	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	2	-	-	-	-	-	-
CO5	-	-	2	-	3	2	-	-	-	-	-	-	-	-	-	-	-
Average	3	-	2	1	3	2	-	-	-	-	2	-	-	-	-	3	2

1 - Low, 2 - Medium, 3 - High, ‘-’; - No Correlation

PREREQUISITE:

- Basic knowledge of a programming language (like Python or R) is helpful for data analysis and automation.

COURSE OBJECTIVES (CO):

- To find the association between attributes.
- To find the correlation between two variables and form regression lines.
- To understand the concepts of sampling

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Analyze the data pertaining to attributes and to interpret the results.	K4
CO2	To recognize and evaluate the relationship between two quantitative variables through simple linear correlation and regression	K5
CO3	Determine the familiar concepts for binomial, Poisson, normal and log-normal probability distributions	K5
CO4	Experiment with linear combinations of random variables and the Central Limit Theorem	K3
CO5	Apply correctly a variety of statistical techniques, both descriptive and inferential	K3
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I - INTRODUCTION **12 HOURS**

Analysis of categorical data: Definition of attributes. Independence, association and partial association of attributes, various measures of association (Yule's) for 2 –way data and coefficient of contingency (Pearson and Tcheprow's) and coefficient of colligation. Bivariate data, Scattered diagram, Principle of least squares, fitting of a straight line, quadratic, exponential and power curves.

12 HOURS

UNIT II – CORRELATION AND REGRESSION

Definition, Types of correlation, Methods of Studying Correlation Scattered diagram, Product moment correlation coefficient and its properties. Computation of correlation coefficient for grouped data, Spearman's Rank correlation coefficient and its properties, Correlation ratio. Partial and multiple correlation coefficients (only for three variables). Simple linear regression, lines of regression, properties of regression coefficients, Correlation verses Regression.

12 HOURS

UNIT III – DISTRIBUTIONS

Concepts of population, parameter, random sample, statistic, sampling distribution and standard error. Standard error of sample mean(s) and sample proportion(s). Exact sampling distributions- statement and properties of χ^2, t and F distributions and their interrelationships. Independence of sample mean and variance in random sampling from normal distributions.

UNIT IV - NORMAL AND EXPONENTIAL **12 HOURS**

Statement of Neyman's Factorization theorem, derivations of sufficient statistics in case of Binomial, Poisson, Normal and Exponential (one parameter only) distributions. Estimation by the method of moments, Maximum likelihood estimation (MLE).

UNIT V – ESTIMATITON OF INTERVALS **12 HOURS**

Statements of asymptotic properties of MLE. Concept of interval estimation. Confidence intervals of parameters of normal population by pivot method. Criteria of a good estimator- consistency, unbiasedness, efficiency and sufficiency with examples. Point estimation of a parameter, concept of bias and mean square error of an estimate.

TOTAL:60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
R (Programming language)	Regression, hypothesis testing, time-series analysis
JASP	Supports both classical and Bayesian statistics

TEXT BOOKS:

1. Goon A.M., Gupta M.K. and Dasgupta B. (2012): *"Fundamentals of Statistics*", Vol. I& II,8th Edn. The World Press, Kolkata.
2. Miller, Irwin and Miller,Marylees John E. (2016*):"Freund's Mathematical Statistics with Applications",* 7th Edn, Pearson Education, Asia.
3. V.K.Kapoor and S.C.Gupta(2010): *"Fundamentals of Mathematical Statistics*", Seventh Edition, Sultan Chand & Sons, New Delhi

REFERENCE BOOKS:

1. William Feller (2016) *"Introduction to Probability theory and its applications"*. Volume- I, Fifthe Edition, Asian Publishing House, Bombay.
2. Hoel P.G(2017), *"Introduction to mathematical Statistics*", Third Edition Revised, Asia Publishing house.

WEBSITES:

1. http://www.bu.edu/met/metropolitan_college_people/student/resources/conduct/code.html
2. https://www.openintro.org/stat/

JOURNALS:

1. Cleveland, W.S., Robust, "Locally Weighted Regression and Smoothing Scatterplots", Journal of The American Statistical Association *74*: 829, 2015.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**3**	-	**2**	-	-	-	-	-	-	-	**3**	-	-	-	-	**2**	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	**2**
CO3	-	-	**2**	-	**2**	-	-	-	-	-	-	-	-	-	-	-	-
CO4	**3**	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	**3**	-	-	-	**2**	-	-	**1**	-	-	**3**	-	-	-	-	-	-
Average	**3**	-	**2**	-	**2**	-	-	**1**	-	-	**3**	-	-	-	-	**2**	**2**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER II**

26CSP202B **Edge and Fog Computing** **4H - 4C**

Instruction Hours/week: L:4 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

PREREQUISITE:

- Basic knowledge of computer networks, cloud computing, IoT, distributed systems, and operating systems.

COURSE OBJECTIVES (CO):

- To understand the students about edge computing, an important branch of distributed computing and IoT with significant applications in Data Science.
- To implement the concepts of fog and cloud computing and exposes students to modern tools and API to deploy relevant infrastructures.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply fog computing principles and IoT architectural models to develop scalable and interoperable IoT solutions.	K3
CO2	Apply fog computing techniques in intelligent transportation systems to manage traffic and improve transportation efficiency.	K3
CO3	Identify potential technical challenges of the transition process and suggest solutions.	K4
CO4	Analyze data and application requirements and pertaining issues.	K4
CO5	Evaluate the effectiveness of IoT, edge, machine-to-machine (M2M), and SCADA architectures in addressing real-world application requirements.	K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I: EDGE COMPUTING **12 HOURS**

Fog computing requirements when applied to IoT: Scalability, Interoperability, Fog IoT architectural model, Challenges on IoT Stack Model via TCP/IP Architecture, Data Management, filtering, Event Management, Device Management, cloudification, virtualization, security and privacy issues. Integrating IoT, Fog, Cloud Infrastructures: Methodology, Integrated C2F2T Literature by Modelling Technique re by Use-Case Scenarios, Integrated C2F2T Literature by Metrics.

UNIT II: FOG COMPUTING IN HEALTH MONITORING **12 HOURS**

Exploiting Fog Computing in Health Monitoring: An Architecture of a Health Monitoring IoT-Based System with Fog Computing, Fog Computing Services in Smart E-Health Gateways, Discussion of Connected Components. Fog Computing Model for Evolving Smart Transportation Applications: Introduction, Data-Driven Intelligent Transportation Systems, Fog Computing for Smart Transportation Applications Case Study: Intelligent Traffic Lights Management (ITLM) System.

UNIT III: FOG COMPUTING APPLICATION **12 HOURS**

Software Defined Networking and application in Fog Computing: Open Flow Protocol, Open Flow Switch, SDN in Fog Computing, Home Network using SDN. Security and Privacy issues: Trust and privacy issues in IoT Network, web Semantics and trust Management for Fog Computing, Machine Learning based security in Fog Computing, Cyber- Physical Energy Systems over Fog Computing

UNIT IV: INTRODUCTION TO EDGE COMPUTING **12 HOURS**

Introduction to Edge Computing Scenarios and Use cases - Edge computing purpose and definition, Edge computing use cases, Edge computing hardware architectures, Edge platforms, Edge vs Fog Computing, Communication Models - Edge, Fog, and M2M.

UNIT V: IOT ARCHITECTURE AND CORE IOT MODULES **12 HOURS**

IoT Architecture and Core IoT Modules-A connected ecosystem, IoT versus machine-to-machine versus, SCADA, The value of a network and Metcalfe's and Beckstrom's laws, IoT and edge architecture, Role of an architect, Understanding Implementations with the examples- Edge computing with RaspberryPi, Industrial, and Commercial IoT and Edge, and Edge computing and solutions.

TOTAL:60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
EdgeX Foundry	Open-source framework for building interoperable edge computing systems
Azure IoT Edge	Deploy and manage AI and cloud workloads on edge devices
AWS IoT Greengrass	Run cloud services and process data locally on edge devices
KubeEdge	Extends Kubernetes for managing edge computing infrastructure
TensorFlow Lite	Enables lightweight AI model deployment on edge devices

TEXT BOOKS:

1. Fog Computing: Theory and Practice by Assad Abbas, Samee U. Khan, Albert Y. Zomaya IoT and Edge Computing for Architects - Second Edition, by Perry Lea, Publisher: Packt Publishing, 2020, ISBN: 9781839214806
2. Raspberry Pi Cookbook, 3rd Edition, by Simon Monk, Publisher: O'Reilly Media, Inc., 5. 2019, ISBN: 978149204322
3. David Jensen, “Beginning Azure IoT Edge Computing: Extending the Cloud to the Intelligent Edge, MICROSOFT AZURE

REFERENCE BOOKS:

1. Flavio Bonomi, Rodolfo Milito, Jiang Zhu, Sateesh Addepalli, Fog Computing and Its Role in the Internet of Things, MCC’12, August 17, 2012, Helsinki, Finland. Copyright 2012 ACM 978- 1-4503-1519-7/12/08.
2. Shanhe Yi, Cheng Li, Qun Li, A Survey of Fog Computing: Concepts, Applications and Issues, Mobidata’15, ACM 978-1-4503-3524-9/15/06, DOI: 10.1145/2757384.2757397, June 21, 2015, Hangzhou, China

WEBSITES:

1. https://www.geeksforgeeks.org/software-engineering/difference-between-edge-computing-and-fog-computing/
2. https://www.akamai.com/blog/edge/fog-computing-edge-computing-roles-modern-technology
3. https://www.techtarget.com/searchnetworking/answer/Fog-computing-vs-edge-computing-Whats-the-difference

JOURNALS:

1. Mansouri W, Alohali MA, Alqahtani H, Alruwais N, Hamza MA, El-Latif AAA. Deep convolutional neural network-based enhanced crowd density monitoring for intelligent urban planning on smart cities. Sci Rep. 2025;15(1):1847. doi:10.1038/s41598-025-90430-4.
2. Pan Z, Liang Y, Wang W, Yu Y, Zheng Y, Zhang J. Urban traffic prediction from spatio-temporal data using deep meta learning. In: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining; 2019 Aug 4–8; Anchorage, AK, USA. New York, NY, USA: ACM; 2019. p. 1720–30.
3. Guo S, Lin Y, Feng N, Song C, Wan H. Attention-based spatial-temporal graph convolutional networks for traffic flow forecasting. In: Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 33. Palo Alto, CA, USA: AAAI Press; 2019. p. 922–9.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	-	-	**2**	-	**3**	-	-	**3**	-	-	-	-	-	-	-	**3**	-
CO2	**2**	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	**2**
CO3	-	-	-	**1**	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	**2**	-	**2**	-	-	**3**	-	-	-	-	-	-	-	-	-	-	-
CO5	-	-	-	-	**3**	-	-	**3**	-	-	-	-	-	-	**1**	-	-
Average	**2**	-	**2**	**1**	**3**	**3**	-	**3**	-	-	-	-	-	-	**1**	**3**	**2**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

M. Sc. Computer Science **2026-2027**

Category: Major **SEMESTER II**

26CSP202C **Soft Computing** **4H - 4C**

Instruction Hours / Week: L: 4 T: 0 P: 0 **Marks: Internal: 40 External: 60 Total: 100**

End Semester Exam: 3 Hours

PREREQUISITE:

- Basic concepts of Artificial Intelligence and Knowledge of Algorithms.

COURSE OBJECTIVES (CO):

- To understand the fundamentals of soft computing techniques.
- To learn about the problem-solving techniques efficiently.
- To learn about generation of Neural Networks.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply fundamental concepts of soft computing including fuzzy computing, neural computing, and genetic algorithms to model and solve complex problems.	K3
CO2	Analyze ANN Algorithms for solving Optimization Problems.	K4
CO3	Evaluate Advanced Neural Network Models and Hybrid Soft Computing Techniques.	K5
CO4	Apply Fuzzy Logic to Solve Real-World Problems	K3
CO5	Analyze the working principles of genetic algorithms, schema theorem, and evolutionary strategies to solve optimization problems.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I - INTRODUCTION TO SOFT COMPUTING **12 HOURS**

Introduction of soft computing, soft computing vs. hard computing, various types of soft computing techniques, Fuzzy Computing, Neural Computing, Genetic Algorithms, Associative Memory, Adaptive Resonance Theory, Classification, Clustering, Bayesian Networks, Probabilistic reasoning, applications of soft computing.

UNIT II – ARTIFICIAL NEURAL NETWORK **12 HOURS**

Fundamental concept, Evolution of Neural Networks, Basic Models, McCulloh-Pitts Neuron, Linear Separability, Hebb Network. Special Networks: Simulated annealing, Boltzman machine, Gaussian Machine, Cauchy Machine, Probabilistic neural net, cascade correlation network, cognition network, neo cognition network, cellular neural network, optical neural network

UNIT III – THIRD GENERATION NEURAL NETWORKS **12 HOURS**

Spiking Neural networks, convolutional neural networks, deep learning neural networks, extreme learning machine model. Supervised Learning Network: Perceptron Networks, Adaptive Linear Neuron, Multiple Adaptive Linear Neurons, Backpropagation Network, Radial Basis Function, Time Delay Network, Functional Link Networks, Tree Neural Network-Un Supervised Learning Networks: Fixed weight competitive nets, Kohonen self-organizing feature maps, learning vectors quantization, counter propagation networks, adaptive resonance theory networks.

UNIT IV - INTRODUCTION TO FUZZY LOGIC **12 HOURS**

Classical sets, Fuzzy Sets-Classical Relations and Fuzzy Relations-Cartesian Product of relation-classical relation-fuzzy relations-tolerance and equivalence relations-non-iterative fuzzy sets-Defuzzification: Lambda-cuts for fuzzy sets-Lambda-cuts for fuzzy relations-Defuzzification methods.

UNIT V – GENETIC ALGORITHM **12 HOURS**

Traditional optimization and search techniques-genetic algorithm and search space-genetic algorithm vs. traditional algorithms-basic terminologies-problem solving using genetic algorithm-the schema theorem-classification of genetic algorithm- genetic programming-advantages and limitations and applications of genetic algorithm-Differential Evolution Algorithm-Hybrid soft computing techniques – neuro – fuzzy hybrid, genetic neuro-hybrid systems, genetic fuzzy hybrid and fuzzy genetic hybrid systems.

TOTAL:60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS

AI Tools / Software Tools	Purpose
Octave	Open-source alternative to MATLAB for soft computing applications.
Keras	High-level API for designing neural networks easily.

TEXT BOOKS:

1. Roy, S., & Chakraborty, U. (2024). "*Introduction to Soft Computing: Neuro-Fuzzy and Genetic Algorithms"*. 1st edition, Pearson India.
2. Hassoun, M. H. (2023). "*Fundamentals of Artificial Neural Networks"* (2nd ed.). MIT Press.
3. Van der Aalst, W. M. P., & Kurniawan, S. (2023). "Soft Computing for Complex Problem Solving" 1st edition, Springer.

REFERENCE BOOKS:

1. S. N. Sivanandam, S. N.Deepa, (2007). *"Principles of soft computing"*, 1st edition, John Wiley & Sons.
2. Timothy J. Ross. (2016) *"Fuzzy Logic with engineering applications",* 2nd edition, John Wiley & Sons.
3. Driankov D., Hellendoorn H. and Reinfrank M.(2001). *"An Introduction to Fuzzy Control",* 2nd edition, Narosa Publications

WEBSITES:

1. www.myreaders.info/html/soft_computing.html
2. https://archive.nptel.ac.in/courses/106/105/106105173/

JOURNALS:

1. Sourabh Katoch1 & Sumit Singh Chauhan1 & Vijay Kumar, "A review on genetic algorithm: past, present, and future", Multimedia Tools and Applications, Springer, 80:8091–8126, 2021.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	2	-	-	-	-	3	-
CO2	-	-	3	2	-	-	-	-	-	-	-	-	-	-	-	-	2
CO3	-	-	-	-	-	-	-	-	-	-	2	-	-	-	-	-	-
CO4	3	-	2	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	3	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-
Average	3	-	2	2	1	-	-	-	-	-	2	-	-	-	-	3	2

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER II**

26CSP241 **Web Development Using Angular JS & MongoDB** **5H - 3C**

Instruction Hours/week: L:1 T:0 P:4 **Marks: Internal:50 External:50 Total:100**

End Semester Exam: 3 Hours

PREREQUISTICS:

- Basic knowledge of HTML, CSS, & JavaScript
- Basic understanding of Databases, Frontend development, Text editor and execution of programs, etc.

COURSE OBJECTIVES (CO):

- To Construct a basic website using HTML and Cascading Style Sheets
- To learn the fundamentals of document databases technologies and to understand the significance of using MongoDB as a database system
- To understand the design issues in the development of backend components using Node.js and Express

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Classify the concepts of client- s i d e programming using CSS and Java Script	K4
CO2	Classify the concepts of Angular JS to extend basic HTML features	K4
CO3	Construct and build dynamic server-side applications	K4
CO4	Build Responsive Web application using Angular Typescript	K3
CO5	Use Mongo DB queries, tools and apply CRUD operations.	K3
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I: REFRESHING JAVA SCRIPT AND CSS **3T+12P = 15 HOURS**

Advanced Web Interface Design Using CSS and Bootstrap- JavaScript Programming and Asynchronous Web Application Development

PRACTICAL:

- Responsive E-Commerce Website Interface Using CSS and Bootstrap
- Interactive Student Portal with DOM Manipulation and Event Handling
- Design a student registration form using Bootstrap form controls and validation styles.
- Create a to-do list application using AngularJS directives.

UNIT II: ANGULAR JS APPLICATION **3T+12P = 15 HOURS**

AngularJS Fundamentals and Application Architecture- Single Page Application Development Using AngularJS

PRACTICAL:

- Create a program to display a list of items using ng-repeat directive.
- Develop a user registration form with AngularJS form validation.

- Develop a structured AngularJS application with multiple controllers and views.
- Create an SPA for student management with add, view, and delete operations.

UNIT III: NODE JS AND REST API DEVELOPMENT **3T+12P = 15 HOURS**

Node.js Runtime Environment - Network Programming, REST API Development, and Scalable Web Applications in Node.js

PRACTICAL:

- Real-Time Chat Application Using Node.js Event Loop and Web Sockets
- Develop a program to demonstrate Node.js modules
- Develop a REST API to perform CRUD operations (Create, Read, Update, Delete).
- Write a program to implement schema validation using Mongoose.

UNIT IV: INTRODUCTION TO NoSQL **3T+12P = 15 HOURS**

Fundamentals and Architecture of NoSQL Databases- NoSQL Database Models and Applications

PRACTICAL:

- To program to create and connect to a NoSQL database using MongoDB shell.
- Develop a program to query documents using conditions.
- Develop a program to compare relational model vs NoSQL document model.
- XML-Based Healthcare Record Management System
- Social Network Relationship Analysis Using Graph Database
- Create a program to analyze query performance using explain () method.

UNIT V: NODE JS and MONGODB **3T+12P = 15 HOURS**

MongoDB Fundamentals and Node.js Integration- MongoDB Administration and Scalable Database Management

PRACTICAL:

- Student Information Management System Using Node.js and MongoDB CRUD Operations
- Create collections and insert single and multiple JSON documents.
- Hospital Patient Record Storage and Retrieval System Using MongoDB Projections
- Real-Time Inventory Monitoring System Using MongoDB Update and Aggregation Operations
- Create a program to sort, limit, and skip documents in query results.
- Banking Transaction Analysis System Using Aggregation Framework in MongoDB
- Distributed Big Data Storage System Using MongoDB Replication and Sharding
- Automated Backup and Recovery Management System for MongoDB Databases
- Develop a scalable REST API using Node.js and MongoDB for high-volume data processing.

TOTAL: 75 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

Tools	Purpose
Visual Studio Code	A free code editor with AI-powered extensions (like IntelliSense) for developing JavaScript, AngularJS, Node.js, and MongoDB applications
GitHub Copilot	AI-powered coding assistant that helps in writing JavaScript, Node.js APIs,

	debugging, and learning syntax faster
Postman	Used for testing REST APIs, handling HTTP requests, and validating Node.js backend services
MongoDB Compass	Provides a visual interface to manage MongoDB databases, perform CRUD operations, indexing, and aggregation
Firebase	A cloud-based NoSQL platform used for real-time database management, authentication, and scalable web app backend development

TEXT BOOKS:

1. Ruebbelke,A(2013), "Angularjs in Action", ISBN 9789351198383, 3rd Edition. Wiley Publication.
2. Alex Young, Bradley Meck, Mike Cantelon, Tim Oxley, Marc Harter, T.J. Holowaychuk, Nathan Rajlich (2018), "Node.js in Action", ISBN 9789386052049, Second Editon,Wiley Publication.
3. Alex Young, Marc Harter, Ben Noordhuis (2017), "Node.Js in Practice", ISBN 9789351197744 3rd Edition. Wiley Publication.
4. Jeffrey C and Jackson,(2011), "Web Technologies A Computer Science Perspective", 3rd Edition. Pearson Education., New Delhi

REFERENCE BOOKS:

1. Shyam Seshadri (2019), "Angular: Up and Running: Learning Angular, Step by Step", 1st edition, O′Reilly Publisher.
2. Pro MERN (2015), "Stack, Full Stack Web App Development with Mongo, Express, React, and Node,
3. First Edition, Pearson Education.

WEBSITES:

1. https://www.tutorialspoint.com/mongodb/mongodb_tutorial.pdf
2. hadoop.apache.org/
3. https://neo4j.com/
4. https://nptel.ac.in/courses/106104189/
5. www.mongodb.com

JOURNALS:

1. S. Sachar, Kamini, & L. Suneja, "Review Paper on Mean Stack for Web Development". International Journal for Scientific Research & Development, vol. 5, no.1, pp.497-498, April 2017.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**1**	**-**	**-**	**-**	**3**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**2**	**-**
CO2	**-**	**-**	**3**	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2**
CO3	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**-**	**-**	**-**	**3**	**-**	**-**	**2**	**1**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO5	**-**	**-**	**3**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
Average	**1**	**-**	**3**	**3**	**3**	**-**	**2**	**1**	**1**	**-**	**2**	**-**	**-**	**-**	**-**	**2**	**2**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER II**

26CSP242 **Machine Learning Operations** **5H - 4C**

Instruction Hours/week: L:3 T:0 P:2 **Marks: Internal:50 External:50 Total:100**

End Semester Exam: 3T+2P=5 Hours

PREREQUISITE:

- Basic understanding of Machine Learning.

COURSE OBJECTIVES (CO):

- To understand the fundamentals of MLOps and its significance in the machine learning (ML) lifecycle.
- To Learn various tools and technologies used in MLOps to design and build scalable ML pipelines.
- To Evaluate hands-on exposure to deploying ML models in production environments.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply MLOps principles to implement ML model version tracking and lifecycle management in Python-based environments.	K3
CO2	Apply preprocess data for ML workflows.	K3
CO3	Analyze Docker and Kubernetes-based deployment architectures for scalable machine learning model serving.	K4
CO4	Monitor, optimize, and secure ML systems.	K4
CO5	Design and manage MLOps infrastructure.	K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I INTRODUCTION **9T+6P = 15 HOURS**

Overview of MLOps and its importance- Understanding the challenges in deploying and managing ML models-ML development lifecycle-Role of MLOps in the ML development lifecycle-Introduction to DevOps and its application to ML.

PRACTICAL:

- ML Model Version Tracking using Python
- Simple ML Lifecycle Stage Display

UNIT II DATA MANAGEMENT **9T+6P = 15 HOURS**

Data versioning and reproducibility-Data preprocessing and feature engineering pipelines-Data validation and monitoring-Data quality assurance and governance-Model versioning and tracking-Model training pipelines and automation-Hyperparameter tuning and model selection-Model evaluation and validation techniques.

PRACTICAL:

- Implementation of Data Versioning and Reproducibility for Student Dataset
- Implement data versioning using DVC (Data Version Control) for ML datasets.

UNIT III MODEL DEPLOYMENT AND SERVING **9T+6P = 15 HOURS**

Model packaging and containerization (e.g., Docker)-Infrastructure provisioning and orchestration (e.g., Kubernetes)-Deploying models as scalable services-managing model endpoints and versioning-Version control and collaboration (e.g., Git)-Building reproducible ML pipelines-Automated testing and code quality checks-Continuous integration and deployment strategies.

PRACTICAL:

- Docker based Model packaging
- Create a GitHub repository for ML project and manage version history.

UNIT IV MONITORING AND PERFORMANCE OPTIMIZATION **9T+6P = 15 HOURS**

Monitoring model performance and behavior-Real-time and batch monitoring techniques-Logging and error tracking in ML systems-Performance optimization and scalability considerations- Data privacy and protection in ML systems-Access control and authentication mechanisms-Security considerations for model deployment-Compliance with industry regulations (e.g., GDPR, HIPAA).

PRACTICAL:

- Monitoring Model Performance
- Logging and Error Tracking
- Access Control in ML Systems

UNIT V INFRASTRUCTURE FOR MLOPS **9T+6P = 15 HOURS**

Model versioning and governance-Retraining and revalidation strategies-Model deployment and retirement-Ensuring fairness, transparency and accountability. Introduction to cloud platforms (e.g., AWS, Azure, GCP)-Deploying ML models on cloud infrastructure, Managing resources and scaling ML workloads, Cost optimization strategies for ML systems.

PRACTICAL:

- Demonstrate Model Retraining Strategy
- Demonstrate Cloud Resource Scaling
- Compare old and new model versions using evaluation metrics (accuracy, F1-score).

TOTAL: 75 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS

AI Tools / Software Tools	Purpose
MLflow	Lifecycle & model management
Kubeflow	Pipeline automation & deployment
Data Version Control	Monitoring & optimization

TEXT BOOKS:

1 Noah Gift. (2021). Practical MLOps: A Guide to Building Real-World Machine Learning Systems (1st ed.). O'Reilly Media.

2 Mark Treveil, & Nicolas Omont. (2021). Introducing MLOps: How to Scale Machine Learning in the Enterprise (1st ed.). O'Reilly Media.

3 Emmanuel Raj. (2021). Engineering MLOps: Rapidly build, test, and manage production-ready machine learning life cycles at scale (1st ed.). Packt Publishing Limited.

REFERENCE BOOKS:

1 Hannes Hapke, & Catherine Nelson. (2020). Building Machine Learning Pipelines: Automating Model Life Cycles with TensorFlow (1st ed.). O'Reilly Media.

2 Chris Fregly, & Antje Barth. (2021). Data Science on AWS: Implementing End-to-End Continuous Machine Learning Pipelines (1st ed.). O'Reilly Media.

3 Sridhar Alla, & Suman Kalyan Adari. (2020). Beginning MLOps with MLFlow: Deploy Models in AWS SageMaker, Google Cloud, and Microsoft Azure (1st ed.). Apress.

WEBSITES:

1 https://cloud.google.com/architecture/mlops-continuous-delivery-and-automation-pipelines

2 https://aws.amazon.com/what-is/mlops/

3 https://buildmlops.com/resources

JOURNALS:

1 https://doi.org/10.1109/ICSE.2019.00039

2 https://doi.org/10.1016/j.jss.2020.110724

3 https://doi.org/10.1145/3132747

CO, PO, PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	3	-	-	-	-	1	-	-	3	3	2	-	-	-	-	-	-
CO2	3	-	3	2	2	1	-	2	-	-	-	-	-	-	-	3	3
CO3	3	-	-	2	2	1	-	2	-	-	-	-	-	-	-	-	2
CO4	3	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-
CO5	3	-	-	-	-	1	-	1	-	-	-	-	-	-	-	-	1
Average	3	-	3	2	2	1	-	2	3	3	1	-	-	-	-	3	2

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER II**

26CSP243 **Cyber Security** **5H - 4C**

Instruction Hours/week: L:3 T:0 P:2 **Marks: Internal:50 External:50 Total:100**

End Semester Exam: 3T+2P=5 Hours

PREREQUISITE:

- Basic knowledge of computer systems, internet usage, and fundamental programming concepts along with an understanding of basic networking concepts.

COURSE OBJECTIVES (CO):

- To understand the fundamental concepts of cybersecurity, cyberspace architecture, and security challenges in modern digital systems.
- To gain knowledge of cybercrime types, cyber laws (IT Act 2000), and methods for cybercrime prevention, detection, and reporting.
- To develop practical skills in implementing cybersecurity solutions such as network security tools, social media security analysis, and secure digital systems.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Analyze the structure and architecture of cyberspace, including communication and web technologies used in modern internet infrastructure.	K4
CO2	Analyze the nature, scope, and evolution of cyber-crime and classify different types of cyber-criminal activities in digital environments.	K4
CO3	Apply basic cyber security tools and techniques to detect threats such as unsafe URLs, IP validation issues, and malware indicators.	K3
CO4	Implement simple security analysis systems for social media, digital transactions, and network communication using practical tools.	K3
CO5	Develop and demonstrate basic cybersecurity solutions for real-world applications such as device security, authentication systems, and cybercrime tracking.	K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I PRINCIPLES OF CYBER SECURITY **9T+6P = 15 HOURS**

Introduction to Cyber security - Defining cyberspace – Architecture of cyberspace - Communication and web technology - Internet infrastructure for data transfer and governance - Regulation of cyber space - Concept of Cyber security - Issue and challenges of cyber security.

PRACTICAL:

- Setting, configuring and managing three password policy in the computer (BIOS, Administrator and Standard User).
- Security patch management and updates in Computer and Mobiles.

UNIT II CYBER CRIME **9T+6P = 15 HOURS**

Introduction to Cyber-crime - Classification of Cyber-crimes – Cyber-crime against women and children – Financial frauds - Social engineering attacks – Malware - Zero day and zero click attacks

PRACTICAL:

- Installation and configuration of Computer Host Firewall.
- To configure virtual networks using network simulator

UNIT III CYBER LAW **9T+6P = 15 HOURS**

Cyber Criminals modus-operandi – Reporting of cybercrimes – remedial and mitigation measures – Legal perspective of cybercrime– IT Act 2000 and its amendments – Organization dealing with cyber crimes and cyber security in India.

PRACTICAL:

- To install and exploit security tools for protecting a network
- To implement cryptographic algorithm for building a secure communication network

UNIT IV SOCIAL MEDIA SECURITY **9T+6P = 15 HOURS**

Introduction to social network – Types of social media – Social media platform – Hashtag – Viral content – Security issues related to social media. – Cyber Security tools: N map – Introduction to Nmap – Nmap scan types- Nmap command list. Digital Transaction: Introduction to digital payments – Components of digital payments – Modes of digital payments – Banking cards – UPI (Unified Payment Interface) – e-Wallets.

PRACTICAL:

- To exploit the vulnerabilities in a LAN environment and launch attacks
- To analyze the network packet using Wireshark
- To perform the web penetration testing using Burp suite

UNIT V DIGITAL DEVICES SECURITY **9T+6P = 15 HOURS**

Digital Devices Security: End point device and Mobile phone security – Password policy –Security patch management – Data backup – Device security policy – Cyber security best practices. Installation and configuration of Computer Anti-Virus. **Case studies:** Illustrations of Financial frauds – Digital Signature. Prepare a checklist for secure net banking

PRACTICAL:

- To perform vulnerability assessment of wireless devices
- To exploit vulnerabilities in the systems
- To perform the log analysis using Splunk
- To find vulnerable apps in play store and perform static and dynamic analysis on it

TOTAL: 75 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Wireshark	Detect suspicious network traffic
VirusTotal	Detect malware files using AI and global threat intelligence
Nmap	Identify vulnerabilities in systems

TEXT BOOKS:

1 Stallings, W. (2017). Cryptography and Network Security: Principles and Practice. Pearson Education.

2 Whitman, M. E., & Mattord, H. J. (2021). Principles of Information Security. Cengage Learning.

3 Easttom, C. (2020). Computer Security Fundamentals. Pearson Education.

REFERENCE BOOKS:

1 Daswani, N., & Gupta, C. (2016). Foundations of Cyber Security. Springer.

2 Furnell, S. (2019). *Cybercrime: Vandalizing the Information Society*. Springer.

3 Gupta, B. B. (2021). *Handbook of Computer Networks and Cyber Security*. CRC Press.

WEBSITES:

1 https://www.cisa.gov

2 https://owasp.org

3 https://www.cert-in.org.in

JOURNALS:

1 IEEE Editorial Board. (2014–present). IEEE Internet of Things Journal. IEEE Internet of Things Journal. IEEE, USA.

2 Elsevier Editorial Board. (1982–present). Computers & Security. Computers & Security Journal. Elsevier, Netherlands.

3 Wang, L., & Liu, J. (Eds.). (2015–present). Journal of Cyber Security Technology. Journal of Cyber Security Technology. Taylor & Francis.

CO, PO, PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	3	-	-	-	-	1	-	-	3	3	2	-	-	-	-	-	-
CO2	3	-	3	2	2	1	-	2	-	-	-	-	-	-	-	3	3
CO3	3	-	-	2	2	1	-	2	-	-	-	-	-	-	-	-	2
CO4	3	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-
CO5	3	-	-	-	-	1	-	1	-	-	-	-	-	-	-	-	1
Average	3	-	3	2	2	1	-	2	3	3	1	-	-	-	-	3	2

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

26CSP203 Community Engagement and Social Responsibility 2H - 2C

Instruction Hours / Week: L: 2 T: 0 P: 0 **Marks: Internal: 40 External: 60 Total: 100**

End Semester Exam: 3 Hours

PREREQUISITE:

- Not Required.

COURSE OBJECTIVES (CO):

- To provide students with an understanding of rural society, culture, values, and the significance of community engagement in sustainable social development.
- To develop knowledge of rural economy, livelihoods, agriculture, entrepreneurship, and resource management for promoting inclusive development.
- To familiarize students with rural and local governance institutions and their role in community participation, local administration, and social responsibility.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Explain the characteristics of rural society, community values, and rural infrastructure in the context of community engagement.	K2
CO2	Describe the structure and functioning of the rural economy, including agriculture, water management, animal husbandry, and rural entrepreneurship.	K2
CO3	Examine the role of rural livelihoods, rural markets, artisans, and migrant labour in local economic development.	K3
CO4	Illustrate the functions of Panchayati Raj Institutions, Gram Sabha, municipalities, Self-Help Groups, and other rural institutions in local governance.	K2
CO5	Analyze the importance of community participation, local governance, and social responsibility in achieving sustainable rural development.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I APPRECIATION OF RURAL SOCIETY 06 HOURS

Rural lifestyle, rural society, caste and gender relations, rural values with respect to community, nature and resources, elaboration of "soul of India lies in villages' (Gandhi), rural infrastructure.

UNIT II UNDERSTANDING RURAL AND LOCAL ECONOMY AND LIVELIHOOD 06 HOURS

Agriculture, farming, land ownership, water management, animal husbandry, non-farm livelihoods and artisans, rural entrepreneurs, rural markets, migrant labour.

UNIT III RURAL AND LOCAL INSTITUTIONS **06 HOURS**

Traditional rural and community organisations, Self-help Groups, Panchayati raj institutions (Gram Sabha, Gram Panchayat, Standing Committees), Nagarpalikas and municipalities, local civil society, local administration.

ACTIVITIES **12 HOURS**

- ❖ Village Socio-Economic Survey (Field-Based)
- ❖ Financial Literacy and Digital Banking Awareness Camp
- ❖ Government Scheme Awareness Drive
- ❖ Rural Entrepreneurship Development Workshop
- ❖ SHG Performance Evaluation
- ❖ Village Resource Mapping
- ❖ Waste Management and Green Campus–Green Village Initiative
- ❖ Social Business Plan Competition
- ❖ Community Market Survey
- ❖ Adopt-a-Village Development Project (Semester Activity)

Each student group adopts a village or urban locality and undertakes one measurable intervention, such as:

- Financial literacy
- Entrepreneurship mentoring
- Digital literacy
- Waste management
- Skill development
- Government scheme awareness

TOTAL: 30 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS

AI Tools / Software Tools	Purpose
ChatGPT	Learn concepts of rural society, local governance, community engagement, rural entrepreneurship, government schemes, and prepare awareness materials, reports, and field activity documentation.
Canva AI	Design posters, pamphlets, infographics, social media content, and presentations for financial literacy, digital literacy, government scheme awareness, and environmental campaigns

REFERENCE BOOKS:

1 Singh Katar. Rural Development: Principles, Policies and Management. Sage Publications, New Delhi, 2015.

2 A Handbook on Village Panchayat Administration. Rajiv Gandhi Chair for Panchayati Raj Studies, 2002.

3 G. Mahida, Rinkesh Kumar. Rural Development in India: Policies and Programmes. Red Shine Publication, 2024.

4 M. P. Boraian. Best Practices in Rural Development. Shanlax Publishers, 2016.

WEBSITES:

1. https://rural.gov.in
2. https://www.mygov.in
3. https://vikaspedia.in

CO, PO, PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	-	-	-	-	-	3	-	-	-	-	2	3	3	-	-	-	2
CO2	-	-	-	-	-	3	-	-	-	-	2	3	3	-	-	-	2
CO3	-	-	-	-	-	3	-	-	-	-	2	3	3	-	-	-	2
CO4	-	-	-	-	-	3	-	-	-	-	2	3	3	-	-	-	2
CO5	-	-	-	-	-	3	-	-	-	-	2	3	3	-	-	-	2
Average	-	-	-	-	-	3	-	-	-	-	2	3	3	-	-	-	2

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER II**

26CSP244 **Professional Certification – UI/UX Design** **3H - 2C**

Instruction Hours/week: L:1 T:0 P:2 **Marks: Internal:- External:- Total:-**

End Semester Exam: -

PREREQUISITES:

Basic knowledge of computers, web technologies, and an interest in designing user-friendly digital interfaces.

COURSE OBJECTIVES(CO):

- To provide a sound knowledge in UI & UX.
- To understand the need for UI and UX.
- To understand the various research methods used in Design.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Build user interfaces for applications.	K3
CO2	Evaluate UX design for products and applications.	K3
CO3	Demonstrate UX skills in product development.	K4
CO4	Apply sketching principles in UI/UX design.	K5
CO5	Create wireframes and interactive prototypes.	K6
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I FOUNDATIONS OF DESIGN **3T+6P = 09 HOURS**

UI vs. UX Design - Core Stages of Design Thinking - Divergent and Convergent Thinking - Brainstorming and Game Storming - Observational Empathy

PRACTICAL:

- Designing a responsive layout for a societal application.
- Exploring various UI interaction patterns.
- Developing an interface with proper UI style guides.
- Developing a wireflow diagram for an application using open-source software.

UNIT II FOUNDATIONS OF UI DESIGN **3T+6P = 09 HOURS**

Visual and UI Principles - UI Elements and Patterns - Interaction Behaviors and Principles - Branding - Style Guides

PRACTICAL:

- Exploring various open-source collaborative interface platforms.
- Hands-on Design Thinking process for a new product.
- Brainstorming features for a proposed product.
- Defining the look and feel of the new project.

UNIT III FOUNDATIONS OF UX DESIGN **3T+6P = 09 HOURS**

Introduction to User Experience - Why You Should Care about User Experience - Understanding User Experience - Defining the UX Design Process and its Methodology - Research in User

Experience Design - Tools and Methods Used for Research - User Needs and its Goals - Know about Business Goals

PRACTICAL:

- Creating a sample pattern library for the product (Mood Board, Fonts, Colors based on UI principles).
- Identifying a customer problem to solve.
- Sketching and designing with a popular tool, building a prototype, performing usability testing, and identifying improvements.

UNIT IV WIREFRAMING, PROTOTYPING AND TESTING 3T+6P = 09 HOURS

Sketching Principles - Sketching Red Routes - Responsive Design - Wireframing - Creating Wireflows - Building a Prototype - Building High-Fidelity Mockups - Designing Efficiently with Tools - Interaction Patterns - Conducting Usability Tests - Other Evaluative User Research Methods - Synthesizing Test Findings - Prototype Iteration

PRACTICAL:

- Draw a paper wireframe for the home page of a website.
- Create a simple mobile app wireframe using Figma or Penpot.
- Design a responsive login page for desktop and mobile screens.
- Create a wireflow showing the navigation between three application screens.

UNIT V RESEARCH, DESIGNING, IDEATING, & INFORMATION ARCHITECTURE 3T+6P = 09 HOURS

Identifying and Writing Problem Statements - Identifying Appropriate Research Methods - Creating Personas - Solution Ideation - Creating User Stories - Creating Scenarios - Flow Diagrams - Flow Mapping - Information Architecture

PRACTICAL:

- Prepare three user stories for the application.
- Draw a simple user flow diagram for completing a task.
- Create a sitemap (Information Architecture) for the application.
- Present the final UI/UX design with a brief explanation.

TOTAL: 45 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS

AI Tools / Software Tools	Purpose
Figma AI	Generate UI layouts, design suggestions, and automate design tasks.
Khroma	Generate AI-based color palettes for interface design.
Adobe Firefly	Generate images, icons, and visual assets using AI.
Maze AI	Analyze usability testing results and improve user experience.

TEXT BOOKS:

1. Joel Marsh, UX for Beginners, O'Reilly Media, 2022.
2. Jon Yablonski, Laws of UX: Using Psychology to Design Better Products & Services, O'Reilly Media, 2021.

REFERENCE BOOKS:

1. Jenifer Tidwell, Charles Brewer, and Aynne Valencia, *Designing Interface: Patterns for Effective Interaction Design*, 3rd Edition, O'Reilly Media, 2020.
2. Steve Schoger and Adam Wathan, *Refactoring UI*, 2018.

3. Steve Krug, *Don't Make Me Think, Revisited: A Common Sense Approach to Web and Mobile Usability*, 3rd Edition, New Riders, 2015.
4. https://www.nngroup.com/articles/
5. https://www.interaction-design.org/literature/

WEBSITES:

1. https://www.nngroup.com
2. https://www.interaction-design.org
3. https://www.figma.com
4. https://penpot.app

JOURNALS:

1. Association for Computing Machinery. (n.d.). *ACM Transactions on Computer-Human Interaction (TOCHI).* https://dl.acm.org/journal/tochi
2. Elsevier. (n.d.). *International Journal of Human-Computer Studies.* https://www.sciencedirect.com/journal/international-journal-of-human-computer-studies
3. Taylor & Francis. (n.d.). *International Journal of Human–Computer Interaction.* https://www.tandfonline.com/journals/hihc20

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**3**	**-**	**-**	**1**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**2**	**-**
CO2	**-**	**-**	**3**	**-**	**-**	**-**	**2**	**3**	**-**	**-**	**2**	**-**	**-**	**-**	**3**	**-**	**-**
CO3	**-**	**-**	**3**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**-**	**-**	**-**	**1**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO5	**-**	**-**	**-**	**1**	**2**	**-**	**2**		**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
Average	**3**	**-**	**3**	**1**	**2**	**-**	**2**	**1**	**2**	**-**	**2**	**-**	**-**	**-**	**3**	**2**	**-**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

M.Sc Computer Science

Category: Major **SEMESTER II**

26CSP211 **Data Analytics using Power BI** **0H - 1C**

Instruction Hours/week: L:0 T:0 P:0 **Marks: Internal:- External: 50 Total:50**

End Semester Exam: 3 Hours

PREREQUISITE:

- Basic knowledge of statistics, data handling, and computer fundamentals.

COURSE OBJECTIVES (CO):

- Understand the importance of data cleaning and preprocessing.
- Apply data transformation techniques using Power BI.
- Create visualizations to analyze and present cleaned data.

COURSE OUTCOMES (COs):

COs	Course Outcomes	Blooms Level
CO1	Apply the importance of data preprocessing techniques such as handling missing values, duplicates, and inconsistent data in business analytics.	K3
CO2	Apply Power Query transformation techniques to clean datasets by replacing null values, removing duplicate records, and correcting data types.	K3
CO3	Analyze data transformation operations, including splitting and merging columns, standardizing text formats, trimming spaces, and correcting inconsistent values.	K4
CO4	Analyze datasets to identify outliers, invalid records, and data quality issues, and evaluate the impact of data cleaning on dataset reliability.	K4
CO5	Evaluate interactive visualizations and summary reports in Power BI to compare raw and cleaned datasets and communicate data quality improvements effectively.	K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

PRACTICAL:

1. Sales Performance Dashboard using Power BI
2. Design an Interactive Sales Performance Dashboard
3. Develop a Customer Analytics Dashboard with Filters and Slicers
4. Build an HR Analytics Dashboard to Analyze Employee Performance and Attrition
5. Develop an Inventory Management Dashboard with Stock and Supplier Analysis
6. Create a Student Performance Analytics Dashboard
7. Develop a Healthcare Analytics Dashboard for Patient Data
8. Build an E-Commerce Sales and Customer Insights Dashboard
9. Design a Retail Store Performance Dashboard
10. Create a Banking Analytics Dashboard for Deposits and Loans

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Power BI AI Insights	Automated pattern detection
OpenRefine	Data cleaning

TEXT BOOKS:

1. Knight, B., & Knight, D. (2019). Microsoft Power BI complete reference: Bring your data to life. McGraw-Hill Education.
2. Powell, B. (2019). Power BI cookbook: Creating business intelligence solutions of analytical data models, reports, and dashboards. Packt Publishing.
3. Ferrari, A., & Russo, M. (2018). Introducing DAX in Power BI: Basic concepts of data analysis expressions. SQLBI Publishing.

REFERENCE BOOKS:

1. Kimball, R., & Ross, M. (2013). The data warehouse toolkit: The definitive guide to dimensional modeling (3rd ed.). Wiley.
2. Howson, C. (2014). Successful business intelligence: Secrets to making BI a killer app. McGraw-Hill Education.
3. Clark, D. (2018). Beginning Power BI with Excel 2016. Apress.

WEBSITES:

1. https://learn.microsoft.com/power-bi
2. https://www.microsoft.com/power-platform
3. https://www.kaggle.com (datasets)
4. https://www.datacamp.com
5. https://www.edx.org

JOURNALS:

1. Sharma, S., & Singh, R. (2020). Data analytics and business intelligence: A systematic review. International Journal of Data Science and Analytics, 10(2), 85–102.
2. Wang, Y., & Wang, Y. (2015). Power BI and modern business intelligence systems. IEEE Transactions on Knowledge and Data Engineering, 27(9), 2400–2412.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	3	3	-	-	3	-	-	-	-	-	-	-	-	-	2	-	-
CO2	3	2	-	-	3	-	2	1	-	2	-	-	-	-	-	-	1
CO3	3	-	2	3	-	2	-	-	-	-	-	-	-	-	-	-	-
CO4	3	-	2	-	-	2	-	1	-	3	-	-	-	-	2	-	1
CO5	3	-	2	1	-	-	-	-	-	-	-	-	-	-	-	-	-
Average	**3**	**2.5**	**2**	**2**	**3**	**2**	**2**	**1**	**-**	**2.5**	**-**	**-**	**-**	**-**	**2**	**-**	**1**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category:- **SEMESTER II**

Journal Paper Analysis and Presentation **2H - 0C**

Instruction Hours/week: L:2 T:0 P:0 **Marks: Internal: - External: - Total: -**

End Semester Exam: -

Category: Major **SEMESTER III**

26CSP301 **Blockchain Technology** **4H - 4C**

Instruction Hours/week: L:4 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

PREREQUISTICS:

- Familiarity with fundamental concepts like algorithms, data structures, and networking.

COURSE OBJECTIVES (CO):

- To understand the concepts of block chain technology
- To understand the consensus and hyper ledger fabric in block chain technology.
- To assess blockchain applications in a structured manner.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Make use of basic concepts and technology used for blockchain.	K3
CO2	Determine the various technologies and protocols that make block chain possible.	K5
CO3	Analyze the block chain applications in a structure manner.	K4
CO4	Analyze the positive and negative implications of using blockchain technology in various industries.	K4
CO5	Apply security features in blockchain technologies.	K3
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I: INTRODUCTION **12 HOURS**

History: Digital Money to Distributed Ledgers-Design Primitives: Protocols, Security, Consensus, Permissions, Privacy- Block chain Architecture and Design-Basic crypto primitives: Hash, Signature Hash chain to Blockchain-Basic consensus mechanisms.

UNIT II: **BLOCK CHAIN PROTOCOLS** **12 HOURS**

Requirements for the consensus protocols-Proof of Work (PoW)-Scalability aspects of Block chain consensus protocols: Permissioned Blockchains-Design goals-Consensus protocols for Permissioned Blockchains.

UNIT III: COMPONENTS OF CHAIN **12 HOURS**

Decomposing the consensus process-Hyper ledger fabric components-Chain code Design and Implementation: Hyper ledger Fabric II-Beyond Chain code: fabric SDK and Front End-Hyper ledger composer tool.

UNIT IV: BLOCK CHAIN IN FINANCIAL SOFTWARE AND SYSTEMS **12 HOURS**

Blockchain in Financial Software and Systems (FSS): Settlements-KYC-Capital Markets-Insurance Block chain in trade/supply chain: Provenance of goods, visibility, trade/supply chain finance, invoice management/discounting.

UNIT V: BLOCK CHAIN FOR GOVERNMENT **12 HOURS**

Blockchain for Government: Digital identity, land records and other kinds of record keeping between government entities, public distribution system/social welfare systems: Blockchain Cryptography: Privacy and Security on Blockchain.

TOTAL: 60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
PyCharm	Specialized IDE for Python development.
TensorFlow	Used for deep learning and AI model building.
Scikit-learn	Supports classification, regression, and clustering.

TEXT BOOKS:

1. Mark Gates, (2017), "*Block chain: Ultimate guide to understanding block chain, bit coin, crypto currencies, smart contracts and the future of money*", Third Edition, Wise Fox Publishing and Mark Gates.
2. Salman Baset, Luc Desrosiers, Nitin Gaur, Petr Novotny, Anthony O'Dowd, Venkatraman Ramakrishna, (2018), "*Hands-On Block chain with Hyper ledger: Building decentralized applications with Hyperledger Fabric and Composer*". Fourth Edition, Wise Fox Publishing and Mark Gates.
3. Bahga, Vijay Madisetti, (2017), "*Blockchain Applications: A Hands-On Approach*", Second Edition, Arshdeep Bahga, Vijay Madisetti publishers.

REFERENCE BOOKS:

1. Andreas Antonopoulos, (2014), "*Mastering Bitcoin: Unlocking Digital Crypto Currencies*", Fourth Edition, O'Reilly Media, Inc.
2. Melanie Swa, (2014), "*Blockchain"*, Third Edition, O'Reilly Media.

WEBSITES:

1. https://nptel.ac.in/courses/106105184/
2. www.tutorials.point

JOURNALS:

1. Saha, Satarupa and Jana, Bappaditya and Poray, Jayanta, A Study on Blockchain Technology (October 15, 2019), Available at SSRN: https://ssrn.com/abstract=3477373.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**3**	**-**	**2**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2**	**-**
CO2	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**-**	**-**	**-**	**-**
CO3	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**3**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO5	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**-**	**-**
Average	**3**	**-**	**2**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**1**	**-**	**-**	**2**	**-**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER III**

26CSP302 **Digital Image Processing and Applications** **4H - 4C**

Instruction Hours/week: L:4 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

PREREQUISTICS:

- Basic knowledge of signals and systems will help in understanding image

COURSE OBJECTIVES (CO):

- To understand the image fundamentals and mathematical transforms necessary for image processing and to study the image enhancement techniques.
- To understand the image segmentation and representation techniques.
- To analyze object representation, description and recognition.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Analyze the principles of digital image formation, acquisition, digitization, and image representation techniques.	K4
CO2	Apply image transformation, enhancement, restoration, and segmentation techniques for improving image quality and extracting meaningful information.	K3
CO3	Analyze color image processing methods, feature extraction techniques, and image registration approaches for image analysis applications.	K4
CO4	Examine object representation, description, and pattern recognition techniques for solving computer vision and image understanding problems.	K4
CO5	Evaluate intelligent image processing solutions using feature-based classification and recognition techniques for real-world applications.	K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I: FUNDAMENTALS OF DIGITAL IMAGE PROCESSING **12 HOURS**

Introduction to Digital Image Processing- Applications of Digital Image Processing- Image Acquisition and Formation- Signal Digitization and Quantization- Sampling and Resolution- Pixel Relationships: Neighbors, Connectivity, Distance Measures- Camera Models and Imaging Geometry- Image Interpolation Techniques.

UNIT II: **IMAGE TRANSFORMATIONS AND ENHANCEMENT** **12 HOURS**

Image Transformation Fundamentals-Geometric Transformations- Intensity Transformations- Histogram Processing- Spatial Domain Enhancement Techniques- Image Enhancement using

Point Processing- Image Enhancement using Neighborhood Processing- Frequency Domain Enhancement Methods.

UNIT III: IMAGE RESTORATION AND REGISTRATION 12 HOURS

Image Degradation Models- Noise Models in Digital Images- Image Restoration Fundamentals- Spatial Filtering for Noise Reduction- Frequency Domain Restoration- Inverse Filtering- Wiener Filtering- Image Registration Concepts and Techniques- Applications of Image Registration.

UNIT IV: COLOR IMAGE PROCESSING AND SEGMENTATION 12 HOURS

Color Fundamentals and Color Models- Colour Transformations- Colour Image Enhancement- Image Segmentation Fundamentals- Thresholding Methods- Edge-Based Segmentation- Region-Based Segmentation-Object Description- Shape Descriptors- Boundary Descriptors- Regional Descriptors- Geometric Features- Texture Descriptors- Moments and Invariant Moments.

UNIT V: OBJECT REPRESENTATION AND RECOGNITION 12 HOURS

Representation of Regions and Boundaries- Boundary Representation Techniques- Chain Codes- Polygonal Approximation- Signatures- Skeleton Representation- Boundary Segments- Fundamentals of Pattern Recognition- Feature Extraction and Selection- Classification Techniques- Template Matching- Statistical Pattern Recognition- Structural Pattern Recognition.

TOTAL: 60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Pixlane AI Tools	OCR, object detection, segmentation, depth estimation
OpenCV	Used for filtering, segmentation, edge detection, morphology, object tracking
Hugging Face Spaces	Ready-made AI demos for segmentation, restoration, OCR, super-resolution

TEXT BOOKS:

1. Gonzalez, R. C., & Woods, R. E. (2018). Digital image processing (4th ed.). Pearson Education.
2. Jain, A. K. (1989). Fundamentals of digital image processing. Prentice Hall.
3. Gonzalez, R. C., Woods, R. E., & Eddins, S. L. (2020). Digital image processing using MATLAB (4th ed.). Pearson.

REFERENCE BOOKS:

1. Image Processing, Analysis, and Machine Vision Sonka, M., Hlavac, V., & Boyle, R. (2014). Image processing, analysis, and machine vision (4th ed.). Cengage Learning.
2. Digital Image Processing and Analysis Chanda, B., & Dutta Majumder, D. (2011). Digital image processing and analysis. PHI Learning.

WEBSITES:

1. https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_ee178
2. https://www.imageprocessingplace.com

JOURNALS:

1. Benoit Macq, "IEEE Transactions on Image Processing", IEEE Signal Processing Society, Vol. 33, 2024.
2. Jay Kumar Boyat, Brijendra Kumar Joshi, "A Review Paper: Noise Models in Digital Image Processing", arXiv Publication, 2015.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**3**	**-**	**2**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2**	**-**
CO2	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**-**	**-**	**-**	**-**
CO3	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**3**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO5	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**-**	**-**
Average	**3**	**-**	**2**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**1**	**-**	**-**	**2**	**-**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER III**

noc26_ee178 **Digital Image Processing** **4H - 4C**

Instruction Hours/week: L:4 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

PREREQUISITE:

- Basic knowledge of mathematics, signals and systems, linear algebra, and programming fundamentals.

COURSE OBJECTIVES (CO):

- To understand the fundamental concepts of digital image formation, image representation, enhancement, restoration, and segmentation techniques.
- To Apply image processing algorithms and tools to analyze, transform, and improve digital images for various applications.
- To Evaluate advanced image processing techniques for object recognition, computer vision, and real-world image analysis problems.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply the fundamentals of digital image processing, image acquisition, digitization, and image representation.	K3
CO2	Apply image transformation, enhancement, and filtering techniques to improve image quality.	K3
CO3	Analyze image degradation models and implement image restoration and registration methods.	K4
CO4	Analyze color image processing and segmentation techniques for image interpretation.	K4
CO5	Evaluate morphological operations, feature extraction, and object recognition methods for computer vision applications.	K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I: FUNDAMENTALS OF DIGITAL IMAGE PROCESSING **12 HOURS**

Introduction to Digital Image Processing- Applications of Digital Image Processing- Image Acquisition and Formation- Signal Digitization and Quantization- Sampling and Resolution- Pixel Relationships: Neighbors, Connectivity, Distance Measures- Camera Models and Imaging Geometry- Image Interpolation Techniques.

UNIT II: **IMAGE TRANSFORMATIONS AND ENHANCEMENT** **12 HOURS**

Image Transformation Fundamentals-Geometric Transformations- Intensity Transformations- Histogram Processing- Spatial Domain Enhancement Techniques- Image Enhancement using Point Processing- Image Enhancement using Neighborhood Processing- Frequency Domain Enhancement Methods.

UNIT III: IMAGE RESTORATION AND REGISTRATION **12 HOURS**

Image Degradation Models- Noise Models in Digital Images- Image Restoration Fundamentals- Spatial Filtering for Noise Reduction- Frequency Domain Restoration- Inverse Filtering- Wiener Filtering- Image Registration Concepts and Techniques- Applications of Image Registration.

UNIT IV: COLOR IMAGE PROCESSING AND SEGMENTATION **12 HOURS**

Color Fundamentals and Color Models- Colour Transformations- Colour Image Enhancement- Image Segmentation Fundamentals- Thresholding Methods- Edge-Based Segmentation- Region-Based Segmentation.

UNIT V: MORPHOLOGICAL PROCESSING **12 HOURS**

Morphological Image Processing Fundamentals- Structuring Elements- Dilation, Erosion, Opening and Closing- Morphological Algorithms- Object Representation and Description- Shape Representation and Feature Extraction- Boundary and Region Descriptors- Object Recognition Techniques.

TOTAL: 60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Pixlane AI Tools	OCR, object detection, segmentation, depth estimation
OpenCV	Used for filtering, segmentation, edge detection, morphology, object tracking
GNU Octave	Implementing image processing algorithms and experiments.

TEXT BOOKS:

1. Gonzalez, R. C., & Woods, R. E. (2018). Digital image processing (4th ed.). Pearson.
2. Jain, A. K. (1989). Fundamentals of digital image processing. Prentice Hall.
3. Pratt, W. K. (2007). Digital image processing: PIKS scientific inside (4th ed.). Wiley-Inder science.

REFERENCE BOOKS:

1. Burger, W., & Burge, M. J. (2022). Principles of digital image processing: Core algorithms. Springer.
2. Davies, E. R. (2018). Computer vision: Principles, algorithms, applications, learning (5th ed.). Academic Press.

WEBSITES:

1. https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_ee178
2. https://www.imageprocessingplace.com

JOURNALS:

1. Benoit Macq, “IEEE Transactions on Image Processing”, IEEE Signal Processing Society, Vol. 33, 2024.

2. Jay Kumar Boyat, Brijendra Kumar Joshi, "A Review Paper: Noise Models in Digital Image Processing", arXiv Publication, 2015.

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**3**	**-**	**2**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2**	**-**
CO2	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**-**	**-**	**-**	**-**
CO3	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**3**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO5	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**-**	**-**
Average	**3**	**-**	**2**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**1**	**-**	**-**	**2**	**-**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER III**

26CSP303A **Quantum Artificial Intelligence** **4H - 4C**

Instruction Hours/week: L:4 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

PREREQUISITE:

- Fundamentals of linear algebra, quantum mechanics, algorithms, and basic AI with programming knowledge.

COURSE OBJECTIVES (CO):

- To understand how the physical nature, as described by quantum physics, can lead to algorithms that imitate human behavior.
- To explore possibilities for the realization of artificial intelligence by means of quantum computation.
- To learn computational algorithms as described by quantum computation.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply the principles of quantum mechanics such as superposition, entanglement, and measurement to explain quantum system behavior in computing contexts.	K3
CO2	Apply quantum mechanics principles to qubit computation and quantum systems.	K3
CO3	Analyze entropy, information measures, and knowledge structures.	K4
CO4	Evaluate quantum algorithms and circuit models for problem-solving.	K5
CO5	Apply quantum computing models and algorithms to explore applications in machine learning, pattern recognition, and data processing tasks.	K3
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I INTRODUCTION **12 HOURS**

Physics in Quantum: The Journey to Quantum - Quantum Physics Essentials. Quantum theory: Quantum Mechanics – Quantum Decoherence – Quantum Entanglement – Quantum Measurement - Copenhagen Interpretation.

UNIT II QUANTUM COMPUTING **12 HOURS**

Qubits – classical computer vs Quantum computer – Superposition – entanglement – Multi-Qubit systems. Quantum Gates: Hadamard Gates – Phase Shift Gates – Pauli Gates – Toffoli Gates – Controlled Gates. Quantum Circuits – D-Wave Quantum Architecture.

UNIT III QUANTUM ALGORITHM **12 HOURS**

Introduction – Deutsch's Algorithm – Deutsch-Jozsa Algorithm - Simon's Algorithm – Shor's Algorithm – Grover's Algorithm-Quantum Fourier Transform (QFT)- Quantum Phase Estimation (QPE)- Variational Quantum Eigensolver (VQE)- Quantum Approximate Optimization Algorithm (QAOA).

UNIT IV QUANTUM COGNITIVE COMPUTING **12 HOURS**

A General Model of a Quantum Computer - Cognitive architecture - Representation – Quantum Cognition - Decision making - Unpacking Effects - Quantum walk on a graph – Quantum annealing - Optimization problems - Quantum Neural Computation - Applications on Quantum annealing Computer - Development libraries - Quantum Computer simulation tool kits.

UNIT V APPLICATIONS **12 HOURS**

A General Model of a Quantum Computer – Machine Learning Tasks Using Quantum Algorithms (Quantum Neural Networks) –Potential Speedups for Certain Learning Tasks and Data Processing– Applications in Pattern Recognition, Data Analysis, and Decision-Making-Other Emerging Applications-Quantum Computing in Communication Networks and the Quantum Web-Sensing and Quantum Metrology-Applications in Aerospace-Applications in Defence-Challenges and Opportunities.

TOTAL: 60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Qiskit	Circuits & Algorithms
QuTiP	Quantum Physics Simulation
TensorFlow Quantum	Quantum AI & Applications

TEXT BOOKS:

1 Wiley. Easttom, W.C. (2021). Quantum computing fundamentals. (1st ed.). Addison-Wesley Professional.

2 Stuart Russell, S., & Peter Norvig. (2021). Artificial intelligence: A modern approach (4th ed.). Pearson.

3 Tyagi, A. K., Tiwari, S., & Nagaraj, S. V. (2025). Quantum computing: The future of information processing (1st ed.). CRC Press, Taylor & Francis Group.

REFERENCE BOOKS:

1 Nielsen, M.A., & Chuang, I. L. (2010). Quantum computation and quantum information (10th anniversary ed.). Cambridge University Press.

2 Biamonte, G., Wittek, P., Pancotti, N., Rebentrost, P., Wiebe, N., & Lloyd, S. (2017). Quantum machine learning. Nature Publishing Group.

3 Sipser, M. (2012). Introduction to the theory of computation (3rd ed.). Cengage Learning.

WEBSITES:

1 https://quantum.ibm.com
2 https://azure.microsoft.com/en-us/solutions/quantum-computing/
3 https://www.dwavesys.com

JOURNALS:

1 https://www.nature.com/npjqi/
2 https://www.jair.org/
3 http://www.rintonpress.com/journals/qic

CO, PO, PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	3	-	-	-	-	1	-	-	3	3	2	-	-	-	-	-	-
CO2	3	-	3	2	2	1	-	2	-	-	-	-	-	-	-	3	3
CO3	3	-	-	2	2	1	-	2	-	-	-	-	-	-	-	-	2
CO4	3	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-
CO5	3	-	-	-	-	1	-	1	-	-	-	-	-	-	-	-	1
Average	**3**	**-**	**3**	**2**	**2**	**1**	**-**	**1.7**	**3**	**3**	**1.5**	**-**	**-**	**-**	**-**	**3**	**2**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

M.Sc Computer Science

Category: Major		**SEMESTER III**
26CSP303B	**Robotic Process Automation**	**4H - 4C**
Instruction Hours/week: L:4 T:0 P:0		**Marks: Internal:40 External:60 Total:100**
		End Semester Exam: 3 Hours

PREREQUISITE:

- Basic understanding of computers, programming concepts, and familiarity with web and office applications.

COURSE OBJECTIVES (CO):

- Explain and analyze the fundamental concepts of RPA, process methodologies, vendor evaluation, bot development, and future automation trends.
- Apply RPA tools and process improvement techniques to design, develop, and deploy automation solutions.
- Evaluate and design strategic RPA implementations, including vendor selection, governance models, and intelligent automation solutions.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply RPA concepts and development methodologies to identify and design automation solutions for business workflows.	K3
CO2	Analyze business processes using Lean and Six Sigma methodologies to identify suitable automation opportunities.	K4
CO3	Apply RPA tools and techniques to design, develop, test, and deploy software bots.	K3
CO4	Evaluate RPA vendors, governance models, and deployment strategies for effective enterprise implementation.	K5
CO5	Analyze leading RPA vendors (UiPath, Automation Anywhere, Blue Prism, etc.) and open-source RPA platforms to compare their features and capabilities.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I INTRODUCTION **12 HOURS**

Introduction to RPA Foundations- What is RPA - Flavors of RPA- History of RPA- The Benefits of RPA- The downsides of RPA- RPA Compared to BPO, BPM and BPA- Consumer Willingness for Automation- The Workforce of the Future- RPA Skills-On-Premise Vs. the Cloud- Web Technology- Programming Languages and Low Code- OCR-Databases-APIs- Al-Cognitive Automation-Agile, Scrum, Kanban and Waterfall-DevOps- Flowcharts.

12 HOURS

UNIT II PROCESS METHODOLOGIES

Introduction to Process Methodologies-Lean-Six Sigma-Implementation of Six Sigma-Six Sigma Roles and Levels-Lean Six Sigma-Finding the Right Balance-Applying Lean and Six Sigma to RPA-Planning-The Preliminaries-Use a Consulting Firm-RPA Consulting: Some Case Studies-What to Automate? ROI for RPA-RPA Use Cases-The Plan.

UNIT III RPA VENDOR EVALUATION **12 HOURS**

Introduction to RPA Vendor Evaluation-Be Realistic-Check Out Third Parties-Minimum Capabilities-Who is the user-Funding-Ecosystem-Costs-Training and Education-Support-Best-of-Breed vs. End-to-End-Thought Leadership and Vision-Industry Expertise-Security, Monitoring, and Deployment-What Type of RPA? -The Design-Next-Generation Technologies-Center of Excellence: What is the CoE? -Why have a CoE? -Forming the Team-Business Analyst-Developer-RPA Solution Architect-RPA Supervisor-What should a CoE do? -Communication-Change Management-CoE Case Study: Intuit.

UNIT IV BOT DEVELOPMENTS **12 HOURS**

Introduction to Bot Developments-Preliminaries-Installation of UiPath-getting Started-Activities-Flowcharts and Sequences-Log Message-Variables-Loops and Conditionals-Switch-Debug-Common UiPath Functions-The UiPath Orchestrator-Best Practices for Bot Development-Deployment and Monitoring: Testing-Going into Production-Monitoring-Security-Scaling. Introduction to Data Preparation-Types of Data-Big Data-The issues with Big data-The data process-Types of algorithms-The Perils of the Moonshot-Bias.

UNIT V OPEN SOURCE AND FUTURE OF RPA **12 HOURS**

Introduction to RPA Vendors: UiPath-Automation Anywhere Bot-Bot Store-Automation Anywhere Mobile App-Automation Anywhere University-Validation of Automation Anywhere-Blue Prism-EdgeVerve-PEGA-Verint-WorkFusion-Nintex-Kofax-Open-Source RPA: Importance of Open-Source Software-The Business model of open-source software-The pros and cons of open-source software-Process mining: Background on Process mining-Celonis-ProM-Future of RPA: Consolidation and IPOs-Microsoft-Software-as-a-Service (SaaS)and Open Source-Chatbots-Artificial Intelligence.

TOTAL: 60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
UiPath Process Mining	RPA + Business optimization
QPR ProcessAnalyzer	Process visualization & analysis
Google OR-Tools	AI-based optimization algorithms

TEXT BOOKS:

1 Taulli, T. (2020). The Robotic Process Automation Handbook: A Guide to Implementing RPA Systems. Apress.

2 Tripathi, A. M. (2018). Learning Robotic Process Automation. Packt Publishing.

3 Lacity, M., & Willcocks, L. (2015). Robotic Process Automation and Cognitive Automation: The Next Phase (1st ed.). SB Publishing.

REFERENCE BOOKS:

1 Casale, F., Dilla, R., Jaynes, I., & Livingston, L. (2015). Introduction to robotic process automation: A primer (1st ed.). Institute of Robotic Process Automation.

2 Murdoch, R. (2018). Robotic process automation: Guide to building software robots,

automate repetitive tasks & become an RPA consultant (1st ed.). Independently published.
3 Merianda,S.(2018). Robotic process automation tools, process automation and their benefits: Understanding RPA and intelligent automation (1st ed.). Independently published.

WEBSITES:

1 https://www.uipath.com
2 https://www.automationanywhere.com
3 https://www.blueprism.com

JOURNALS:

1 https://iaeme.com/Home/journal/IJRPA
2 https://ace.ewapub.com/article/view/19845
3 https://jtar.org/index.php/JTAR/article/view/3

CO, PO, PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	3	-	-	-	-	1	-	-	3	3	2	-	-	-	-	-	-
CO2	3	-	3	2	2	1	-	2	-	-	-	-	-	-	-	3	3
CO3	3	-	-	2	2	1	-	2	-	-	-	-	-	-	-	-	2
CO4	3	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-
CO5	3	-	-	-	-	1	-	1	-	-	-	-	-	-	-	-	1
Average	**3**	**-**	**3**	**2**	**2**	**1**	**-**	**1.7**	**3**	**3**	**1.5**	**-**	**-**	**-**	**-**	**3**	**2**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlations

M.Sc Computer Science		2026-2027
Category: Major		**SEMESTER III**
26CSP303C	**Intelligent Agents**	**4H - 4C**
Instruction Hours/week: L:4 T:0 P:0	**Marks: Internal:40 External:60 Total:100**	
	End Semester Exam: 3 Hours	

PREREQUISITE:

- Understanding of basic artificial intelligence concepts is crucial. This includes knowledge of search algorithms, knowledge representation and reasoning, machine learning algorithms, and decision-making under uncertainty**.**

COURSE OBJECTIVES (CO):

- To gain insight about automation using Intelligent Agents
- To understand the learning behavior and functioning of Agents
- To develop knowledge in the application domains of Agents

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Determine the basic theoretical foundations of common intelligent systems	K5
CO2	Applied artificial intelligence techniques	K3
CO3	Analyze various agent negotiation strategies	K4
CO4	Build the multiagent systems	K3
CO5	Evaluate current trends and applications of intelligent agents	K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I INTRODUCTION **12 HOURS**

Introduction to Intelligent Autonomous Agents- Motivations for agent-based computing - Abstract Architectures for Intelligent Agents - Key concepts and models of reasoning agents – deductive reasoning - symbolic reasoning - reactive reasoning - practical reasoning - Rational decision making and handling uncertainty.

UNIT II AGENT INTERACTIONS **12 HOURS**

Agent Interactions – Communication and cooperation – Ontology fundamentals – Building blocks – Ontology Languages – Software tools for ontologies – Agent Communication Languages. Conceptual Foundations of Communication in Multiagent systems - Traditional Software Engineering Approaches - Traditional AI Approaches - Commitment-Based Multiagent Approaches - Engineering with Agent Communication

UNIT III COOPERATIVE DISTRIBUTED PROBLEM SOLVING **12 HOURS**

Cooperative Distributed Problem Solving - Task Sharing and Result Sharing - Coordination - Multiagent Planning and Synchronization - Negotiation and Bargaining - Aspects of Negotiation - Game-Theoretic Approaches for Single-Issue Negotiation - Game-Theoretic Approaches for Multi-Issue Negotiation - Heuristic Approaches for Multi-Issue Negotiation -Argumentation-Based Negotiation

UNIT IV MULTI AGENT LEARNING **12 HOURS**

Multiagent Learning - Introduction - Challenges in Multiagent Learning - Reinforcement Learning for Multiagent Systems - Evolutionary Game Theory as a Multiagent Learning paradigm - Swarm Intelligence as a Multiagent Learning Paradigm -Neuro-Evolution as a Multiagent Learning Paradigm - Case Study in Multiagent Learning

UNIT V AGENT APPLICATIONS **12 HOURS**

Agent Applications - Agents for Workflow and Business Process Management - Agents for Distributed Sensing - Agents for Information Retrieval and Management - Agents for Electronic Commerce - Agents for Human–Computer Interfaces - Agents for Virtual Environments - Agents for Social Simulation - Deploying agents within a simulated environment - Practical reasoning strategies for computational markets

TOTAL: 60 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
JADE	Widely used for building multi-agent systems and distributed applications.
TensorFlow	Used to build learning agents with neural networks.
AnyLogic	Supports agent-based modeling and simulation.

TEXT BOOKS:

1 Michael Wooldridge (2009), *"An Introduction to Multi Agent Systems"* 2nd ed"Wiley.

2 G. Weiss (2013), *"Multi-Agent Systems - A Modern Approach to Distributed Artificial Intelligence*",2nd ed. MIT Press.

3 M. Wooldridge (2000*), "Reasoning about Rational Agents".* Second Edition, MIT Press.

REFERENCE BOOKS:

1 Yoav Shoham, Kevin Leyton-Brown, (2008), *"Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations",* Second Edition, MIT Press.

WEBSITES:

1 https://nptel.ac.in/courses/106/105/106105077

JOURNALS:

1 Ralf Schleiffer, "An intelligent agent model, European Journal of Operational Research, Volume 166, Issue 3, 1 November 2005, Pages 666-693.

CO, PO, PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	-	-	2	-	3	-	-	3	-	-	-	-	-	-	-	3	-
CO2	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2
CO3	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	-	2	-	-	3	-	-	-	-	-	-	-	-	-	-	-
CO5	-	-	-	-	3	-	-	3	-	-	-	-	-	-	1	-	-
Average	2	-	2	1	3	3	-	3	-	-	-	-	-	-	1	3	2

1 - Low, 2 - Medium, 3 - High, '-' - No Correlations

Category: Open Elective **SEMESTER III**

26EGPOE301 **English for Competitive Examinations** **3H - 3C**

Instruction Hours/week: L:3 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

COURSE OBJECTIVES (CO):

- To train learners to crack competitive exams
- To enhance their ability to speak in English and face an interview.
- To make the student apply, prepare and clear the competitive exams.
- To prepare the student to concentrate, stay positive and confident.
- To take even failure at ease and continue the target of clearing competitive exams.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply basic grammar rules to form correct sentences and eliminate common errors.	K3
CO2	Use vocabulary effectively through synonyms, antonyms, idioms, and word forms.	K2
CO3	Develop structured writing skills for brochures, SOPs, and business blogs.	K6
CO4	Demonstrate fluency in spoken English in conversations, discussions, and interviews.	K3
CO5	Apply soft skills like self-management, communication, and leadership in real-life situations.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I GRAMMAR **09 HOURS**

Parts of Speech, Modal Verbs and Sentence Improvement, Number-Subject, Verb and Agreement-Articles-Sequences of Tenses-Common Errors

UNIT II WORD POWER **09 HOURS**

Idioms and Phrases-One word substitution-Synonyms-Antonyms-Words often confused- Analogy, Homophones and Homonyms and Vocabulary Building Techniques.

UNIT III WRITING **09 HOURS**

Brochure and Pamphlet Writing, Statement of Purpose (SOP) Writing, and Blog Writing for Business

UNIT IV FLUENCY DEVELOPMENT **09 HOURS**

Speaking without Hesitation, thinking in English, Using Connections in Speech (and, but, because, however), Avoiding Mother Tongue Influence (MTI) and Improving Speed and Accuracy with Situational Conversation- Public Speaking-Group Discussion-Interview-Spoken English

UNIT V SOFT SKILLS **09 HOURS**

Self-management Skills, Emotional Intelligence Skills, Professional Skills, Leadership Skills, Creative and Cognitive Skills, and Life Skills.

TOTAL: 45 HOURS

REFERENCE BOOKS:

1 Murphy, R. (2019). English grammar in use (5th ed.). Cambridge University Press.
2. Swan, M. (2016). Practical English usage (4th ed.). Oxford University Press.
3. Lewis, N. (2014). Word power made easy. Anchor Books.
4. Martin, R. C., & Wren, P. C. (2018). High school English grammar and composition. S. Chand Publishing.
5. Hewings, M. (2013). Advanced grammar in use (2nd ed.). Cambridge University Press.
6. Redman, S. (2017). English vocabulary in use: Upper-intermediate (4th ed.). Cambridge University Press.
7. McCarthy, M., & O'Dell, F. (2017). English idioms in use: Advanced (2nd ed.). Cambridge University Press.
8. McCarthy, M., & O'Dell, F. (2016). Academic writing skills (2nd ed.). Cambridge University Press.
9. Carnegie, D. (2009). The quick and easy way to effective speaking. Pocket Books.
10. Goleman, D. (2006). Emotional intelligence: Why it can matter more than IQ. Bantam Books.

E-RESOURCES:

1 https://tie.uic.edu/student-services/esl-practice-sites/
2 https://www.spring.edu/webroom/grammar.html
3 https://www.spring.edu/webroom/vocabulary.html
4 https://www.wccnet.edu/succeed/international/esl/resources.php
5 https://library.ctstate.edu/c.php?g=1476834&p=11001710
6 https://usm.maine.edu/english-speakers-other-languages/english-skills-practice-websites/
7. https://resources.library.lemoyne.edu/ell/websites

CO, PO, CO, PO, PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	–	2	2	–	–	–	–	–	–	2	1	_	_	_	1	_	_
CO2	–	3	2	–	–	–	–	–	–	2	1	_	_	_	1	1	_
CO3	–	3	3	1	–	–	–	–	–	2	1	_	_	_	1	_	_
CO4	–	3	2	1	–	–	2	–	1	2	1	2	_	1	1	_	1
CO5	–	2	2	2	–	–	3	–	2	1	2	2	2	3	2	_	_
Average	–	2.6	2.2	1.3	–	–	2.5	–	1.5	1.8	1.2	**2**	**2**	**2**	**1.2**	**1**	**1**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlations

Category: Open Elective **SEMESTER III**

26CMPOE301 **Personal Finance and Planning** **3H - 3C**

Instruction Hours/week: L:3 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

COURSE OBJECTIVES (CO):

- To understand financial planning basics, including goal setting, personal loans, digital banking, and fraud prevention.
- To learn investment strategies, focusing on asset risk and return, portfolio management, and various investment avenues.
- To master in personal tax planning, covering tax structures, exemptions, deductions, and the distinction between tax avoidance and evasion.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Familiarize with regard to the concept of Investment Planning and its methods	K2
CO2	Examine the scope and ways of Personal Tax Planning	K4
CO3	Analyze Insurance Planning and its relevance	K4
CO4	Develop an insight in to retirement planning and its relevance.	K6
CO5	Construct an optimal portfolio in real life situations	K6
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I INTRODUCTION TO FINANCIAL PLANNING **09 HOURS**

Computer forensics and investigations as a profession – Preparing for computer investigations – Taking a systematic approach–Procedures for corporate high-tech investigations–Data recovery work stations and software– Conducting an investigation.

UNIT II INTRODUCTION TO INVESTMENT PLANNING **09 HOURS**

Investment Planning - Process and objectives of investment, Concept and measurement of return & risk for various assets class, Measurement of portfolio risk and return, Diversification & Portfolio formation. Gold Bond; Real estate; Investment in Greenfield and brownfield Projects; Investment in fixed income instruments- financial derivatives& Commodity market in India. Mutual fund schemes including SIP; International investment avenues.

UNIT III PERSONAL TAX PLANNING **09 HOURS**

Personal Tax Planning - Tax Structure in India for personal taxation, Scope of Personal tax planning, Exemptions and deductions available to individuals under different heads of income and gross total income, Special provision u/s 115BAC vis-à-vis General provisions of the Income-tax Act, 1961. Tax avoidance versus tax evasion.

UNIT IV INSURANCE PLANNING **09 HOURS**

Insurance Planning - Need for Protection planning. Risk of mortality, health, disability and property. Importance of Insurance: life and non-life insurance schemes. Deductions available under the Income-tax Act for premium paid for different policies.

UNIT V RETIREMENT BENEFITS PLANNING **09 HOURS**

Retirement Benefits Planning - Retirement Planning Goals, Process of retirement planning, Pension plans available in India, Reverse mortgage, New Pension Scheme. Exemption available under the Income-tax Act, 1961 for retirement benefits.

TOTAL: 45 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
ChatGPT	AI-assisted learning in financial planning, taxation, insurance, and retirement planning.
Microsoft Excel with Copilot	Budget preparation, financial forecasting, portfolio analysis, and retirement corpus estimation.
Tally Prime	To understand accounting records, taxation basics, GST, and financial reporting relevant to personal finance.

TEXT BOOKS:

1 Indian Institute of Banking & Finance. (2017). Introduction to Financial Planning, Taxmann Publication., New Delhi.

2 Pandit, A. (2014). The Only Financial Planning Book that You Will Ever Need, Network Publications Ltd., Mumbai.

REFERENCE BOOKS:

1 Madhu Sinha. (2017). Financial Planning: A Ready Reckoner (2nd ed.). McGraw Hill Education (India) Pvt. Ltd., New Delhi.

2. Halan, M. (2023). Let's Talk Money: You've Worked Hard for It, Now Make It Work for You, HarperCollins Publishers, India

3. Tripathi, V. (2017). Fundamentals of Investment, Taxmann Publication, New Delhi.

4. Indian Institute of Banking & Finance, (2025), Financial Planning and Tax Planning. Macmillan Education India Pvt. Ltd.

JOURNALS:

1 International Journal of Economics and Financial Issues. (n.d.). EconJournals. https://www.econjournals.com

2 Journal of Financial Services Research. (n.d.). Springer. https://link.springer.com/journal/10693

3 The Journal of Wealth Management. (n.d.). Portfolio Management Research. https://www.pm-research.com

WEBSITES:

1 https://www.coursera.org/learn/international-taxation

2 https://www.incometax.gov.in

CO, PO,PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	3	-	-	-	3	-	-	-	-	-	3	-	-	-	3	3	3
CO2	3	-	2	-	3	-	-	-	-	-	3	-	-	-	3	3	2
CO3	3	-	2	-	3	-	-	-	2	-	3	-	-	-	3	3	3
CO4	3	-	2	-	3	-	-	-	2	-	3	-	-	-	3	3	3
CO5	3	-	3	-	3	-	-	-	2	-	3	-	-	-	3	3	3
Average	**3**	**-**	**2**	**-**	**3**	**-**	**-**	**-**	**2**	**-**	**3**	**-**	**-**	**-**	**3**	**3**	**3**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlations

Category: Open Elective **SEMESTER III**

26MBAPOE301 **Organizational Behaviour** **3H - 3C**

Instruction Hours/week: L:3 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

COURSE OBJECTIVES (CO):

- To understand the basic concepts of organizational behavior.
- To analyze the individual behavior traits required for performing as an individual or group.
- To obtain the perceiving skills to judge the situation and communicate the thoughts and ideas.
- To evaluate how to perform in group and team and how to manage the power, politics and conflict.
- To recognize the importance of organizational culture and organizational change, group and team work to managing the conflict between members of the organization.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Connect organizational behavior issues in the context of the organizational behavior theories and concepts.	K2
CO2	Assess the behavior of the individuals and groups in organization and manage the stress.	K3
CO3	Categorize team, power, politics and conflict arising between the members.	K4
CO4	Explain how organizational change and culture affect the working relationship within organizations.	K5
CO5	Plan and exhibit the communication skills to convey the thoughts and ideas of case analysis to the individuals and group.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I ORGANIZATION BEHAVIOR INTRODUCTION 09 HOURS

Organization Behavior: Meaning and definition - Fundamental concepts of Organization Behavior - Contributing disciplines to the Organization Behavior field – Organization Behavior Model - Significance of Organization Behavior in the organization success - Challenges and Opportunities for Organization Behavior.

UNIT II BEHAVIOUR AND PERSONALITY 09 HOURS

Attitudes – Sources - Types - Functions of Attitudes – Attitude and Job satisfaction, Emotions and Moods – Emotional Intelligence – Organization Behavior Applications of Emotions and Moods, Learning – Theories of Learning. Personality – Determinants of personality- Theories of Personality - psycho-analytical, social learning, job-fit, and trait theories.

UNIT III PERCEPTION 09 HOURS

Perception – factors influencing perception - Person Perception – Attribution Theory – Frequently Used Shortcuts in Judging Others- Perceptual Process- Perceptual Selectivity - Organization Errors of perception – Linkage between perception and Decision making.

UNIT IV GROUP AND STRESS MANAGEMENT 09 HOURS

Foundation of Group Behavior - Concept of Group - Types of Groups - Stages of Group Development - Group Norms - Group Cohesiveness – Stress- Causes of Stress- Effects of Occupational stress- Coping strategies for

stress.

UNIT V ORGANIZATION CULTURE AND CHANGE AND STRESS MANAGEMENT **09 HOURS**

Organizational culture- Definitions and Characteristics of Culture- Types of Culture – Creating and Maintaining an Organizational Culture. Organizational change –Meaning- Forces for Change- Managing Planned Change - Factors in Organizational Change - Resistance to change- Overcoming resistance to change.

TOTAL: 45 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Psymetrics	AI based behavioral assessments for recruitment and understanding personality traits
Slack	Demonstrates real-time group communication and teamwork

TEXT BOOKS:

1 Fred Luthans. (2017). Organizational Behavior: An Evidence-Based Approach, 12th Edition, Mcgraw Hill Education, New Delhi.

2 Steven Mcshane and Mary Ann Von Glinow (2017), Organizational Behavior, 6th Edition, McGraw Hill Education, New Delhi.

3 Robbins, S. P., and Judge, T.A. (2016). Organizational Behaviour, 16th Edition, Prentice Hall of India, New Delhi.

REFERENCE BOOKS:

1 Laurie J. Mullins (2016). Management and Organisational behaviour, 10th Edition, Pearson Education, New Delhi.

2. Robbins, S. P., and Judge, T.A. (2016). Essentials of Organizational Behavior, 13th Edition, Pearson Education.

E-RESOURCES:

1 https://nptel.ac.in/courses/110/105/110105033/

CO, PO, PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	-	-	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	-	-	-	-	2	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	3	-	-	-	-	-	-	-	-	-	-
CO5	-	-	-	3	-	-	-	-	-	-	-	-	-	-	-	2	-
Average	**-**	**-**	**2**	**3**	**2**		**3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2.5**	**-**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlations

Category: Open Elective **SEMESTER III**

26CAPOE301 **Data Intelligence Systems** **3H - 3C**

Instruction Hours/week: L:3 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

COURSE OBJECTIVES (CO):

- To provide a comprehensive understanding of the concepts and architecture of Data Intelligence Systems.
- To develop skills in intelligent data processing and knowledge discovery techniques.
- To equip students with advanced analytics and artificial intelligence methodologies.
- To enable the design of scalable and distributed data intelligence solutions.
- To familiarize students with emerging applications, research trends, and ethical practices in Data Intelligence Systems.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Explain the concepts, architecture, and ethical principles of Data Intelligence Systems.	K2
CO2	Apply intelligent data processing techniques for data cleaning, transformation, feature selection, and knowledge discovery.	K3
CO3	Analyze complex datasets using statistical learning, machine learning, graph analytics, and explainable AI techniques.	K4
CO4	Design and develop scalable data intelligence solutions using distributed computing frameworks, real-time analytics, and MLOps practices.	K6
CO5	Evaluate emerging applications, governance frameworks, and research trends in Data Intelligence Systems for real-world decision-making.	K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I FOUNDATIONS OF DATA INTELLIGENCE SYSTEMS 09 HOURS

Evolution of Data Intelligence - Data Ecosystem and Architecture - Data Acquisition Techniques - Data Integration and ETL Processes - Metadata Management - Data Privacy and Ethical Considerations

UNIT II INTELLIGENT DATA PROCESSING 09 HOURS

Data Cleaning and Transformation-Feature Selection and Dimensionality Reduction-Pattern Recognition-Knowledge Discovery in Databases (KDD)-Data Fusion Techniques-Intelligent Decision Support Systems

UNIT III ADVANCED ANALYTICS AND AI 09 HOURS

Statistical Learning Methods-Ensemble Learning Techniques-Time Series Analysis and Forecasting-Graph Analytics and Network Analysis-Reinforcement Learning Fundamentals-Explainable and Trustworthy AI

UNIT IV SCALABLE DATA INTELLIGENCE 09 HOURS

Distributed Computing Frameworks-Real-Time Data Analytics-Edge and Fog Computing-Data Lake and Lakehouse Architectures-MLOps and Model Deployment-AI Model Monitoring and Performance Optimization

UNIT V EMERGING APPLICATIONS AND RESEARCH **09 HOURS**

Digital Twins and Intelligent Systems-Autonomous Decision-Making Systems-Cybersecurity Analytics-Social Network and Sentiment Analytics-Responsible AI and Governance-Current Research Trends and Industrial Case Studies

TOTAL: 45 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Claude AI	Document analysis, research support, and technical writing.
PyTorch	AI model development for machine learning and reinforcement learning.
Microsoft Copilot	AI-powered coding, data analysis, and productivity assistance.

TEXT BOOKS:

1 Han, J., Kamber, M., & Pei, J. (2022). *Data Mining: Concepts and Techniques* (4th ed.). Morgan Kaufmann Publishers.
2 Loshin, D. (2012). *Business Intelligence: The Savvy Manager's Guide* (2nd ed.). Morgan Kaufmann Publishers..
3 Buyya, R., Calheiros, R. N., & Dastjerdi, A. V. (2016). *Big Data: Principles and Paradigms*. Morgan Kaufmann Publishers.

REFERENCE BOOKS:

1 Aggarwal, C. C. (2015). Data Mining: The Textbook. Springer.
2. Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
3. Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd ed.). O'Reilly Media.
4. Erl, T., Khattak, W., & Buhler, P. (2016). Big Data Fundamentals: Concepts, Drivers & Techniques. Prentice Hall.
5. O'Neil, C., & Schutt, R. (2013). Doing Data Science: Straight Talk from the Frontline. O'Reilly Media.

WEBSITES:

1 https://www.sciencedirect.com/book/9780123814791/data-mining-concepts-and-techniques
2 https://mitpress.mit.edu/9780262035613/deep-learning/

CO, PO,PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	3	-	-	-	-	2	-	-	2	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	3	-	-	3	-	-	-	-	-	-	3	-
CO3	-	-	-	3	-	-	2	-	3	-	-	-	-	-	-	-	-
CO4	2	2	-	-	-	-	2	1	2	-	-	-	-	-	-	-	-
CO5		2	1	2	-	-	-	-	-	-	-	-	-	-	-	-	-
Average	**2.5**	**2**	**1**	**2.5**		**2.5**	**2**	**1**	**2.5**	-	-	-	-	-	-	-	-

1 - Low, 2 - Medium, 3 - High, '-' - No Correlations

Category: Open Elective **SEMESTER III**

26CSPOE301 **CYBER FORENSICS** **3H - 3C**

Instruction Hours/week: L:3 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

PREREQUISITE:

- Basics of Cyber Security.

COURSE OBJECTIVES (CO):

- To understand about computer forensics and investigations.
- To know about digital evidence, e-mail investigation, and Mobile device forensics.
- To analyze and validate forensics data.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Explain various investigation procedures and summarize duplication of digital evidence.	K5
CO2	Apply the knowledge of digital evidences.	K3
CO3	Design and develop various forensics tools and analyze the network forensics.	K4
CO4	Determine the systematic study of high-tech forensics	K5
CO5	Analyze and validate digital evidence data	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I COMPUTER FORENSICS AND INVESTIGATIONS **09 HOURS**

Computer forensics and investigations as a profession – Preparing for computer investigations – Taking a systematic approach–Procedures for corporate high-tech investigations–Data recovery work stations and software– Conducting an investigation.

UNIT II DATA ACQUISITION **09 HOURS**

Data acquisition – Storage formats for digital evidence – Validating data acquisitions – Processing crime and incident scenes–Identifying digital evidence–Collecting evidence in private sector incident scenes – Preparing for search-seizing digital evidence at the scene-storing digital evidence –Reviewing a case.

UNIT III COMPUTER FORENSICS TOOLS **09 HOURS**

Current computer forensics tools–Software tools–Hardware tools–The Macintosh file structure and boot process – Computer forensics analysis and validation – Addressing data –Hiding techniques.

UNIT IV NETWORK FORENSICS **09 HOURS**

Virtual machines – Network forensics – Developing standard procedures – Live acquisitions – email investigations – Investigating e-mail crimes and violations – Understanding e-mail servers – Cell phone and mobile device forensics.

09 HOURS

UNIT V MOBILE DEVICE FORENSICS

Understanding mobile device forensics – Acquisition procedures –Report writing for high-tech investigations – Importance of reports – Guidelines for writing reports –Expert testimony in high-tech investigations.

TOTAL: 45 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
EnCase	Widely used commercial tool for digital investigations.
FTK (Forensic Toolkit)	Helps in data recovery, indexing, and analysis.
Volatility	Analyzes RAM dumps to detect malicious activity.

TEXT BOOKS:

1 Bill Nelson, Amelia Phillips and Christopher Steuart, (2018). *Computer Forensics and Investigations*, Cengage Learning, 5th Edition.

2 Eoghan Casey. (2017). *Handbook of Digital Forensics and Investigation*, 1st Edition, Academic Press.

3 John R Vacca, (2016). *Computer Forensics*, 2nd Edition, Cengage Learning.

REFERENCE BOOKS:

1 John R. Vacca, (2005) *Computer Forensics: Computer Crime Scene Investigation*, 2nd Edition Cengage Learning.

2. Marjie T Britz, (2008), *Computer Forensics and Cyber Crime: An Introduction*, 2nd Edition, Pearson Education.

3. Mari E-Helen Maras, (2014). *Computer Forensics: Cybercriminals, Laws, and Evidence*, 2nd Edition Jones & Bartlett Learning.

WEBSITES:

1 www.cps.brockport.edu/~shen/cps301/figures/figure1.pdf

2 www.forensicsguru.com/devicedataextractionsimcell.php

3 www.nptel.ac.in/courses/106101060

4 www.samsclass.info/121/ppt/ch11.ppt

5 www.garykessler.net/library/role_of_computer_forensics.html

6 www.ukessays.com/essays/information-technology/computer-forensics-and-crime investigations-information-technology-essay.php.

JOURNALS:

1. Casino, F., Dasaklis, T. K., Spathoulas, G., Anagnostopoulos, M., Ghosal, A., Borocz, I., Solanas, A., Conti, M., & Patsakis, C. (2021). Research trends, challenges, and emerging topics of digital forensics: A review of reviews.

2. Nayerifard, T., Amintoosi, H., Bafghi, A. G., & Dehghantanha, A. (2023). Machine learning in digital forensics: A systematic literature review.

CO, PO, PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	-	-	3	-	-	-	-	-	-	-	2	-	-	-	-	2	-
CO2	-	-	3	-	1	-	-	-	-	-	-	-	-	-	-	-	2
CO3	3	-	-	-	-	1	2	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-		2	-	-	-	-	-	-	-	-	-	-
CO5	3	-	3	-	-	-	-	-	-	-	2	-	-	-	-	-	-
Average	**3**	**-**	**3**	**-**	**1**	**1**	**2**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**2**	**2**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlations

Category: Open Elective **SEMESTER III**

26MMPOE301 **Applied Statistics and Data Analysis** **3H - 3C**

Instruction Hours/week: L:3 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

PREREQUISITE:

Basic understanding of arithmetic, statistics and probability

COURSE OBJECTIVES (CO):

- To introduce fundamental concepts of data, probability, sampling, and statistical analysis.
- To develop skills in summarizing, visualizing, and interpreting data using statistical tools and software.
- To provide foundational understanding of regression, hypothesis testing, and introductory multivariate techniques.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Understand the types of data, measurement scales, and basic data handling techniques.	K4
CO2	Apply suitable graphical tools to visualize and interpret datasets using Excel/R/Python/SPSS.	K4
CO3	Explain basic probability concepts and select appropriate probability and non-probability sampling methods.	K4
CO4	Perform simple regression analysis and conduct hypothesis tests such as z-test, t-test, and chi-square test.	K4
CO5	Demonstrate introductory skills in multivariate analysis techniques like PCA, factor analysis.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I DATA AND DATA HANDLING **09 HOURS**

Introduction to Data Analytics and Analytical Workflow – Types of data: qualitative, quantitative, cross-sectional, and time-series – Measurement scales: nominal, ordinal, interval, and ratio – Data sources and methods of data collection – Data preprocessing: data cleaning, handling missing values, outlier detection and treatment – Data coding, classification, and tabulation – Introduction to exploratory data analysis (EDA).

UNIT II DATA VISUALIZATION **09 HOURS**

Principles of data visualization - Graphical tools: pie charts, bar charts, histograms, boxplots, stem–leaf plots, scatterplots, heatmaps – Basic concepts of correlation and correlation plots – Data visualization using Excel / R / Python / SPSS.

UNIT III PROBABILITY BASICS AND SAMPLING TECHNIQUES **09 HOURS**

Concepts of population, sample, parameters and statistic – Principles of sample surveys – Basics of probability: events, outcomes, sample space, random variables – Probability sampling methods: simple random, stratified, systematic and cluster sampling – Non-probability sampling methods: convenience, purposive, and quota.

UNIT IV REGRESSION ANALYSIS AND TESTING OF HYPOTHESIS **09 HOURS**

Regression analysis: introduction and assumptions – Simple linear regression – Multiple regression (basic concepts) – logistic regression – Hypothesis testing: null and alternative hypotheses – Types of errors, significance levels, p-values – Parametric tests: z-test, t-test, chi-square test.

UNIT V MULTIVARIATE AND MODERN DATA ANALYSIS TECHNIQUES **09 HOURS**

Introduction to multivariate data –Principal Component Analysis (PCA) – Factor analysis (conceptual) – Introduction to clustering: K-means (conceptual) – Modern analytical approaches: classification vs. prediction – Applications and interpretation using software.

TOTAL: 45 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
MS-Excel	The use of Microsoft Excel helps in organizing, analyzing, and presenting statistical data accurately and effectively.
PSPP/SPSS	PSPP helps to easily and accurately calculate and analyse statistical data.
R-Programming	To identify patterns and trends in time-based data to forecast future outcomes and support better decision-making.

TEXT BOOKS:

1 Gupta, S. P. (2012). Statistical Methods, Sultan Chand & Sons, New Delhi.
2 Navaneetham, P. A. (2019). Business Mathematics and Statistics, Jai Publishers, Trichy.
3 Gupta, O.P, Chaudhary, S.S., Ansari, M.A. (1984). Applied Statistics, Kedar Nath Ram Nath Publishers.
4. Pillai R.S.N., and Bagavathi V., (2002). Statistics, S. Chand & Company Ltd, New Delhi
5 Anderson, T.W. (2003). An Introduction to Multivariate Statistical Analysis, Third Edition, Wiley – Interscience, New York.

REFERENCE BOOKS:

1 Agarwal, B. L. (2006). Basic Statistics, New Age International Private Limited, New Delhi, India.
2. Cochran, W.G. (1977). Sampling Techniques, Third Edition, John Wiley & Sons, NY.
3. Hair, J. F., Jr., Black, W. C., Babin, B. J., & Anderson, R. E. (2010). Multivariate Data Analysis (7th ed.). Pearson.
4. Denis, D. J. (2020). Univariate, bivariate, and multivariate statistics using R: Quantitative tools for data analysis and data science. John Wiley & Sons.

WEBSITES:

1 https://onlinecourses.nptel.ac.in/noc21_ma34/preview
2 https://nptel.ac.in/courses/106107220

CO, PO, PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	2	1	-	-	-	-	1	-	-	-	-	-	-	-	-	-	1
CO2	3	2	1	-	-	-	-	-	-	-	-	-	-	-	-	-	1
CO3	2	1	-	-	-	-	1	-	-	-	-	-	-	-	-	-	1
CO4	2	1		-	-	-		-	-	-	-	-	-	-	-	-	1
CO5	3	2	1	-	-	-	1	-	-	-	-	-	-	-	-	-	1
Average	**2.4**	**1.4**	**1**	-	-	-	**1**	-	-	-	-	-	-	-	-	-	**1**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlations

Category: Open Elective **SEMESTER III**

26PHPOE301 **Electrical Appliances and Servicing** **3H - 3C**

Instruction Hours/week: L:3 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

PREREQUISITE:

Not Required

COURSE OBJECTIVES (CO):

- To create awareness about types and handling of domestic appliances
- To acquire knowledge about principle of operation, working and application of various domestic appliances.
- To gain the skills in assembly, repair, installation, testing and maintenance of domestic appliances.
- To acquire skills in entrepreneurship

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Repair maintenance of the basic electrical and electronics appliances	K3
CO2	Identification to protective devices	K2
CO3	Repair and maintenance of the split Vacuum Cleaner and washing machine	K4
CO4	Repair and maintenance of the electric fan & hair drier	K3
CO5	Acquire knowledge about tools, equipment and Instruments	K2
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I INSTRUMENTS AND TESTING **09 HOURS**

Introduction – voltage tester screwdriver – continuing test – insulation test – measurement of Power for dc & ac circuits.

Electrical Cooking Appliances introduction – types – construction – electric toaster – types – automatic and non-automatic.

Electric Iron Box types – non-automatic – automatic – construction and working – comparison – trouble shooting – Steam iron box.

UNIT II WATER HEATERS & COFFEE MAKERS **09 HOURS**

Water heater – function – types – electric kettle – immersion water heater – construction and working – storage water heaters – non pressure type – pressure type – construction and working – repairs & remedies – coffee maker – types – construction and working of percolator type.

UNIT III ELECTRIC MIXER & EGG BEATERS **09 HOURS**

Electric maker – function and its construction – general operating instruction – caution – cleaning – repairs and remedies – egg beaters – hand operated crank type – electric type and its construction.

UNIT IV VACUUM CLEANER AND WASHING MACHINE **09 HOURS**

Vacuum cleaner – function – principle – main components – features – types - working – accessories - filters – repairing. washing machine – function – types – semi and fully automatic – top and front

loading – washing technique – working cycle – construction and working of washing machine – comparison of top and front-loading machines – problems and remedies.

UNIT V ELECTRIC FAN & HAIR DRIER **09 HOURS**

Fan – function – terminology – construction and working of ceiling & table fans –exhaust fan – general fault and remedy. hair drier – function – types – construction and working – safety features – repairs & remedies.

TOTAL: 45 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Proteus / Multisim	Simulates basic electrical and electronic circuits.
Tinkercad Circuits	Designs and tests simple electrical circuits.

TEXT BOOKS:

1 Electrical Practical, Directorate General of employment & training (DGET),(2018) .Arihant Publisher.

2 Handbook of Repair and Maintenance of Domestic Electronics Appliances handbook By Shashi Bhushan Sinha, BPB Publications.

REFERENCE BOOKS:

1 Dixon and Graham, Electrical Appliance Manual–Hardcover, ISBN 13: 9781859608005.

2. Graham and Dixon, (1995). Electrical Appliances: The Complete Guide to the Maintenance and Repair of Domestic Electrical Appliances (Haynes for Home DIY S.).

3. Shashi Bhushan Sinha, Handbook of Repair and Maintenance of Domestic Electronics Appliances.

WEBSITES:

1 https://alison.com/courses?query=Electrical%20Appliance%20and%20Servicings#.

2 https://www.scribd.com/document/269725441/Electrical-Appliances-PDF.

3 https://www.unitec.ac.nz/career-and-study-options/electrical-and-electronics-engineering/electrical-appliance-serviceperson-eas.

CO, PO, PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	-	-	3	-	-	-	-	-	1	-	2	-	2	-	-	2	-
CO2	-	-	3	-	1	-	-	-	1	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	1	2	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-		2	-	-	-	-	-	2	-	-	-	2
CO5	3	-	3	-	-	-	-	-	-	-	2	-	-	-	-	-	-
Average	**3**	**-**	**3**	**-**	**1**	**1**	**2**	**-**	**1**	**-**	**2**	**-**	**2**	**-**	**-**	**2**	**2**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlations

Category: Open Elective **SEMESTER III**

26CHPOE301 **Industrial Chemistry** **3H - 3C**

Instruction Hours/week: L:3 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

COURSE OBJECTIVES (CO):

- To understand the comprehensive process of cane sugar, cement and rubber production.
- To gain the understanding of paint classification, constituents and diverse applications
- To understand the physical and chemical properties of glass, its characteristics and the manufacturing processes.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Explain the process of cane sugar production.	K5
CO2	Classify paints, constituents and diverse applications.	K4
CO3	Examine the physical and chemical properties of glass.	K4
CO4	Analyze the manufacturing processes of cement, including the wet and dry processes,	K4
CO5	Discuss the rubber fabrication, including refining processes, fabrication methods, and vulcanization techniques.	K6
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I SUGAR **10 HOURS**

Introduction, manufacture of cane sugar-extraction of juice, purification of juice, defecation, sulphitation, carbonation, concentration or evaporation. Crystallization-separation of crystals, drying, refining, recovery of sugar from molasses, bagasse. Manufacture of sucrose from beet root. Estimation of sugar, double sulphitation process, double carbonation.

UNIT II PAINTS **10 HOURS**

Classification, constituents, setting of paints, requirements of a good paint. Emulsion, latex, luminescent, fire retardant and heat resistant paints. Methods of applying paints. Special applications and failures of paint. Varnishes- introduction – raw materials-manufacture of varnishes.

UNIT III GLASS **09 HOURS**

Introduction, physical/chemical properties, characteristics of glass. Raw materials, methods of manufacture - formation of batch material, melting, shaping, annealing and finishing of glass.

UNIT IV CEMENT **09 HOURS**

Introduction, raw materials, manufacture–wet process, dry process, reactions in kiln, setting of cement, properties and uses of cement. Plaster of paris, gypsum, lime.

UNIT V RUBBER **07 HOURS**

Introduction, importance, types and properties of rubber. Refining of crude rubber, drawbacks of raw rubber. Rubber fabrication, vulcanization techniques.

TOTAL: 45 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
ChatGPT	Concept explanation, note preparation, question answering, MCQs, assignments, summaries
ChemDraw	Drawing sucrose structure and process diagrams
MolView	Visualization of polymer molecules

TEXT BOOKS:

1 Sharma, B.K. (2014). *Industrial Chemistry* (14th Edition). Meerut: Goel Publishing House.

2 Jain, P.C. & Monika Jain. (2016). *Engineering Chemistry* (16th Edition). New Delhi: Dhanpat Rai Publishing Co. (Pvt) Ltd.

REFERENCE BOOKS:

1 Kent, J. A. (1997). *Riegel's Handbook of Industrial Chemistry*. New Delhi: CBS Publishers.

WEBSITES:

1 https://www.fao.org/?utm_source=chatgpt.com

2 https://www.sciencedirect.com/topics?utm_source=chatgpt.com

3 https://www.acs.org/?utm_source=chatgpt.com

JOURNALS:

1 Cement and Concrete Research

2 Construction and Building Materials

3 Rubber Chemistry and Technology

4 Polymer Testing

CO, PO, PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	3	-	-	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	2	-	-	-	-	-	-	-	-	-	-		-	-
CO4	3	-	-	2	-	2	-	-	2	-	2	-	-	-	-	-	-
CO5	3	-	-	2	-	2	-	-	2	-	2	-	-	-	-	-	-
Average	**3**	-	-	**2**	-	**2**	-	-	**2**	-	**2**	-	-	-	-	-	-

1 - Low, 2 - Medium, 3 - High, '-' - No Correlations

Category: Open Elective **SEMESTER III**

26MBPOE301 **Probiotics** **3H - 3C**

Instruction Hours/week: L:3 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

PREREQUISITE:

COURSE OBJECTIVES (CO):

- To introduce the fundamental concept, scope, and historical development of probiotics along with the selection and characterization of candidate strains.
- To train students in the classification, identification, and assessment of probiotic microorganisms, emphasizing mechanisms of action and therapeutic applications.
- To evaluate the health benefits, safety considerations, regulatory issues, and quality assurance practices in probiotic research and industry.

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Explain the definition, scope, and historical development of probiotics, including commonly used strains	K2
CO2	Comprehend the mechanisms of probiotic action in the gastrointestinal tract and microbiota	K3
CO3	Evaluate the therapeutic applications and health effects of probiotics on the GI tract and immune system	K4
CO4	Analyze the safety of probiotics and regulatory issues	K5
CO5	Explain the definition, scope, and prebiotic ingredients and their health effects, and synbiotics and postbiotics	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I THE PROBIOTIC CONCEPT **09 HOURS**

Definition, Scope and history of probiotics, Characteristics and classification of Probiotics. Properties of Probiotics.

UNIT II PROBIOTIC -MECHANISM OF ACTION **09 HOURS**

Bacterial fermentation and metabolism. Overall mechanism of action – GI tract and its microbiota. Common probiotics – *Lactobacillus, Bifidobacterium*. Testing methods for probiotics.

UNIT III HEALTH EFFECTS OF PROBIOTICS **09 HOURS**

Antibiotic-Associated Diarrhoea, Infectious Diarrhoea, Lactose Intolerance. Probiotics and Allergy, Probiotics and Cardiovascular diseases, Immunomodulatory properties of probiotics. Application of probiotics in food.

UNIT IV SAFETY CONSIDERATION AND QUALITY ASSURANCE **09 HOURS**

Antimicrobial resistance profiles of probiotics. Safety of probiotics - Regulatory issues, appropriate labelling. Manufacturing and handling of probiotics. Post market surveillance.

UNIT V PREBIOTICS, SYNBIOTIC AND POSTBIOTICS **09 HOURS**

Definition and history. Characterization of prebiotic ingredients. Criteria for prebiotic selection. Health effects of prebiotics. Application of prebiotics in food. Synbiotics and Postbiotics.

TOTAL: 45 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
metaProbiotics	Language modelbased metaProbiotics to rapidly detect probiotic bins from metagenomes, demonstrating superior performance in simulated benchmark datasets
ProbioMinServer	Intended to help researchers access information on probiotics about a wide variety of characteristics, such as safety (e.g. antimicrobial resistance, virulence, pathogenic, plasmid, and prophage genes) and functionality (e.g. functional classes, carbohydrate-active enzyme, and metabolite gene cluster profile)

TEXT BOOKS:

1 Parmjit Singh Panesar, Anil Kumar Anal (2022). Probiotics, Prebiotics and Synbiotics. John Wiley & Sons Ltd.

2 Bamforth C.W. (2005) Food, Fermentation and Microorganisms, Blackwell Science.

REFERENCE BOOKS:

1 Doyle M.P. and Buchanan R.L. (Ed.) (2013) Food Microbiology: Fundamentals and Frontiers, 4th Edn.ASM press.

2 Frazier W.C. and Westhoff D.C. (2008) Food Microbiology, 4th Edn. Tata McGraw Hill Publishing Co.,New Delhi.

WEBSITES:

1 https://my.clevelandclinic.org/health/treatments/14598-probiotics

2 https://pmc.ncbi.nlm.nih.gov/articles/PMC3714544/

3 https://pmc.ncbi.nlm.nih.gov/articles/PMC4006993/

JOURNALS:

1 Probiotics and Antimicrobial Proteins. https://link.springer.com/journal/

2 Probiotics. https://link.springer.com/collections/bjgjgigheb

CO, PO, PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	3	3	2	-	-	-	-	-	-	-	-	-	-	-	3	3	-
CO2	3	3	2	2	2	3	2	2	-	-	-	-	-	-	3	3	3
CO3	3	3	2	2	2	3	2	2	-	-	-	-	-	-	3	3	3
CO4	3	3	2	2	2	3	2	3	2	-	-	-	-	-	3	3	3
CO5	3	3	2	-	2	3	2	3	3	-	2	-	-	-	3	3	3
Average	**3**	**3**	**2**	**2**	**2**	**3**	**2**	**3**	**2**	-	**2**	-	-	-	**3**	**3**	**3**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlations

Category: Open Elective **SEMESTER III**

26BTPOE301 **Nutrition and Dietetics** **3H - 3C**

Instruction Hours/week: L:3 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam: 3 Hours

PREREQUISITE: Student should know about basics of food, its nutrients and their relationship to health

COURSE OBJECTIVES (CO):

- To understand the fundamentals of food, nutrients and their relationship to health
- To respect to deriving maximum benefit from available food resources
- To understand of the consequences of vitamin and mineral deficiency/excess of vitamin

COURSE OUTCOMES (COs):

Upon completion of this course, the student will be able to:

COs	Course Outcomes	Blooms Level
CO1	Name the fundamentals of nutrition and their relationship to health	K1
CO2	Learn to derive maximum benefits from available food resources	K2
CO3	Identify the consequences of vitamin and mineral deficiency/excess of vitamin	K3
CO4	Analyze the importance of nutrition in various stages of human life	K4
CO5	Assess about nutrition deficiency diseases and their consequences	K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I BASIC CONCEPTS IN FOOD AND NUTRITION **09 HOURS**

Understanding relationship between food, nutrition and health, Functions of food-Physiological, psychological and social. food pyramid, Basic food Groups, Recommended Dietary Allowances: Factors affecting RDA, general principles, determination of RDA, Requirements and RDA, Reference man and reference woman

UNIT II NUTRIENTS **09 HOURS**

Functions, dietary sources and clinical manifestations of deficiency/ excess of the following nutrients: Carbohydrates, lipids and proteins, Fat soluble vitamins-A, D, E and K, Water soluble vitamins – thiamin, riboflavin, niacin, pyridoxine, folate, vitamin B12 and vitamin C, Minerals – calcium, iron and iodine

UNIT III NUTRITION IN LIFE CYCLE **09 HOURS**

Menu planning – Objectives, planning balanced diets, food exchange lists., Physiological changes, RDA, nutritional guidelines, nutritional concerns and healthy food choices - Infants, Preschool children, School children, Adolescents. Pregnant woman, Lactating mother, Elderly. Child health and morbidity, neonatal, infant and child mortality

UNIT IV INTRODUCTION TO NUTRITIONAL DEFICIENCY DISEASES **09 HOURS**

Causes, symptoms, treatment, prevention of the following: Protein Energy Malnutrition (PEM), Vitamin A Deficiency (VAD), Iron Deficiency Anemia (IDA), Iodine Deficiency Disorders (IDD), Zinc Deficiency, Flurosis Nutritional needs during pregnancy, common disorders of pregnancy (Anemia, HIV infection, Pregnancy induced hypertension), relationship between maternal diet and birth. Maternal health and nutritional status, maternal mortality and issues relating to maternal health.

UNIT V DIET THERAPEUTICS **09 HOURS**

Dietary and stress management. Dietary recommendations of WHO. Diet for diabetes mellitus-

Nutrition recommendations for patient with diabetes, Meal planning, Diet for Cardiovascular Diseases -Dietary management and general guidelines for coronary heart disease, Diet for cancers at various sites in the human body, diet therapy, managing eating problems during treatment. Hormonal imbalance - Poly cystic ovarian syndrome, causes of hormonal imbalance. Diet management.

TOTAL: 45 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
SciSpace	Explains complex nutrition research papers, interprets statistical results, and summarizes journal articles.
SnapCalorie	Uses AI image recognition to estimate calories and macronutrients from food images.
Lifesum	Offers personalized diet recommendations and healthy eating plans.

TEXT BOOKS:

1 Srilakshmi. B. (2015) Food Science: New Age International (P) Ltd. Publishers. 6nd Edition., New Delhi

2 Swaminathan. M. (2008). Essential of Food and Nutrition Vol II The Bangalore Printing and Publishing Co. Ltd., Bangalore.

REFERENCE BOOKS:

1 Garrow, J.S., and James, W.P.T., (2000). Human Nutrition & Dietetics, Longman Group, UK.

2 Gordon M, Wardlaw and Paul M. (2012). Perspectives in Nutrition: U.S.A. McGraw Hil Publishers. 9th Edition. New Delhi

3 Sharma, R (2004). Diet Management,3rd ed, Reed Elsevier India Private Limited, Chennai.

4 Srilakshmi. B. (2014) Nutrition Science: New Age International(P)Ltd. Publishers. 4th ed. New Delhi.

WEBSITES:

1 he American Journal of Clinical Nutrition. (n.d.). Oxford University Press on behalf of the American Society for Nutrition. https://academic.oup.com/ajcn

2 The Journal of Nutrition. (n.d.). Oxford University Press on behalf of the American Society for Nutrition. https://academic.oup.com/jn

3 Public Health Nutrition. (n.d.). Cambridge University Press. https://www.cambridge.org/core/journals/public-health-nutrition.

JOURNALS:

1 World Health Organization. (n.d.). *Nutrition*. https://www.who.int/health-topics/nutrition

2 Food and Agriculture Organization of the United Nations. (n.d.). *Nutrition*. https://www.fao.org/nutrition/en/

3 Indian Council of Medical Research–National Institute of Nutrition. (n.d.). *National Institute of Nutrition*. https://www.nin.res.in/

CO, PO, PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PO 13	PO 14	PO 15	PSO 1	PSO 2
CO1	3	-	-	-	-	-	-	2	2	-	2	2	2	-	2	2	2
CO2	3	-	-	-	-	-	-	2	2	-	2	2	2	-	2	2	2
CO3	3	-	-	-	-	-	-	2	2	-	2	2	2	-	2	2	2
CO4	3	-	-	-	-	-	-	2	2	-	2	2	2	-	2	2	2
CO5	3	-	-	-	-	-	-	2	2	-	2	2	2	-	2	2	2
Average	3	-	-	-	-	-	-	2	2	-	2	2	2	-	2	2	2

1 - Low, 2 - Medium, 3 - High, '-' - No Correlations

Category: Major **SEMESTER III**

26CSP341 **Deep Learning** **5H - 3C**

Instruction Hours/week: L:1 T:0 P:4 **Marks: Internal:50 External:50 Total:100**

End Semester Exam: 3 Hours

PREREQUISTICS:

- Basic understanding of linear algebra, calculus, probability theory, programming (preferably Python), and machine learning fundamentals.

COURSE OBJECTIVES (CO):

- To Understand the fundamentals of neural networks, including their architecture, types of neurons, and the limitations of traditional computing.
- To Gain proficiency in TensorFlow for managing variables, operations, sessions, and visualizing machine learning graphs.
- To Learn the basics of Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), including their applications in sequence analysis and reinforcement learning.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply basic neural network architectures by designing and implementing feedforward neural networks with soft max output layers.	K3
CO2	Apply TensorFlow computational graph structures to design and implement deep learning models for real-world applications.	K3
CO3	Analyze CNN architectures and feature extraction mechanisms to evaluate their performance in image recognition and classification tasks.	K4
CO4	Apply recurrent neural networks (RNNs) and LSTM for sequence data analysis and memory-augmented models.	K3
CO5	Analyze reinforcement learning concepts, including Markov Decision Processes (MDP), Q-learning, and their use in decision-making systems.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I: BASICS OF NEURAL NETWORK **3T+12P = 15 HOURS**

Fundamentals of Neural Networks and Machine Learning- Neural Network Architectures and Output Processing

PRACTICAL:

- Write a program to implement XOR gates using Perceptron.
- Write a program to implement Wine Classification using Back propagation.
- Medical Disease Classification System Using SoftMax Output Layer
- Design & implement a simple deep learning network for classification of images use MNIST dataset.

UNIT II: FUNDAMENTALS OF TENSORFLOW AND COMPUTATIONAL GRAPHS **3T+12P = 15 HOURS**

Fundamentals of TensorFlow and Computational Graphs- Graph Construction, Visualization, and Deep Learning Applications

PRACTICAL:

- Image Classification System Using TensorFlow Computational Graphs
- Real-Time Stock Market Prediction Model Using TensorFlow Variables and Sessions
- Design a deep learning network for fine tuning of convolution networks use MNIST dataset.
- Smart Recommendation Engine Using TensorFlow Operations and Graph Visualization

UNIT III: BASICS OF CNN **3T+12P = 15 HOURS**

Fundamentals of Convolutional Neural Networks and Feature Extraction- Pooling Techniques and Real-World Applications of CNN

PRACTICAL:

- Face Recognition System Using Convolutional Neural Networks
- Implement Data Augmentation in Deep Learning use any medical image dataset.
- Implement RNN for handwriting digit recognition.
- Traffic Sign Detection and Recognition Using Convolution Layers and Feature Maps

UNIT IV: BASICS OF RNN **3T+12P = 15 HOURS**

Recurrent Neural Networks and Sequence Learning Techniques- Memory Augmented Neural Networks and Advanced Deep Learning Applications.

PRACTICAL:

- Implement Bidirectional RNNs for music generation.
- Implement Bidirectional LSTM for sentiment analysis.
- Implement Region Based CNN for object detection.
- Implement Variational Auto encoders for image denoising.
- Implement Generative Adversarial Networks to generate realistic photographs

UNIT V: REINFORCEMENT LEARNING **3T+12P = 15 HOURS**

Reinforcement Learning –MDP–Q Learning–Applications

PRACTICAL:

- Inventory Management Optimization Using Reinforcement Learning
- Dynamic Resource Allocation System Using MDP Models
- Autonomous Drone Path Planning Using Q-Learning
- Recommendation System Using Reinforcement Learning Techniques
- Energy-Efficient Smart Grid Control Using Reinforcement Learning
- Implement basic policy evaluation for a given MDP.
- Simulate exploration vs exploitation trade-off using ε-greedy strategy.
- Implement RL-based recommendation system simulation.
- Implement inventory management optimization using reinforcement learning.

TOTAL: 75 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
TensorFlow	Build, train, and deploy deep learning models at scale
PyTorch	Flexible and widely used for research and production deep learning
Google Colab	Run deep learning code with free GPU/TPU support

TEXT BOOKS:

1. François Chollet. (2021). "*Deep Learning with Python*", 2nd Edition. Manning Publications.
2. NikhilBuduma, NicholasLocascio. (2018). "*Fundamentals of Deep Learning: Designing Next Generation Machine Intelligence Algorithms",* 1stedtion, O'ReillyMedia.
3. Ian Good fellow, Yoshua Bengio, Aaron Courville, (2017). *"Deep Learning (Adaptive computation And Machine Learning series"*, 1st edition, MITPress.

REFERENCE BOOKS:

1. Andrew W. Trask (2019). "*Grokking Deep Learning"*, 1st Edition. Manning Publications.
2. Charu C. Aggarwal. (2018). *"Neural Networks and Deep Learning".* 1st Edition. Springer.
3. Joseph, Josh Patterson and Adam Gibson (2017). "*Deep Learning- a practitioner's approach"*, 1st Edition, O'Reilly Media.

WEBSITES:

1. https://onlinecourses.swayam2.ac.in/aic20_sp06/preview
2. https://onlinecourses.swayam2.ac.in/arp19_ap79/preview

JOURNALS:

1. Iqbal H. Sarker, "Deep Learning: A Comprehensive Overview on Techniques, Taxonomy, Applications and Research Directions", SN Computer Science, 2:420, 2021. https://doi.org/10.1007/s42979-021-00815-1

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**2**	**-**	**3**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**3**	**-**
CO2	**-**	**-**	**3**	**-**	**-**	**-**	**-**	**-**	**3**	**-**	**-**	**-**	**2**	**-**	**-**	**-**	**2**
CO3	**-**	**-**	**-**	**-**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO4	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO5	**-**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**3**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**-**
Average	**2**	**-**	**3**	**-**	**1**	**-**	**2**	**-**	**2**	**-**	**2**	**-**	**2**	**-**	**2**	**3**	**2**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

M. Sc. Computer Science		2026-2027
Category: Major		**SEMESTER III**
26CSP342	**Software Testing and Quality Assurance**	**5H - 4C**

Instruction Hours/week: L:3 T:0 P:2 **Marks: Internal:50 External:50 Total:100**

End Semester Exam: 3T+2P=5 Hours

PREREQUISTICS:

- Understanding the phases of SDLC (Requirements gathering, Design, Development, Testing, Deployment, Maintenance).
- Understanding of test planning, test case design, test execution, defect reporting, and tracking.

COURSE OBJECTIVES (CO):

- To understand the basics of testing, test planning &design and test team organization
- To study the various types of tests in the life cycle of the software product.
- To learn the techniques for quality assurance and applying for applications.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Determine the quality assurance process and its role in software development.	K5
CO2	Determine variety of testing techniques, methods, and tools	K5
CO3	Plan the test cases for different applications.	K3
CO4	Identify defect prevention techniques and software quality assurance metrics.	K3
CO5	Apply systematic procedure for software design and deployment.	K3
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I: SOFTWARE TESTING - CONCEPTS, ISSUES, AND TECHNIQUES **9T+6P = 15 HOURS**

Quality Revolution, Verification and Validation, Failure, Error, Fault, and Defect, Objectives of Testing, Testing Activities, Test Case Selection White-Box and Black, test Planning and design, Test Tools and Automation, Power of Test. Test Team Organization and Management-Test Groups, Software Quality Assurance Group, System Test Team Hierarchy, Team Building.

PRACTICAL:

- Conduct a test suite for any two WEB LINKS.
- Program to Select Checkbox in Selenium Driver.

UNIT II: SYSTEM TESTING **9T+6P = 15 HOURS**

System Testing - System Integration Techniques-Incremental, Top-Down Bottom-Up Sandwich and Big Bang, Software and Hardware Integration, Hardware Design Verification Tests, Hardware and Software Compatibility Matrix Test Plan for System Integration. Built- in Testing. functional testing - Testing a Function in Context. Boundary Value Analysis, Decision Tables. Acceptance testing -

Selection of Acceptance Criteria, Acceptance Test Plan, Test Execution Test. software reliability - Fault and Failure, Factors Influencing Software, Reliability Models.

PRACTICAL:

- Program to Select Radio Button in Selenium Driver.
- Program to Scroll a web page in Selenium WebDriver.

UNIT III: SYSTEM TEST CATEGORIES **9T+6P = 15 HOURS**

System test categories Taxonomy of System Tests, Interface Tests Functionality Tests. GUI Tests, Security Tests Feature Tests, Robustness Tests, Boundary Value Tests Power Cycling Tests Interoperability Tests, Scalability Tests, Stress Tests, Load and Stability Tests, Reliability Tests, Regression Tests, Regulatory Tests. Test Generation from FSM models- State-Oriented Model. Finite-State Machine Transition Tour Method, Testing with State Verification. Test Architectures-Local, distributed, Coordinated, Remote. system test design- Test Design Factors Requirement Identification, modeling a Test Design Process Test Design Preparedness, Metrics, Test Case Design Effectiveness. system test execution- Modeling Defects, Metrics for Monitoring Test Execution. Defect Reports, Defect Causal Analysis, Beta testing, measuring Test Effectiveness.

PRACTICAL:

- Create Locators in Selenium using IDE.
- Design and execute GUI testing for a login or registration system.

UNIT IV: SOFTWARE QUALITY **9T+6P = 15 HOURS**

Software quality - People ‘s Quality Expectations, Frameworks and ISO-9126, McCall ‘s Quality Factors and Criteria – Relationship. Quality Metrics. Quality Characteristics ISO 9000:2000 Software Quality Standard. Maturity models- Test Process Improvement, Testing Maturity Model-Estimation Techniques.

PRACTICAL:

- Selenium Program for Test Process Validation
- Create Locators in Selenium using IDE.
- Find Element and Find Elements in Web using Selenium WebDriver.

UNIT V: SOFTWARE QUALITY ASSURANCE **9T+6P = 15 HOURS**

Quality Assurance - Root Cause Analysis, modeling, technologies, standards and methodologies for defect prevention. Fault Tolerance and Failure Containment - Safety Assurance and Damage Control, Hazard analysis using fault-trees and event-trees. Comparing Quality Assurance Techniques and Activities. QA Monitoring and Measurement, Risk Identification for Quantifiable Quality Improvement. Case Study: FSM-Based Testing of Web-Based Applications.

PRACTICAL:

- Write a Program to Scroll a web page in Selenium WebDriver.
- Write and test a program to login a specific webpage.
- Conduct a test suite for any two WEB LINKS.

TOTAL: 75 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
Testim	Uses AI to create stable automated tests
Functionize	Natural language test creation
Applitools	Detects UI bugs using visual comparison

TEXT BOOKS:

1. Stephan Goericke.(2019). *"The Future of Software Quality Assurance",* Stephan Goericke,2nd Edition, Springer International Publishing.
2. Sandeep Desai, Abhishek Srivastava. (2018). "*Software Testing: A Practical Approach*", Second Edition, PHI, New Delhi.
3. Kshirasagar Nak Priyadarshi Tripathy. (2017). *"Software Testing and Quality Assurance-Theory and Practice"*, Second Edition, John Wiley & Sons Inc.

REFERENCE BOOKS:

1. Milind Limaye. (2016). *"Software Quality Assurance"*, Second Edition, TMH, New Delhi.
2. Jeff Tian. (2016). *"Software Quality Engineering: Testing, Quality Assurance, and Quantifiable Improvement"*, Second Edition, John Wiley & Sons, Inc., Hoboken, New Jerse.
3. Daniel Galin. (2016). *"Software Quality Assurance - From Theory to Implementation",*

WEBSITES:

1. https://www.tutorialspoint.com/software_testing/software_testing_qa_qc_testing.htm
2. https://www.altexsoft.com/whitepapers/quality-assurance-quality-control-and-testing-the-basics-of-software-quality-management/
3. http://softwaretestingfundamentals.com/software-quality-assurance/
4. https://www.softwaretestinghelp.com/software-quality-assurance/
5. https://nptel.ac.in/courses/106105150/
6. http://ijiet.com/wp-content/uploads/2016/04/15.pdf
7. https://www.springer.com/journal/11219

JOURNALS:

1. Manuela Tuteja, Gaurav Dubey " A Research Study on importance of Testing and Quality Assurance in Software Development Life Cycle (SDLC) Models." International Journal of Soft Computing and Engineering (IJSCE) ISSN: 2231-2307, Volume-2, Issue-3, July 2012

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**2**	-	**3**	-	**3**	-	**1**	-	-	-	-	-	-	-	-	**3**	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	**2**
CO3	-	-	**3**	**2**	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	**2**	-	-	-	-	**1**	-	-	-	-	-	-	-	-	-	-	-
CO5	-	-	**3**	**2**	**3**	-	-	-	-	-	-	-	**1**	-	-	-	-
Average	**2**	-	**3**	**2**	**3**	**1**	**1**	-	-	-	-	-	**1**	-	-	**3**	**2**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major **SEMESTER III**

26CSP343 **Natural Language Processing** **5H - 4C**

Instruction Hours/week: L:3 T:0 P:2 **Marks: Internal:50 External:50 Total:100**

End Semester Exam: 3T+2P=5 Hours

PREREQUISTICS:

- Basic knowledge of Python programming, probability, statistics, machine learning fundamentals, and linguistic concepts.

COURSE OBJECTIVES (CO):

- To understand and apply fundamental concepts and techniques in natural language processing.
- To implement morphological, syntactic, and semantic analysis for real-world text processing.
- To make students to develop NLP applications using machine learning and deep learning models with modern NLP tools.

COURSE OUTCOMES (COs):

Upon the completion of this course, the students will be able to:

COs	Course Outcomes	Blooms Level
CO1	Apply NLP concepts such as phonology, morphology, syntax, and semantics to perform basic text processing tasks using Python.	K3
CO2	Apply morphological, syntactic, and parsing techniques for processing and analyzing natural language data.	K3
CO3	Analyze and implement semantic processing techniques, including word sense disambiguation and knowledge representation.	K4
CO4	Demonstrate the ability to work with discourse structures, coreference resolution, and pragmatic aspects in NLP tasks.	K3
CO5	Analyze NLP techniques used in machine translation, question answering systems, and chatbots to evaluate their performance in real-world applications.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create		

UNIT I: INTRODUCTION **9T+6P = 15 HOURS**

Introduction to NLP- Applications of NLP- Challenges in NLP- Natural Language Components: Phonology, Morphology, Syntax, Semantics, Pragmatics- NLP Approaches: Rule-based, Statistical, and Neural Network-based- Overview of Linguistic Resources – Corpora, WordNet, FrameNet.

PRACTICAL:

- Python program to demonstrate NLP Tokenization
- Python program for implementation of Word Frequency

UNIT II: **MORPHOLOGICAL AND SYNTACTIC ANALYSIS 9T+6P = 15 HOURS**

Morphological Processing: Inflection, Derivation- Lemmatization and Stemming- Part-of-Speech (POS) Tagging – Rule-based, HMM, and Neural Models- Syntax and Parsing Techniques- Context-Free Grammars (CFG)- Dependency Parsing- Transition-based Parsing.

PRACTICAL:

- Python program for Stemming and Lemmatization
- Python program for POS Tagging

UNIT III: SEMANTIC ANALYSIS AND REPRESENTATION 9T+6P = 15 HOURS

Word Sense Disambiguation (WSD)- Lexical Semantics – Synonymy, Antonymy, Polysemy- Semantic Role Labeling- Knowledge Representation and Meaning- Ontologies in NLP- Ambiguity in Natural Language- Distributional Semantics and Word Embeddings (Word2Vec, GloVe)

PRACTICAL:

- Synonym and Antonym Finder using WordNet
- Word Embedding using Word2Vec

UNIT IV: DISCOURSE AND PRAGMATICS 9T+6P = 15 HOURS

Discourse Structure and Coherence- Coreference Resolution- Dialogue Systems- Discourse Parsing- Pragmatics: Speech Acts, Implicature- Sentiment Analysis- Emotion Detection.

PRACTICAL:

- Python program for Sentiment Analysis
- Implement a Program for Text-Based Emotion Analysis
- Simple Chatbot using Dialogue System

UNIT V: APPLICATIONS AND RECENT TRENDS IN NLP 9T+6P = 15 HOURS

Machine Translation (MT) – Statistical and Neural Models- Question Answering Systems- Text Summarization – Extractive and Abstractive Methods- Text Classification and Clustering- Chatbots and Conversational Agents.

PRACTICAL:

- Python program for Text Summarization
- Implement a python program for Text Classification
- Text Clustering using K-Means

TOTAL: 75 HOURS

SUGGESTED AI TOOLS / SOFTWARE TOOLS:

AI Tools / Software Tools	Purpose
NLTK (Natural Language Toolkit)	Beginner-friendly Python library
Stanford CoreNLP	Complete NLP suite (POS tagging, NER, parsing)
Apache OpenNLP	Java-based ML NLP toolkit

TEXT BOOKS:

1. Daniel Jurafsky and James H.Martin, (2023), "*Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition",* Pearson Education.

2. Daniel Jurafsky, James H. Martin (2014), "*Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech*", Pearson Publication.
3. T V Geetha (2024), "*Understanding Natural Language Processing – Machine Learning and Deep Learning Perspectives*", Pearson.

REFERENCE BOOKS:

1. M. Bikel and Imed Zitouni, Daniel, "*Multilingual natural Language Processing Applications: From Theory to Practice",* Pearson Publication, 2012.
2. Akshay Kulkarni, Adarsha Shivananda, "*Natural Language Processing Recipes– Unlocking Text Data with Machine Learning and Deep Learning using Python*", Apress. 2021.

WEBSITES:

1. http://web.stanford.edu/class/cs224n/
2. https://www.pvierth.com/pca
3. https://link-springer-com.proxy.uchicago.edu/book/10.1007%2F978-1-4842-2388-8
4. http://www.nltk.org/book/
5. https://archive.nptel.ac.in/courses/106/106/106106211/
6. https://onlinecourses.nptel.ac.in/noc23_cs45/preview

JOURNALS:

1. Khurana, D., Koli, A., Khatter, K. et al. Natural language processing: state of the art, current trends and challenges. Multimed Tools Appl 82, 3713–3744 (2023). https://doi.org/10.1007/s1 042-022-13428-4

CO, PO, PSO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15	PSO1	PSO2
CO1	**1**	-	-	-	-	-	-	-	-	-	-	-	-	-	-	**1**	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	**1**
CO3	-	-	-	**3**	-	**1**	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	**3**	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	-	-	**3**	**3**	-	-	**1**	-	-	-	-	-	-	-	-	-	-
Average	**1**	-	**3**	**3**	-	**1**	**1**	-	-	-	-	-	-	-	-	**1**	**1**

1 - Low, 2 - Medium, 3 - High, '-' - No Correlation

Category: Major		**SEMESTER III**
26CSP391	**Internship**	**0H - 2C**
Instruction Hours/week: L:0 T:0 P:0		**Marks: Internal:100 External:- Total:100**
		End Semester Exam: -

Category: Project		**SEMESTER IV**
26CSP491	**Project and Viva Voce**	**0H - 15C**
Instruction Hours/week: L:0 T:0 P:0		**Marks: Internal:80 External: 120 Total:200**
		End Semester Exam: 3 Hours